# MS-GARCH Likelihood Approximation Study

## Complete Model Parameter Reference

Joshua Ogundairo[a] and Oluwadare O. Ojo[a]

[a]Department of Statistics, Federal University of Technology, Akure (FUTA), Nigeria

ogundairosta154470@futa.edu.ng ojooo@futa.edu.ng

Companion document to: *Impact of Likelihood Approximation Choices on the Performance of MS-GARCH Models*

June 2026

---

**About this document.** This file contains the complete estimated parameter sets for all 220 MS-GARCH(1,1) model runs reported in the companion article. Models are organized by market archetype (Section 1–5), then by asset, and within each asset by specification (Baseline and Exogenous) and approximation method (Gray's collapsing, Plug-in, Gaussian Quadrature, Particle Filter). Parameter tables present the twelve-element vector $\Psi = \{c_1, c_2, \gamma_1, \gamma_2, \omega_1, \alpha_1, \beta_1, \omega_2, \alpha_2, \beta_2, p_{11}, p_{22}\}$ alongside the log-likelihood (LogLik) and convergence status (Conv.). Equation blocks then render the fitted model explicitly for each method. Dashes (–) in the $\gamma$ columns indicate the baseline (internal-dynamics-only) specification where $\gamma_1 = \gamma_2 = 0$ by construction.

## Contents

# 1. Equity Market

Pivot (exogenous conditioning variable): `GSPC`.

## 1.1. Asset: `FTSE`

### Baseline Specification

**Table 1:** FTSE — Baseline Specification

| Method | $c_1$ | $c_2$ | $\gamma_1$ | $\gamma_2$ | $\omega_1$ | $\alpha_1$ | $\beta_1$ | $\omega_2$ | $\alpha_2$ | $\beta_2$ | $p_{11}$ | $p_{22}$ | Conv. | LogLik |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Gray's | 0.0720 | -0.0001 | – | – | 0.2687 | 0.3661 | 0.6337 | 0.0001 | 0.0001 | 0.0001 | 0.7607 | 0.4448 | Yes | -1014.99 |
| Plug-in | -0.0003 | 0.0355 | – | – | 0.0001 | 0.0001 | 0.0001 | 0.1062 | 0.2989 | 0.6954 | 0.5689 | 0.7553 | Yes | -753.01 |
| Quadrature | 0.0517 | -0.0001 | – | – | 0.2244 | 0.3836 | 0.6159 | 0.0001 | 0.0001 | 0.0001 | 0.7258 | 0.4785 | Yes | -1044.07 |
| Particle Filter | 0.0000 | 0.0000 | – | – | 0.1000 | 0.1000 | 0.7000 | 0.2000 | 0.2000 | 0.6000 | 0.9000 | 0.9000 | Yes | -3030.64 |

Baseline spec: $\gamma_1 = \gamma_2 = 0$. Conv. = convergence status.

### Fitted Equations (Baseline):

### Gray's Method

$$\mu_1 = 0.0720 \tag{1}$$
$$\mu_2 = -0.0001 \tag{2}$$

$$\sigma_t^2(1) = 0.2687 + 0.3661\, y_{t-1}^2 + 0.6337\, \sigma_{t-1}^2 \tag{3}$$
$$\sigma_t^2(2) = 0.0001 + 0.0001\, y_{t-1}^2 + 0.0001\, \sigma_{t-1}^2 \tag{4}$$

$$P = \begin{pmatrix} 0.7607 & 0.2393 \\ 0.5552 & 0.4448 \end{pmatrix} \tag{5}$$

LogLik $= -1014.99$; Converged: Yes.

### Plug-in Method

$$\mu_1 = -0.0003 \tag{6}$$
$$\mu_2 = 0.0355 \tag{7}$$

$$\sigma_t^2(1) = 0.0001 + 0.0001\, y_{t-1}^2 + 0.0001\, \sigma_{t-1}^2 \tag{8}$$
$$\sigma_t^2(2) = 0.1062 + 0.2989\, y_{t-1}^2 + 0.6954\, \sigma_{t-1}^2 \tag{9}$$

$$P = \begin{pmatrix} 0.5689 & 0.4311 \\ 0.2447 & 0.7553 \end{pmatrix} \tag{10}$$

LogLik $= -753.01$; Converged: Yes.

### Quadrature Method

$$\mu_1 = 0.0517 \tag{11}$$
$$\mu_2 = -0.0001 \tag{12}$$

$$\sigma_t^2(1) = 0.2244 + 0.3836\, y_{t-1}^2 + 0.6159\, \sigma_{t-1}^2 \tag{13}$$
$$\sigma_t^2(2) = 0.0001 + 0.0001\, y_{t-1}^2 + 0.0001\, \sigma_{t-1}^2 \tag{14}$$

$$P = \begin{pmatrix} 0.7258 & 0.2742 \\ 0.5215 & 0.4785 \end{pmatrix} \tag{15}$$

LogLik $= -1044.07$; Converged: Yes.

### Particle Filter Method

$$\mu_1 = 0.0000 \tag{16}$$
$$\mu_2 = 0.0000 \tag{17}$$

$$\sigma_t^2(1) = 0.1000 + 0.1000\, y_{t-1}^2 + 0.7000\, \sigma_{t-1}^2 \tag{18}$$
$$\sigma_t^2(2) = 0.2000 + 0.2000\, y_{t-1}^2 + 0.6000\, \sigma_{t-1}^2 \tag{19}$$

$$P = \begin{pmatrix} 0.9000 & 0.1000 \\ 0.1000 & 0.9000 \end{pmatrix} \tag{20}$$

LogLik $= -3030.64$; Converged: Yes.

### Exogenous Specification

**Table 2:** FTSE — Exogenous Specification

| **Method** | $c_1$ | $c_2$ | $\gamma_1$ | $\gamma_2$ | $\omega_1$ | $\alpha_1$ | $\beta_1$ | $\omega_2$ | $\alpha_2$ | $\beta_2$ | $p_{11}$ | $p_{22}$ | Conv. | Log |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Gray's | -0.0213 | -0.0001 | 0.2146 | -0.0002 | 0.2535 | 0.3391 | 0.6609 | 0.0001 | 0.0001 | 0.0001 | 0.7389 | 0.4627 | Yes | -985 |
| Plug-in | -0.0001 | 0.0219 | 0.0006 | 0.1418 | 0.0001 | 0.0001 | 0.0001 | 0.1021 | 0.2768 | 0.7229 | 0.4883 | 0.7502 | Yes | -726 |
| Quadrature | 0.0917 | -0.0001 | 0.1461 | -0.0000 | 0.1861 | 0.3089 | 0.6911 | 0.0001 | 0.0001 | 0.0001 | 0.7506 | 0.4853 | Yes | -1012 |
| Particle Filter | 0.0000 | 0.0000 | 0.1000 | 0.1000 | 0.1000 | 0.1000 | 0.7000 | 0.2000 | 0.2000 | 0.6000 | 0.9000 | 0.9000 | No | -3006 |

Exogenous spec; pivot = $^{G}SPC.Conv. = convergencestatus.$

### Fitted Equations (Exogenous):

### Gray's Method

$$\mu_1 = -0.0213 + 0.2146\, x_t \tag{21}$$
$$\mu_2 = -0.0001 - 0.0002\, x_t \tag{22}$$

$$\sigma_t^2(1) = 0.2535 + 0.3391\,y_{t-1}^2 + 0.6609\,\sigma_{t-1}^2 \tag{23}$$
$$\sigma_t^2(2) = 0.0001 + 0.0001\,y_{t-1}^2 + 0.0001\,\sigma_{t-1}^2 \tag{24}$$

$$P = \begin{pmatrix} 0.7389 & 0.2611 \\ 0.5373 & 0.4627 \end{pmatrix} \tag{25}$$

LogLik $= -985.96$; Converged: Yes.

**Plug-in Method**

$$\mu_1 = -0.0001 + 0.0006\,x_t \tag{26}$$
$$\mu_2 = 0.0219 + 0.1418\,x_t \tag{27}$$

$$\sigma_t^2(1) = 0.0001 + 0.0001\,y_{t-1}^2 + 0.0001\,\sigma_{t-1}^2 \tag{28}$$
$$\sigma_t^2(2) = 0.1021 + 0.2768\,y_{t-1}^2 + 0.7229\,\sigma_{t-1}^2 \tag{29}$$

$$P = \begin{pmatrix} 0.4883 & 0.5117 \\ 0.2498 & 0.7502 \end{pmatrix} \tag{30}$$

LogLik $= -726.36$; Converged: Yes.

**Quadrature Method**

$$\mu_1 = 0.0917 + 0.1461\,x_t \tag{31}$$
$$\mu_2 = -0.0001 - 0.0000\,x_t \tag{32}$$

$$\sigma_t^2(1) = 0.1861 + 0.3089\,y_{t-1}^2 + 0.6911\,\sigma_{t-1}^2 \tag{33}$$
$$\sigma_t^2(2) = 0.0001 + 0.0001\,y_{t-1}^2 + 0.0001\,\sigma_{t-1}^2 \tag{34}$$

$$P = \begin{pmatrix} 0.7506 & 0.2494 \\ 0.5147 & 0.4853 \end{pmatrix} \tag{35}$$

LogLik $= -1012.80$; Converged: Yes.

**Particle Filter Method**

$$\mu_1 = 0.0000 + 0.1000\,x_t \tag{36}$$
$$\mu_2 = 0.0000 + 0.1000\,x_t \tag{37}$$

$$\sigma_t^2(1) = 0.1000 + 0.1000\,y_{t-1}^2 + 0.7000\,\sigma_{t-1}^2 \tag{38}$$
$$\sigma_t^2(2) = 0.2000 + 0.2000\,y_{t-1}^2 + 0.6000\,\sigma_{t-1}^2 \tag{39}$$

$$P = \begin{pmatrix} 0.9000 & 0.1000 \\ 0.1000 & 0.9000 \end{pmatrix} \tag{40}$$

LogLik $= -3006.88$; Converged: No.

## 1.2. Asset: `N225`

### Baseline Specification

**Table 3:** N225 — Baseline Specification

| Method | $c_1$ | $c_2$ | $\gamma_1$ | $\gamma_2$ | $\omega_1$ | $\alpha_1$ | $\beta_1$ | $\omega_2$ | $\alpha_2$ | $\beta_2$ | $p_{11}$ | $p_{22}$ | Conv. | LogLik |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Gray's | 0.0588 | 0.0001 | – | – | 0.5097 | 0.1881 | 0.8032 | 0.0001 | 0.0001 | 0.0001 | 0.7047 | 0.4934 | Yes | -1523.48 |
| Plug-in | -0.0002 | 0.0666 | – | – | 0.0001 | 0.0001 | 0.0001 | 0.1062 | 0.1389 | 0.8444 | 0.5296 | 0.7529 | Yes | -1183.10 |
| Quadrature | 0.1001 | 0.0003 | – | – | 0.6615 | 0.2244 | 0.7755 | 0.0001 | 0.0001 | 0.0001 | 0.6441 | 0.6348 | Yes | -1617.53 |
| Particle Filter | -0.0000 | -0.0000 | – | – | 0.1000 | 0.1000 | 0.7000 | 0.2000 | 0.2000 | 0.6000 | 0.9000 | 0.9000 | Yes | -3946.35 |

Baseline spec: $\gamma_1 = \gamma_2 = 0$. Conv. = convergence status.

### Fitted Equations (Baseline):

### Gray's Method

$$\mu_1 = 0.0588 \tag{41}$$
$$\mu_2 = 0.0001 \tag{42}$$

$$\sigma_t^2(1) = 0.5097 + 0.1881\, y_{t-1}^2 + 0.8032\, \sigma_{t-1}^2 \tag{43}$$
$$\sigma_t^2(2) = 0.0001 + 0.0001\, y_{t-1}^2 + 0.0001\, \sigma_{t-1}^2 \tag{44}$$

$$P = \begin{pmatrix} 0.7047 & 0.2953 \\ 0.5066 & 0.4934 \end{pmatrix} \tag{45}$$

LogLik $= -1523.48$; Converged: Yes.

### Plug-in Method

$$\mu_1 = -0.0002 \tag{46}$$
$$\mu_2 = 0.0666 \tag{47}$$

$$\sigma_t^2(1) = 0.0001 + 0.0001\, y_{t-1}^2 + 0.0001\, \sigma_{t-1}^2 \tag{48}$$
$$\sigma_t^2(2) = 0.1062 + 0.1389\, y_{t-1}^2 + 0.8444\, \sigma_{t-1}^2 \tag{49}$$

$$P = \begin{pmatrix} 0.5296 & 0.4704 \\ 0.2471 & 0.7529 \end{pmatrix} \tag{50}$$

LogLik $= -1183.10$; Converged: Yes.

### Quadrature Method

$$\mu_1 = 0.1001 \tag{51}$$
$$\mu_2 = 0.0003 \tag{52}$$

$$\sigma_t^2(1) = 0.6615 + 0.2244\, y_{t-1}^2 + 0.7755\, \sigma_{t-1}^2 \tag{53}$$
$$\sigma_t^2(2) = 0.0001 + 0.0001\, y_{t-1}^2 + 0.0001\, \sigma_{t-1}^2 \tag{54}$$

$$P = \begin{pmatrix} 0.6441 & 0.3559 \\ 0.3652 & 0.6348 \end{pmatrix} \tag{55}$$

LogLik $= -1617.53$; Converged: Yes.

### Particle Filter Method

$$\mu_1 = -0.0000 \tag{56}$$
$$\mu_2 = -0.0000 \tag{57}$$

$$\sigma_t^2(1) = 0.1000 + 0.1000\, y_{t-1}^2 + 0.7000\, \sigma_{t-1}^2 \tag{58}$$
$$\sigma_t^2(2) = 0.2000 + 0.2000\, y_{t-1}^2 + 0.6000\, \sigma_{t-1}^2 \tag{59}$$

$$P = \begin{pmatrix} 0.9000 & 0.1000 \\ 0.1000 & 0.9000 \end{pmatrix} \tag{60}$$

LogLik $= -3946.35$; Converged: Yes.

### Exogenous Specification

**Table 4:** N225 — Exogenous Specification

| Method | $c_1$ | $c_2$ | $\gamma_1$ | $\gamma_2$ | $\omega_1$ | $\alpha_1$ | $\beta_1$ | $\omega_2$ | $\alpha_2$ | $\beta_2$ | $p_{11}$ | $p_{22}$ | Conv. | Log |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Gray's | -0.0379 | -0.0429 | 0.4057 | 0.2095 | 0.7990 | 0.0975 | 0.9000 | 0.0253 | 0.1172 | 0.0074 | 0.2865 | 0.3852 | No | -3164 |
| Plug-in | 0.0257 | -0.0001 | 0.6279 | -0.0002 | 0.1280 | 0.1767 | 0.7711 | 0.0001 | 0.0001 | 0.0001 | 0.7530 | 0.5292 | Yes | -958 |
| Quadrature | 0.0163 | -0.0003 | 0.6281 | 0.0001 | 0.2268 | 0.1732 | 0.8266 | 0.0001 | 0.0001 | 0.0001 | 0.7203 | 0.4536 | Yes | -1282 |
| Particle Filter | -0.0000 | -0.0000 | 0.1000 | 0.1000 | 0.1000 | 0.1000 | 0.7000 | 0.2000 | 0.2000 | 0.6000 | 0.9000 | 0.9000 | Yes | -3844 |

Exogenous spec; pivot = $^{G}SPC.Conv. = convergencestatus.$

### Fitted Equations (Exogenous):

### Gray's Method

$$\mu_1 = -0.0379 + 0.4057\, x_t \tag{61}$$
$$\mu_2 = -0.0429 + 0.2095\, x_t \tag{62}$$

$$\sigma_t^2(1) = 0.7990 + 0.0975\, y_{t-1}^2 + 0.9000\, \sigma_{t-1}^2 \tag{63}$$
$$\sigma_t^2(2) = 0.0253 + 0.1172\, y_{t-1}^2 + 0.0074\, \sigma_{t-1}^2 \tag{64}$$

$$P = \begin{pmatrix} 0.2865 & 0.7135 \\ 0.6148 & 0.3852 \end{pmatrix} \tag{65}$$

LogLik $= -3164.01$; Converged: No.

**Plug-in Method**

$$\mu_1 = 0.0257 + 0.6279\, x_t \tag{66}$$
$$\mu_2 = -0.0001 - 0.0002\, x_t \tag{67}$$

$$\sigma_t^2(1) = 0.1280 + 0.1767\, y_{t-1}^2 + 0.7711\, \sigma_{t-1}^2 \tag{68}$$
$$\sigma_t^2(2) = 0.0001 + 0.0001\, y_{t-1}^2 + 0.0001\, \sigma_{t-1}^2 \tag{69}$$

$$P = \begin{pmatrix} 0.7530 & 0.2470 \\ 0.4708 & 0.5292 \end{pmatrix} \tag{70}$$

LogLik $= -958.26$; Converged: Yes.

**Quadrature Method**

$$\mu_1 = 0.0163 + 0.6281\, x_t \tag{71}$$
$$\mu_2 = -0.0003 + 0.0001\, x_t \tag{72}$$

$$\sigma_t^2(1) = 0.2268 + 0.1732\, y_{t-1}^2 + 0.8266\, \sigma_{t-1}^2 \tag{73}$$
$$\sigma_t^2(2) = 0.0001 + 0.0001\, y_{t-1}^2 + 0.0001\, \sigma_{t-1}^2 \tag{74}$$

$$P = \begin{pmatrix} 0.7203 & 0.2797 \\ 0.5464 & 0.4536 \end{pmatrix} \tag{75}$$

LogLik $= -1282.22$; Converged: Yes.

**Particle Filter Method**

$$\mu_1 = -0.0000 + 0.1000\, x_t \tag{76}$$
$$\mu_2 = -0.0000 + 0.1000\, x_t \tag{77}$$

$$\sigma_t^2(1) = 0.1000 + 0.1000\, y_{t-1}^2 + 0.7000\, \sigma_{t-1}^2 \tag{78}$$
$$\sigma_t^2(2) = 0.2000 + 0.2000\, y_{t-1}^2 + 0.6000\, \sigma_{t-1}^2 \tag{79}$$

$$P = \begin{pmatrix} 0.9000 & 0.1000 \\ 0.1000 & 0.9000 \end{pmatrix} \tag{80}$$

LogLik $= -3844.16$; Converged: Yes.

## 1.3. Asset: `IXIC`

### Baseline Specification

**Table 5:** IXIC — Baseline Specification

| Method | $c_1$ | $c_2$ | $\gamma_1$ | $\gamma_2$ | $\omega_1$ | $\alpha_1$ | $\beta_1$ | $\omega_2$ | $\alpha_2$ | $\beta_2$ | $p_{11}$ | $p_{22}$ | Conv. | LogLik |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Gray's | -0.4857 | 0.1456 | – | – | 0.6551 | 0.0766 | 0.9000 | 0.0001 | 0.0001 | 0.7559 | 0.8691 | 0.9569 | Yes | -4070.86 |
| Plug-in | 0.0001 | 0.1090 | – | – | 0.0001 | 0.0001 | 0.0004 | 0.1851 | 0.2735 | 0.7206 | 0.5118 | 0.7568 | No | -1647.97 |
| Quadrature | 0.0232 | 0.1347 | – | – | 0.2404 | 0.1729 | 0.8240 | 0.0135 | 0.0737 | 0.6214 | 0.8661 | 0.8884 | No | -4186.05 |
| Particle Filter | 0.0000 | 0.0000 | – | – | 0.3000 | 0.0500 | 0.8500 | 0.1000 | 0.0800 | 0.7800 | 0.9500 | 0.8000 | No | -4274.18 |

Baseline spec: $\gamma_1 = \gamma_2 = 0$. Conv. = convergence status.

### Fitted Equations (Baseline):

### Gray's Method

$$\mu_1 = -0.4857 \tag{81}$$
$$\mu_2 = 0.1456 \tag{82}$$

$$\sigma_t^2(1) = 0.6551 + 0.0766\, y_{t-1}^2 + 0.9000\, \sigma_{t-1}^2 \tag{83}$$
$$\sigma_t^2(2) = 0.0001 + 0.0001\, y_{t-1}^2 + 0.7559\, \sigma_{t-1}^2 \tag{84}$$

$$P = \begin{pmatrix} 0.8691 & 0.1309 \\ 0.0431 & 0.9569 \end{pmatrix} \tag{85}$$

LogLik $= -4070.86$; Converged: Yes.

### Plug-in Method

$$\mu_1 = 0.0001 \tag{86}$$
$$\mu_2 = 0.1090 \tag{87}$$

$$\sigma_t^2(1) = 0.0001 + 0.0001\, y_{t-1}^2 + 0.0004\, \sigma_{t-1}^2 \tag{88}$$
$$\sigma_t^2(2) = 0.1851 + 0.2735\, y_{t-1}^2 + 0.7206\, \sigma_{t-1}^2 \tag{89}$$

$$P = \begin{pmatrix} 0.5118 & 0.4882 \\ 0.2432 & 0.7568 \end{pmatrix} \tag{90}$$

LogLik $= -1647.97$; Converged: No.

### Quadrature Method

$$\mu_1 = 0.0232 \tag{91}$$
$$\mu_2 = 0.1347 \tag{92}$$

$$\sigma_t^2(1) = 0.2404 + 0.1729\, y_{t-1}^2 + 0.8240\, \sigma_{t-1}^2 \tag{93}$$
$$\sigma_t^2(2) = 0.0135 + 0.0737\, y_{t-1}^2 + 0.6214\, \sigma_{t-1}^2 \tag{94}$$

$$P = \begin{pmatrix} 0.8661 & 0.1339 \\ 0.1116 & 0.8884 \end{pmatrix} \tag{95}$$

LogLik $= -4186.05$; Converged: No.

### Particle Filter Method

$$\mu_1 = 0.0000 \tag{96}$$
$$\mu_2 = 0.0000 \tag{97}$$

$$\sigma_t^2(1) = 0.3000 + 0.0500\, y_{t-1}^2 + 0.8500\, \sigma_{t-1}^2 \tag{98}$$
$$\sigma_t^2(2) = 0.1000 + 0.0800\, y_{t-1}^2 + 0.7800\, \sigma_{t-1}^2 \tag{99}$$

$$P = \begin{pmatrix} 0.9500 & 0.0500 \\ 0.2000 & 0.8000 \end{pmatrix} \tag{100}$$

LogLik $= -4274.18$; Converged: No.

### Exogenous Specification

**Table 6:** IXIC — Exogenous Specification

| **Method** | $c_1$ | $c_2$ | $\gamma_1$ | $\gamma_2$ | $\omega_1$ | $\alpha_1$ | $\beta_1$ | $\omega_2$ | $\alpha_2$ | $\beta_2$ | $p_{11}$ | $p_{22}$ | Conv. | LogL |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Gray's | 0.0509 | 0.1285 | -0.0714 | -0.0721 | 0.2420 | 0.1588 | 0.8411 | 0.0001 | 0.0949 | 0.6968 | 0.8581 | 0.8672 | No | -4155. |
| Plug-in | 0.0001 | 0.0711 | -0.0013 | -0.1073 | 0.0001 | 0.0001 | 0.0004 | 0.1963 | 0.2551 | 0.7449 | 0.4497 | 0.7301 | Yes | -1647. |
| Quadrature | 0.1181 | 0.0718 | -0.2952 | -0.0036 | 0.5152 | 0.2517 | 0.7483 | 0.0871 | 0.1286 | 0.3366 | 0.8120 | 0.8530 | Yes | -4192. |
| Particle Filter | 0.0000 | 0.0000 | 0.0500 | 0.0500 | 0.3000 | 0.0500 | 0.8500 | 0.1000 | 0.0800 | 0.7800 | 0.9500 | 0.8000 | Yes | -4279. |

Exogenous spec; pivot = $^{G}SPC.Conv. = convergencestatus.$

### Fitted Equations (Exogenous):

### Gray's Method

$$\mu_1 = 0.0509 - 0.0714\, x_t \tag{101}$$
$$\mu_2 = 0.1285 - 0.0721\, x_t \tag{102}$$

$$\sigma_t^2(1) = 0.2420 + 0.1588\, y_{t-1}^2 + 0.8411\, \sigma_{t-1}^2 \tag{103}$$
$$\sigma_t^2(2) = 0.0001 + 0.0949\, y_{t-1}^2 + 0.6968\, \sigma_{t-1}^2 \tag{104}$$

$$P = \begin{pmatrix} 0.8581 & 0.1419 \\ 0.1328 & 0.8672 \end{pmatrix} \tag{105}$$

LogLik $= -4155.59$; Converged: No.

**Plug-in Method**

$$\mu_1 = 0.0001 - 0.0013\, x_t \tag{106}$$
$$\mu_2 = 0.0711 - 0.1073\, x_t \tag{107}$$

$$\sigma_t^2(1) = 0.0001 + 0.0001\, y_{t-1}^2 + 0.0004\, \sigma_{t-1}^2 \tag{108}$$
$$\sigma_t^2(2) = 0.1963 + 0.2551\, y_{t-1}^2 + 0.7449\, \sigma_{t-1}^2 \tag{109}$$

$$P = \begin{pmatrix} 0.4497 & 0.5503 \\ 0.2699 & 0.7301 \end{pmatrix} \tag{110}$$

LogLik $= -1647.60$; Converged: Yes.

**Quadrature Method**

$$\mu_1 = 0.1181 - 0.2952\, x_t \tag{111}$$
$$\mu_2 = 0.0718 - 0.0036\, x_t \tag{112}$$

$$\sigma_t^2(1) = 0.5152 + 0.2517\, y_{t-1}^2 + 0.7483\, \sigma_{t-1}^2 \tag{113}$$
$$\sigma_t^2(2) = 0.0871 + 0.1286\, y_{t-1}^2 + 0.3366\, \sigma_{t-1}^2 \tag{114}$$

$$P = \begin{pmatrix} 0.8120 & 0.1880 \\ 0.1470 & 0.8530 \end{pmatrix} \tag{115}$$

LogLik $= -4192.55$; Converged: Yes.

**Particle Filter Method**

$$\mu_1 = 0.0000 + 0.0500\, x_t \tag{116}$$
$$\mu_2 = 0.0000 + 0.0500\, x_t \tag{117}$$

$$\sigma_t^2(1) = 0.3000 + 0.0500\, y_{t-1}^2 + 0.8500\, \sigma_{t-1}^2 \tag{118}$$
$$\sigma_t^2(2) = 0.1000 + 0.0800\, y_{t-1}^2 + 0.7800\, \sigma_{t-1}^2 \tag{119}$$

$$P = \begin{pmatrix} 0.9500 & 0.0500 \\ 0.2000 & 0.8000 \end{pmatrix} \tag{120}$$

LogLik $= -4279.14$; Converged: Yes.

### 1.4. Asset: `EEM`

### Baseline Specification

**Table 7:** EEM — Baseline Specification

| Method | $c_1$ | $c_2$ | $\gamma_1$ | $\gamma_2$ | $\omega_1$ | $\alpha_1$ | $\beta_1$ | $\omega_2$ | $\alpha_2$ | $\beta_2$ | $p_{11}$ | $p_{22}$ | Conv. | LogLik |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Gray's | -0.0113 | 0.0050 | – | – | 0.7411 | 0.1723 | 0.8276 | 0.0001 | 0.0001 | 0.1966 | 0.6085 | 0.6493 | No | -3672.50 |
| Plug-in | 0.0353 | -0.0000 | – | – | 0.0657 | 0.1191 | 0.8789 | 0.0001 | 0.0001 | 0.0001 | 0.7691 | 0.5141 | Yes | -1451.69 |
| Quadrature | 0.0501 | 0.0001 | – | – | 0.3670 | 0.1404 | 0.8589 | 0.0001 | 0.0001 | 0.0001 | 0.7578 | 0.4318 | Yes | -1804.79 |
| Particle Filter | -0.0000 | -0.0000 | – | – | 0.1000 | 0.1000 | 0.7000 | 0.2000 | 0.2000 | 0.6000 | 0.9000 | 0.9000 | Yes | -3993.87 |

Baseline spec: $\gamma_1 = \gamma_2 = 0$. Conv. = convergence status.

### Fitted Equations (Baseline):

### Gray's Method

$$\mu_1 = -0.0113 \tag{121}$$
$$\mu_2 = 0.0050 \tag{122}$$

$$\sigma_t^2(1) = 0.7411 + 0.1723\, y_{t-1}^2 + 0.8276\, \sigma_{t-1}^2 \tag{123}$$
$$\sigma_t^2(2) = 0.0001 + 0.0001\, y_{t-1}^2 + 0.1966\, \sigma_{t-1}^2 \tag{124}$$

$$P = \begin{pmatrix} 0.6085 & 0.3915 \\ 0.3507 & 0.6493 \end{pmatrix} \tag{125}$$

LogLik $= -3672.50$; Converged: No.

### Plug-in Method

$$\mu_1 = 0.0353 \tag{126}$$
$$\mu_2 = -0.0000 \tag{127}$$

$$\sigma_t^2(1) = 0.0657 + 0.1191\, y_{t-1}^2 + 0.8789\, \sigma_{t-1}^2 \tag{128}$$
$$\sigma_t^2(2) = 0.0001 + 0.0001\, y_{t-1}^2 + 0.0001\, \sigma_{t-1}^2 \tag{129}$$

$$P = \begin{pmatrix} 0.7691 & 0.2309 \\ 0.4859 & 0.5141 \end{pmatrix} \tag{130}$$

LogLik $= -1451.69$; Converged: Yes.

### Quadrature Method

$$\mu_1 = 0.0501 \tag{131}$$
$$\mu_2 = 0.0001 \tag{132}$$

$$\sigma_t^2(1) = 0.3670 + 0.1404\, y_{t-1}^2 + 0.8589\, \sigma_{t-1}^2 \tag{133}$$
$$\sigma_t^2(2) = 0.0001 + 0.0001\, y_{t-1}^2 + 0.0001\, \sigma_{t-1}^2 \tag{134}$$

$$P = \begin{pmatrix} 0.7578 & 0.2422 \\ 0.5682 & 0.4318 \end{pmatrix} \tag{135}$$

LogLik $= -1804.79$; Converged: Yes.

### Particle Filter Method

$$\mu_1 = -0.0000 \tag{136}$$
$$\mu_2 = -0.0000 \tag{137}$$

$$\sigma_t^2(1) = 0.1000 + 0.1000\, y_{t-1}^2 + 0.7000\, \sigma_{t-1}^2 \tag{138}$$
$$\sigma_t^2(2) = 0.2000 + 0.2000\, y_{t-1}^2 + 0.6000\, \sigma_{t-1}^2 \tag{139}$$

$$P = \begin{pmatrix} 0.9000 & 0.1000 \\ 0.1000 & 0.9000 \end{pmatrix} \tag{140}$$

LogLik $= -3993.87$; Converged: Yes.

### Exogenous Specification

**Table 8:** EEM — Exogenous Specification

| **Method** | $c_1$ | $c_2$ | $\gamma_1$ | $\gamma_2$ | $\omega_1$ | $\alpha_1$ | $\beta_1$ | $\omega_2$ | $\alpha_2$ | $\beta_2$ | $p_{11}$ | $p_{22}$ | Conv. | Log |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Gray's | -0.0112 | 0.0002 | -0.0946 | 0.0003 | 0.4381 | 0.1825 | 0.8134 | 0.0001 | 0.0001 | 0.0001 | 0.7508 | 0.4697 | Yes | -1777 |
| Plug-in | -0.0000 | 0.0350 | -0.0001 | -0.0749 | 0.0001 | 0.0001 | 0.0001 | 0.0643 | 0.1195 | 0.8796 | 0.5192 | 0.7688 | Yes | -1448 |
| Quadrature | -0.1344 | 0.0001 | -0.2235 | -0.0039 | 0.8549 | 0.6400 | 0.3581 | 0.0001 | 0.1089 | 0.0001 | 0.5021 | 0.4844 | Yes | -2441 |
| Particle Filter | 0.0000 | 0.0000 | 0.1000 | 0.1000 | 0.1000 | 0.1000 | 0.7000 | 0.2000 | 0.2000 | 0.6000 | 0.9000 | 0.9000 | Yes | -4011 |

Exogenous spec; pivot = ${}^{G}SPC.Conv. = convergencestatus.$

### Fitted Equations (Exogenous):

### Gray's Method

$$\mu_1 = -0.0112 - 0.0946\, x_t \tag{141}$$
$$\mu_2 = 0.0002 + 0.0003\, x_t \tag{142}$$

$$\sigma_t^2(1) = 0.4381 + 0.1825\, y_{t-1}^2 + 0.8134\, \sigma_{t-1}^2 \tag{143}$$
$$\sigma_t^2(2) = 0.0001 + 0.0001\, y_{t-1}^2 + 0.0001\, \sigma_{t-1}^2 \tag{144}$$

$$P = \begin{pmatrix} 0.7508 & 0.2492 \\ 0.5303 & 0.4697 \end{pmatrix} \tag{145}$$

LogLik $= -1777.71$; Converged: Yes.

**Plug-in Method**

$$\mu_1 = -0.0000 - 0.0001\, x_t \tag{146}$$
$$\mu_2 = 0.0350 - 0.0749\, x_t \tag{147}$$

$$\sigma_t^2(1) = 0.0001 + 0.0001\, y_{t-1}^2 + 0.0001\, \sigma_{t-1}^2 \tag{148}$$
$$\sigma_t^2(2) = 0.0643 + 0.1195\, y_{t-1}^2 + 0.8796\, \sigma_{t-1}^2 \tag{149}$$

$$P = \begin{pmatrix} 0.5192 & 0.4808 \\ 0.2312 & 0.7688 \end{pmatrix} \tag{150}$$

LogLik $= -1448.70$; Converged: Yes.

**Quadrature Method**

$$\mu_1 = -0.1344 - 0.2235\, x_t \tag{151}$$
$$\mu_2 = 0.0001 - 0.0039\, x_t \tag{152}$$

$$\sigma_t^2(1) = 0.8549 + 0.6400\, y_{t-1}^2 + 0.3581\, \sigma_{t-1}^2 \tag{153}$$
$$\sigma_t^2(2) = 0.0001 + 0.1089\, y_{t-1}^2 + 0.0001\, \sigma_{t-1}^2 \tag{154}$$

$$P = \begin{pmatrix} 0.5021 & 0.4979 \\ 0.5156 & 0.4844 \end{pmatrix} \tag{155}$$

LogLik $= -2441.71$; Converged: Yes.

**Particle Filter Method**

$$\mu_1 = 0.0000 + 0.1000\, x_t \tag{156}$$
$$\mu_2 = 0.0000 + 0.1000\, x_t \tag{157}$$

$$\sigma_t^2(1) = 0.1000 + 0.1000\, y_{t-1}^2 + 0.7000\, \sigma_{t-1}^2 \tag{158}$$
$$\sigma_t^2(2) = 0.2000 + 0.2000\, y_{t-1}^2 + 0.6000\, \sigma_{t-1}^2 \tag{159}$$

$$P = \begin{pmatrix} 0.9000 & 0.1000 \\ 0.1000 & 0.9000 \end{pmatrix} \tag{160}$$

LogLik $= -4011.86$; Converged: Yes.

### 1.5. Asset: `VTV`

### Baseline Specification

**Table 9:** VTV — Baseline Specification

| Method | $c_1$ | $c_2$ | $\gamma_1$ | $\gamma_2$ | $\omega_1$ | $\alpha_1$ | $\beta_1$ | $\omega_2$ | $\alpha_2$ | $\beta_2$ | $p_{11}$ | $p_{22}$ | Conv. | LogLik |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Gray's | 0.0731 | -0.0000 | – | – | 0.2929 | 0.3022 | 0.6977 | 0.0001 | 0.0001 | 0.0001 | 0.7304 | 0.4624 | Yes | -1125.12 |
| Plug-in | 0.0724 | -0.0000 | – | – | 0.1410 | 0.3708 | 0.6292 | 0.0001 | 0.0001 | 0.0004 | 0.7878 | 0.5170 | Yes | -835.80 |
| Quadrature | 0.0630 | -0.0003 | – | – | 0.2014 | 0.3257 | 0.6729 | 0.0001 | 0.0001 | 0.0001 | 0.7729 | 0.4669 | Yes | -1161.13 |
| Particle Filter | 0.0000 | 0.0000 | – | – | 0.1000 | 0.1000 | 0.7000 | 0.2000 | 0.2000 | 0.6000 | 0.9000 | 0.9000 | No | -3185.81 |

Baseline spec: $\gamma_1 = \gamma_2 = 0$. Conv. = convergence status.

### Fitted Equations (Baseline):

### Gray's Method

$$\mu_1 = 0.0731 \tag{161}$$
$$\mu_2 = -0.0000 \tag{162}$$

$$\sigma_t^2(1) = 0.2929 + 0.3022\, y_{t-1}^2 + 0.6977\, \sigma_{t-1}^2 \tag{163}$$
$$\sigma_t^2(2) = 0.0001 + 0.0001\, y_{t-1}^2 + 0.0001\, \sigma_{t-1}^2 \tag{164}$$

$$P = \begin{pmatrix} 0.7304 & 0.2696 \\ 0.5376 & 0.4624 \end{pmatrix} \tag{165}$$

LogLik $= -1125.12$; Converged: Yes.

### Plug-in Method

$$\mu_1 = 0.0724 \tag{166}$$
$$\mu_2 = -0.0000 \tag{167}$$

$$\sigma_t^2(1) = 0.1410 + 0.3708\, y_{t-1}^2 + 0.6292\, \sigma_{t-1}^2 \tag{168}$$
$$\sigma_t^2(2) = 0.0001 + 0.0001\, y_{t-1}^2 + 0.0004\, \sigma_{t-1}^2 \tag{169}$$

$$P = \begin{pmatrix} 0.7878 & 0.2122 \\ 0.4830 & 0.5170 \end{pmatrix} \tag{170}$$

LogLik $= -835.80$; Converged: Yes.

### Quadrature Method

$$\mu_1 = 0.0630 \tag{171}$$
$$\mu_2 = -0.0003 \tag{172}$$

$$\sigma_t^2(1) = 0.2014 + 0.3257\,y_{t-1}^2 + 0.6729\,\sigma_{t-1}^2 \tag{173}$$
$$\sigma_t^2(2) = 0.0001 + 0.0001\,y_{t-1}^2 + 0.0001\,\sigma_{t-1}^2 \tag{174}$$

$$P = \begin{pmatrix} 0.7729 & 0.2271 \\ 0.5331 & 0.4669 \end{pmatrix} \tag{175}$$

LogLik $= -1161.13$; Converged: Yes.

### Particle Filter Method

$$\mu_1 = 0.0000 \tag{176}$$
$$\mu_2 = 0.0000 \tag{177}$$

$$\sigma_t^2(1) = 0.1000 + 0.1000\,y_{t-1}^2 + 0.7000\,\sigma_{t-1}^2 \tag{178}$$
$$\sigma_t^2(2) = 0.2000 + 0.2000\,y_{t-1}^2 + 0.6000\,\sigma_{t-1}^2 \tag{179}$$

$$P = \begin{pmatrix} 0.9000 & 0.1000 \\ 0.1000 & 0.9000 \end{pmatrix} \tag{180}$$

LogLik $= -3185.81$; Converged: No.

### Exogenous Specification

**Table 10:** VTV — Exogenous Specification

| **Method** | $c_1$ | $c_2$ | $\gamma_1$ | $\gamma_2$ | $\omega_1$ | $\alpha_1$ | $\beta_1$ | $\omega_2$ | $\alpha_2$ | $\beta_2$ | $p_{11}$ | $p_{22}$ | Conv. | Log |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Gray's | 0.0869 | -0.0228 | -0.0107 | 0.0176 | 0.5310 | 0.2426 | 0.7556 | 0.0003 | 0.0064 | 0.0007 | 0.7299 | 0.6145 | Yes | -210 |
| Plug-in | -0.0001 | 0.0459 | -0.0005 | -0.0119 | 0.0001 | 0.0001 | 0.0005 | 0.1423 | 0.3254 | 0.6741 | 0.5232 | 0.7447 | Yes | -82 |
| Quadrature | 0.0734 | -0.0000 | -0.0054 | -0.0003 | 0.2588 | 0.3504 | 0.6496 | 0.0001 | 0.0001 | 0.0001 | 0.7387 | 0.4614 | Yes | -116 |
| Particle Filter | -0.0000 | 0.0000 | 0.1000 | 0.1000 | 0.1000 | 0.1000 | 0.7000 | 0.2000 | 0.2000 | 0.6000 | 0.9000 | 0.9000 | Yes | -319 |

Exogenous spec; pivot = $^{G}SPC.Conv. = convergencestatus.$

### Fitted Equations (Exogenous):

### Gray's Method

$$\mu_1 = 0.0869 - 0.0107\,x_t \tag{181}$$
$$\mu_2 = -0.0228 + 0.0176\,x_t \tag{182}$$

$$\sigma_t^2(1) = 0.5310 + 0.2426\, y_{t-1}^2 + 0.7556\, \sigma_{t-1}^2 \tag{183}$$
$$\sigma_t^2(2) = 0.0003 + 0.0064\, y_{t-1}^2 + 0.0007\, \sigma_{t-1}^2 \tag{184}$$

$$P = \begin{pmatrix} 0.7299 & 0.2701 \\ 0.3855 & 0.6145 \end{pmatrix} \tag{185}$$

LogLik $= -2103.47$; Converged: Yes.

**Plug-in Method**

$$\mu_1 = -0.0001 - 0.0005\, x_t \tag{186}$$
$$\mu_2 = 0.0459 - 0.0119\, x_t \tag{187}$$

$$\sigma_t^2(1) = 0.0001 + 0.0001\, y_{t-1}^2 + 0.0005\, \sigma_{t-1}^2 \tag{188}$$
$$\sigma_t^2(2) = 0.1423 + 0.3254\, y_{t-1}^2 + 0.6741\, \sigma_{t-1}^2 \tag{189}$$

$$P = \begin{pmatrix} 0.5232 & 0.4768 \\ 0.2553 & 0.7447 \end{pmatrix} \tag{190}$$

LogLik $= -829.83$; Converged: Yes.

**Quadrature Method**

$$\mu_1 = 0.0734 - 0.0054\, x_t \tag{191}$$
$$\mu_2 = -0.0000 - 0.0003\, x_t \tag{192}$$

$$\sigma_t^2(1) = 0.2588 + 0.3504\, y_{t-1}^2 + 0.6496\, \sigma_{t-1}^2 \tag{193}$$
$$\sigma_t^2(2) = 0.0001 + 0.0001\, y_{t-1}^2 + 0.0001\, \sigma_{t-1}^2 \tag{194}$$

$$P = \begin{pmatrix} 0.7387 & 0.2613 \\ 0.5386 & 0.4614 \end{pmatrix} \tag{195}$$

LogLik $= -1164.80$; Converged: Yes.

**Particle Filter Method**

$$\mu_1 = -0.0000 + 0.1000\, x_t \tag{196}$$
$$\mu_2 = 0.0000 + 0.1000\, x_t \tag{197}$$

$$\sigma_t^2(1) = 0.1000 + 0.1000\, y_{t-1}^2 + 0.7000\, \sigma_{t-1}^2 \tag{198}$$
$$\sigma_t^2(2) = 0.2000 + 0.2000\, y_{t-1}^2 + 0.6000\, \sigma_{t-1}^2 \tag{199}$$

$$P = \begin{pmatrix} 0.9000 & 0.1000 \\ 0.1000 & 0.9000 \end{pmatrix} \tag{200}$$

LogLik $= -3197.38$; Converged: Yes.

## 1.6. Asset: GSPC

### Baseline Specification

**Table 11:** GSPC — Baseline Specification

| **Method** | $c_1$ | $c_2$ | $\gamma_1$ | $\gamma_2$ | $\omega_1$ | $\alpha_1$ | $\beta_1$ | $\omega_2$ | $\alpha_2$ | $\beta_2$ | $p_{11}$ | $p_{22}$ | Conv. | LogLik |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Gray's | 0.1151 | -0.3419 | – | – | 0.0001 | 0.0787 | 0.7068 | 0.4046 | 0.2466 | 0.7534 | 0.9516 | 0.8362 | Yes | -3333.75 |
| Plug-in | -0.0001 | 0.0739 | – | – | 0.0001 | 0.0001 | 0.0001 | 0.1646 | 0.3090 | 0.6910 | 0.5407 | 0.7831 | No | -1054.82 |
| Quadrature | 0.0765 | -0.0002 | – | – | 0.3227 | 0.3107 | 0.6891 | 0.0001 | 0.0001 | 0.0001 | 0.7184 | 0.4804 | No | -1414.35 |
| Particle Filter | -0.0000 | -0.0000 | – | – | 0.1000 | 0.1000 | 0.7000 | 0.2000 | 0.2000 | 0.6000 | 0.9000 | 0.9000 | Yes | -3516.35 |

Baseline spec: $\gamma_1 = \gamma_2 = 0$. Conv. = convergence status.

### Fitted Equations (Baseline):

### Gray's Method

$$\mu_1 = 0.1151 \tag{201}$$
$$\mu_2 = -0.3419 \tag{202}$$

$$\sigma_t^2(1) = 0.0001 + 0.0787\, y_{t-1}^2 + 0.7068\, \sigma_{t-1}^2 \tag{203}$$
$$\sigma_t^2(2) = 0.4046 + 0.2466\, y_{t-1}^2 + 0.7534\, \sigma_{t-1}^2 \tag{204}$$

$$P = \begin{pmatrix} 0.9516 & 0.0484 \\ 0.1638 & 0.8362 \end{pmatrix} \tag{205}$$

LogLik $= -3333.75$; Converged: Yes.

### Plug-in Method

$$\mu_1 = -0.0001 \tag{206}$$
$$\mu_2 = 0.0739 \tag{207}$$

$$\sigma_t^2(1) = 0.0001 + 0.0001\, y_{t-1}^2 + 0.0001\, \sigma_{t-1}^2 \tag{208}$$
$$\sigma_t^2(2) = 0.1646 + 0.3090\, y_{t-1}^2 + 0.6910\, \sigma_{t-1}^2 \tag{209}$$

$$P = \begin{pmatrix} 0.5407 & 0.4593 \\ 0.2169 & 0.7831 \end{pmatrix} \tag{210}$$

LogLik $= -1054.82$; Converged: No.

**Quadrature Method**

$$\mu_1 = 0.0765 \tag{211}$$
$$\mu_2 = -0.0002 \tag{212}$$

$$\sigma_t^2(1) = 0.3227 + 0.3107\, y_{t-1}^2 + 0.6891\, \sigma_{t-1}^2 \tag{213}$$
$$\sigma_t^2(2) = 0.0001 + 0.0001\, y_{t-1}^2 + 0.0001\, \sigma_{t-1}^2 \tag{214}$$

$$P = \begin{pmatrix} 0.7184 & 0.2816 \\ 0.5196 & 0.4804 \end{pmatrix} \tag{215}$$

LogLik $= -1414.35$; Converged: No.

**Particle Filter Method**

$$\mu_1 = -0.0000 \tag{216}$$
$$\mu_2 = -0.0000 \tag{217}$$

$$\sigma_t^2(1) = 0.1000 + 0.1000\, y_{t-1}^2 + 0.7000\, \sigma_{t-1}^2 \tag{218}$$
$$\sigma_t^2(2) = 0.2000 + 0.2000\, y_{t-1}^2 + 0.6000\, \sigma_{t-1}^2 \tag{219}$$

$$P = \begin{pmatrix} 0.9000 & 0.1000 \\ 0.1000 & 0.9000 \end{pmatrix} \tag{220}$$

LogLik $= -3516.35$; Converged: Yes.

## 2. Bond Market

Pivot (exogenous conditioning variable): `TNX`.

### 2.1. Asset: `IEI`

#### Baseline Specification

**Table 12:** IEI — Baseline Specification

| Method | $c_1$ | $c_2$ | $\gamma_1$ | $\gamma_2$ | $\omega_1$ | $\alpha_1$ | $\beta_1$ | $\omega_2$ | $\alpha_2$ | $\beta_2$ | $p_{11}$ | $p_{22}$ | Conv. | LogLik |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Gray's | 0.0019 | 0.0006 | – | – | 0.0256 | 0.2409 | 0.7561 | 0.0001 | 0.0001 | 0.0001 | 0.7280 | 0.5054 | Yes | 1896.91 |
| Plug-in | 0.0129 | -0.0000 | – | – | 0.0017 | 0.1031 | 0.8969 | 0.0001 | 0.0001 | 0.0001 | 0.7409 | 0.5922 | Yes | 2031.30 |
| Quadrature | 0.0006 | 0.0001 | – | – | 0.0001 | 0.2845 | 0.5719 | 0.0001 | 0.0001 | 0.0001 | 0.7160 | 0.4853 | Yes | 1892.71 |
| Particle Filter | -0.0000 | 0.0000 | – | – | 0.1000 | 0.1000 | 0.7000 | 0.2000 | 0.2000 | 0.6000 | 0.9000 | 0.9000 | Yes | -1455.01 |

Baseline spec: $\gamma_1 = \gamma_2 = 0$. Conv. = convergence status.

#### Fitted Equations (Baseline):

#### Gray's Method

$$\mu_1 = 0.0019 \tag{221}$$
$$\mu_2 = 0.0006 \tag{222}$$

$$\sigma_t^2(1) = 0.0256 + 0.2409\, y_{t-1}^2 + 0.7561\, \sigma_{t-1}^2 \tag{223}$$
$$\sigma_t^2(2) = 0.0001 + 0.0001\, y_{t-1}^2 + 0.0001\, \sigma_{t-1}^2 \tag{224}$$

$$P = \begin{pmatrix} 0.7280 & 0.2720 \\ 0.4946 & 0.5054 \end{pmatrix} \tag{225}$$

LogLik = 1896.91; Converged: Yes.

#### Plug-in Method

$$\mu_1 = 0.0129 \tag{226}$$
$$\mu_2 = -0.0000 \tag{227}$$

$$\sigma_t^2(1) = 0.0017 + 0.1031\, y_{t-1}^2 + 0.8969\, \sigma_{t-1}^2 \tag{228}$$
$$\sigma_t^2(2) = 0.0001 + 0.0001\, y_{t-1}^2 + 0.0001\, \sigma_{t-1}^2 \tag{229}$$

$$P = \begin{pmatrix} 0.7409 & 0.2591 \\ 0.4078 & 0.5922 \end{pmatrix} \tag{230}$$

LogLik = 2031.30; Converged: Yes.

### Quadrature Method

$$\mu_1 = 0.0006 \tag{231}$$
$$\mu_2 = 0.0001 \tag{232}$$

$$\sigma_t^2(1) = 0.0001 + 0.2845\, y_{t-1}^2 + 0.5719\, \sigma_{t-1}^2 \tag{233}$$
$$\sigma_t^2(2) = 0.0001 + 0.0001\, y_{t-1}^2 + 0.0001\, \sigma_{t-1}^2 \tag{234}$$

$$P = \begin{pmatrix} 0.7160 & 0.2840 \\ 0.5147 & 0.4853 \end{pmatrix} \tag{235}$$

LogLik $= 1892.71$; Converged: Yes.

### Particle Filter Method

$$\mu_1 = -0.0000 \tag{236}$$
$$\mu_2 = 0.0000 \tag{237}$$

$$\sigma_t^2(1) = 0.1000 + 0.1000\, y_{t-1}^2 + 0.7000\, \sigma_{t-1}^2 \tag{238}$$
$$\sigma_t^2(2) = 0.2000 + 0.2000\, y_{t-1}^2 + 0.6000\, \sigma_{t-1}^2 \tag{239}$$

$$P = \begin{pmatrix} 0.9000 & 0.1000 \\ 0.1000 & 0.9000 \end{pmatrix} \tag{240}$$

LogLik $= -1455.01$; Converged: Yes.

### Exogenous Specification

**Table 13:** IEI — Exogenous Specification

| **Method** | $c_1$ | $c_2$ | $\gamma_1$ | $\gamma_2$ | $\omega_1$ | $\alpha_1$ | $\beta_1$ | $\omega_2$ | $\alpha_2$ | $\beta_2$ | $p_{11}$ | $p_{22}$ | Conv. | Log |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Gray's | -0.0011 | -0.0000 | 0.0026 | -0.0000 | 0.0270 | 0.3322 | 0.6635 | 0.0001 | 0.0001 | 0.0001 | 0.7347 | 0.4642 | Yes | 1891 |
| Plug-in | 0.0002 | -0.0085 | 0.0002 | -0.0026 | 0.0001 | 0.0001 | 0.0001 | 0.0086 | 0.2878 | 0.7122 | 0.5018 | 0.7526 | No | 1994 |
| Quadrature | 0.0005 | 0.0001 | 0.0022 | 0.0001 | 0.0001 | 0.2840 | 0.5712 | 0.0001 | 0.0001 | 0.0001 | 0.7167 | 0.4850 | Yes | 1893 |
| Particle Filter | 0.0000 | 0.0000 | 0.1000 | 0.1000 | 0.1000 | 0.1000 | 0.7000 | 0.2000 | 0.2000 | 0.6000 | 0.9000 | 0.9000 | No | -1693 |

Exogenous spec; pivot = $^{T}NX.Conv. = convergencestatus.$

### Fitted Equations (Exogenous):

### Gray's Method

$$\mu_1 = -0.0011 + 0.0026\, x_t \tag{241}$$
$$\mu_2 = -0.0000 - 0.0000\, x_t \tag{242}$$

$$\sigma_t^2(1) = 0.0270 + 0.3322\,y_{t-1}^2 + 0.6635\,\sigma_{t-1}^2 \tag{243}$$
$$\sigma_t^2(2) = 0.0001 + 0.0001\,y_{t-1}^2 + 0.0001\,\sigma_{t-1}^2 \tag{244}$$

$$P = \begin{pmatrix} 0.7347 & 0.2653 \\ 0.5358 & 0.4642 \end{pmatrix} \tag{245}$$

LogLik = 1891.71; Converged: Yes.

**Plug-in Method**

$$\mu_1 = 0.0002 + 0.0002\,x_t \tag{246}$$
$$\mu_2 = -0.0085 - 0.0026\,x_t \tag{247}$$

$$\sigma_t^2(1) = 0.0001 + 0.0001\,y_{t-1}^2 + 0.0001\,\sigma_{t-1}^2 \tag{248}$$
$$\sigma_t^2(2) = 0.0086 + 0.2878\,y_{t-1}^2 + 0.7122\,\sigma_{t-1}^2 \tag{249}$$

$$P = \begin{pmatrix} 0.5018 & 0.4982 \\ 0.2474 & 0.7526 \end{pmatrix} \tag{250}$$

LogLik = 1994.27; Converged: No.

**Quadrature Method**

$$\mu_1 = 0.0005 + 0.0022\,x_t \tag{251}$$
$$\mu_2 = 0.0001 + 0.0001\,x_t \tag{252}$$

$$\sigma_t^2(1) = 0.0001 + 0.2840\,y_{t-1}^2 + 0.5712\,\sigma_{t-1}^2 \tag{253}$$
$$\sigma_t^2(2) = 0.0001 + 0.0001\,y_{t-1}^2 + 0.0001\,\sigma_{t-1}^2 \tag{254}$$

$$P = \begin{pmatrix} 0.7167 & 0.2833 \\ 0.5150 & 0.4850 \end{pmatrix} \tag{255}$$

LogLik = 1893.21; Converged: Yes.

**Particle Filter Method**

$$\mu_1 = 0.0000 + 0.1000\,x_t \tag{256}$$
$$\mu_2 = 0.0000 + 0.1000\,x_t \tag{257}$$

$$\sigma_t^2(1) = 0.1000 + 0.1000\,y_{t-1}^2 + 0.7000\,\sigma_{t-1}^2 \tag{258}$$
$$\sigma_t^2(2) = 0.2000 + 0.2000\,y_{t-1}^2 + 0.6000\,\sigma_{t-1}^2 \tag{259}$$

$$P = \begin{pmatrix} 0.9000 & 0.1000 \\ 0.1000 & 0.9000 \end{pmatrix} \tag{260}$$

LogLik $= -1693.27$; Converged: No.

## 2.2. Asset: `TLT`

### Baseline Specification

**Table 14:** TLT — Baseline Specification

| **Method** | $c_1$ | $c_2$ | $\gamma_1$ | $\gamma_2$ | $\omega_1$ | $\alpha_1$ | $\beta_1$ | $\omega_2$ | $\alpha_2$ | $\beta_2$ | $p_{11}$ | $p_{22}$ | Conv. | LogLik |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Gray's | -0.0204 | -0.0001 | – | – | 0.4310 | 0.1625 | 0.5545 | 0.0001 | 0.0001 | 0.0001 | 0.7467 | 0.5053 | Yes | -1173.63 |
| Plug-in | -0.0082 | -0.0001 | – | – | 0.0678 | 0.1965 | 0.8035 | 0.0001 | 0.0001 | 0.0001 | 0.7908 | 0.5119 | Yes | -934.93 |
| Quadrature | -0.0001 | -0.0503 | – | – | 0.0001 | 0.0001 | 0.0001 | 0.1621 | 0.1985 | 0.7949 | 0.4683 | 0.7582 | Yes | -1200.19 |
| Particle Filter | -0.0000 | -0.0000 | – | – | 0.1000 | 0.1000 | 0.7000 | 0.2000 | 0.2000 | 0.6000 | 0.9000 | 0.9000 | Yes | -3249.52 |

Baseline spec: $\gamma_1 = \gamma_2 = 0$. Conv. = convergence status.

### Fitted Equations (Baseline):

### Gray's Method

$$\mu_1 = -0.0204 \tag{261}$$
$$\mu_2 = -0.0001 \tag{262}$$

$$\sigma_t^2(1) = 0.4310 + 0.1625\, y_{t-1}^2 + 0.5545\, \sigma_{t-1}^2 \tag{263}$$
$$\sigma_t^2(2) = 0.0001 + 0.0001\, y_{t-1}^2 + 0.0001\, \sigma_{t-1}^2 \tag{264}$$

$$P = \begin{pmatrix} 0.7467 & 0.2533 \\ 0.4947 & 0.5053 \end{pmatrix} \tag{265}$$

LogLik $= -1173.63$; Converged: Yes.

### Plug-in Method

$$\mu_1 = -0.0082 \tag{266}$$
$$\mu_2 = -0.0001 \tag{267}$$

$$\sigma_t^2(1) = 0.0678 + 0.1965\, y_{t-1}^2 + 0.8035\, \sigma_{t-1}^2 \tag{268}$$
$$\sigma_t^2(2) = 0.0001 + 0.0001\, y_{t-1}^2 + 0.0001\, \sigma_{t-1}^2 \tag{269}$$

$$P = \begin{pmatrix} 0.7908 & 0.2092 \\ 0.4881 & 0.5119 \end{pmatrix} \tag{270}$$

LogLik $= -934.93$; Converged: Yes.

### Quadrature Method

$$\mu_1 = -0.0001 \tag{271}$$
$$\mu_2 = -0.0503 \tag{272}$$

$$\sigma_t^2(1) = 0.0001 + 0.0001\, y_{t-1}^2 + 0.0001\, \sigma_{t-1}^2 \tag{273}$$
$$\sigma_t^2(2) = 0.1621 + 0.1985\, y_{t-1}^2 + 0.7949\, \sigma_{t-1}^2 \tag{274}$$

$$P = \begin{pmatrix} 0.4683 & 0.5317 \\ 0.2418 & 0.7582 \end{pmatrix} \tag{275}$$

LogLik $= -1200.19$; Converged: Yes.

### Particle Filter Method

$$\mu_1 = -0.0000 \tag{276}$$
$$\mu_2 = -0.0000 \tag{277}$$

$$\sigma_t^2(1) = 0.1000 + 0.1000\, y_{t-1}^2 + 0.7000\, \sigma_{t-1}^2 \tag{278}$$
$$\sigma_t^2(2) = 0.2000 + 0.2000\, y_{t-1}^2 + 0.6000\, \sigma_{t-1}^2 \tag{279}$$

$$P = \begin{pmatrix} 0.9000 & 0.1000 \\ 0.1000 & 0.9000 \end{pmatrix} \tag{280}$$

LogLik $= -3249.52$; Converged: Yes.

### Exogenous Specification

**Table 15:** TLT — Exogenous Specification

| **Method** | $c_1$ | $c_2$ | $\gamma_1$ | $\gamma_2$ | $\omega_1$ | $\alpha_1$ | $\beta_1$ | $\omega_2$ | $\alpha_2$ | $\beta_2$ | $p_{11}$ | $p_{22}$ | Conv. | Log |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Gray's | -0.0225 | -0.0001 | -0.0024 | -0.0001 | 0.2859 | 0.1930 | 0.8005 | 0.0001 | 0.0001 | 0.0001 | 0.7471 | 0.4607 | Yes | -1170 |
| Plug-in | 0.0405 | -0.0005 | 0.0136 | 0.0001 | 0.0405 | 0.1273 | 0.8727 | 0.0001 | 0.0001 | 0.0001 | 0.7469 | 0.5184 | Yes | -920 |
| Quadrature | -0.0285 | -0.0003 | 0.0059 | 0.0000 | 0.1776 | 0.1736 | 0.8243 | 0.0001 | 0.0001 | 0.0001 | 0.7407 | 0.4673 | Yes | -119 |
| Particle Filter | 0.0000 | 0.0000 | 0.1000 | 0.1000 | 0.1000 | 0.1000 | 0.7000 | 0.2000 | 0.2000 | 0.6000 | 0.9000 | 0.9000 | Yes | -333 |

Exogenous spec; pivot = $^{T}NX.Conv. = convergencestatus.$

### Fitted Equations (Exogenous):

### Gray's Method

$$\mu_1 = -0.0225 - 0.0024\, x_t \tag{281}$$
$$\mu_2 = -0.0001 - 0.0001\, x_t \tag{282}$$

$$\sigma_t^2(1) = 0.2859 + 0.1930\, y_{t-1}^2 + 0.8005\, \sigma_{t-1}^2 \tag{283}$$
$$\sigma_t^2(2) = 0.0001 + 0.0001\, y_{t-1}^2 + 0.0001\, \sigma_{t-1}^2 \tag{284}$$

$$P = \begin{pmatrix} 0.7471 & 0.2529 \\ 0.5393 & 0.4607 \end{pmatrix} \tag{285}$$

LogLik $= -1170.40$; Converged: Yes.

**Plug-in Method**

$$\mu_1 = 0.0405 + 0.0136\, x_t \tag{286}$$
$$\mu_2 = -0.0005 + 0.0001\, x_t \tag{287}$$

$$\sigma_t^2(1) = 0.0405 + 0.1273\, y_{t-1}^2 + 0.8727\, \sigma_{t-1}^2 \tag{288}$$
$$\sigma_t^2(2) = 0.0001 + 0.0001\, y_{t-1}^2 + 0.0001\, \sigma_{t-1}^2 \tag{289}$$

$$P = \begin{pmatrix} 0.7469 & 0.2531 \\ 0.4816 & 0.5184 \end{pmatrix} \tag{290}$$

LogLik $= -926.61$; Converged: Yes.

**Quadrature Method**

$$\mu_1 = -0.0285 + 0.0059\, x_t \tag{291}$$
$$\mu_2 = -0.0003 + 0.0000\, x_t \tag{292}$$

$$\sigma_t^2(1) = 0.1776 + 0.1736\, y_{t-1}^2 + 0.8243\, \sigma_{t-1}^2 \tag{293}$$
$$\sigma_t^2(2) = 0.0001 + 0.0001\, y_{t-1}^2 + 0.0001\, \sigma_{t-1}^2 \tag{294}$$

$$P = \begin{pmatrix} 0.7407 & 0.2593 \\ 0.5327 & 0.4673 \end{pmatrix} \tag{295}$$

LogLik $= -1193.51$; Converged: Yes.

**Particle Filter Method**

$$\mu_1 = 0.0000 + 0.1000\, x_t \tag{296}$$
$$\mu_2 = 0.0000 + 0.1000\, x_t \tag{297}$$

$$\sigma_t^2(1) = 0.1000 + 0.1000\, y_{t-1}^2 + 0.7000\, \sigma_{t-1}^2 \tag{298}$$
$$\sigma_t^2(2) = 0.2000 + 0.2000\, y_{t-1}^2 + 0.6000\, \sigma_{t-1}^2 \tag{299}$$

$$P = \begin{pmatrix} 0.9000 & 0.1000 \\ 0.1000 & 0.9000 \end{pmatrix} \tag{300}$$

LogLik $= -3339.23$; Converged: Yes.

## 2.3. Asset: BNDX

### Baseline Specification

**Table 16:** BNDX — Baseline Specification

| Method | $c_1$ | $c_2$ | $\gamma_1$ | $\gamma_2$ | $\omega_1$ | $\alpha_1$ | $\beta_1$ | $\omega_2$ | $\alpha_2$ | $\beta_2$ | $p_{11}$ | $p_{22}$ | Conv. | LogLik |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Gray's | 0.0068 | 0.0000 | – | – | 0.0262 | 0.3886 | 0.6108 | 0.0001 | 0.0001 | 0.0001 | 0.7316 | 0.4414 | Yes | 1854.41 |
| Plug-in | 0.0005 | 0.0082 | – | – | 0.0001 | 0.0001 | 0.0020 | 0.0076 | 0.3010 | 0.6990 | 0.5175 | 0.7284 | Yes | 1992.60 |
| Quadrature | 0.0071 | 0.0004 | – | – | 0.0001 | 0.3053 | 0.5660 | 0.0001 | 0.0001 | 0.0001 | 0.7077 | 0.4657 | Yes | 1864.15 |
| Particle Filter | 0.0000 | 0.0000 | – | – | 0.1000 | 0.1000 | 0.7000 | 0.2000 | 0.2000 | 0.6000 | 0.9000 | 0.9000 | No | -1463.24 |

Baseline spec: $\gamma_1 = \gamma_2 = 0$. Conv. = convergence status.

### Fitted Equations (Baseline):

### Gray's Method

$$\mu_1 = 0.0068 \tag{301}$$
$$\mu_2 = 0.0000 \tag{302}$$

$$\sigma_t^2(1) = 0.0262 + 0.3886\, y_{t-1}^2 + 0.6108\, \sigma_{t-1}^2 \tag{303}$$
$$\sigma_t^2(2) = 0.0001 + 0.0001\, y_{t-1}^2 + 0.0001\, \sigma_{t-1}^2 \tag{304}$$

$$P = \begin{pmatrix} 0.7316 & 0.2684 \\ 0.5586 & 0.4414 \end{pmatrix} \tag{305}$$

LogLik = 1854.41; Converged: Yes.

### Plug-in Method

$$\mu_1 = 0.0005 \tag{306}$$
$$\mu_2 = 0.0082 \tag{307}$$

$$\sigma_t^2(1) = 0.0001 + 0.0001\, y_{t-1}^2 + 0.0020\, \sigma_{t-1}^2 \tag{308}$$
$$\sigma_t^2(2) = 0.0076 + 0.3010\, y_{t-1}^2 + 0.6990\, \sigma_{t-1}^2 \tag{309}$$

$$P = \begin{pmatrix} 0.5175 & 0.4825 \\ 0.2716 & 0.7284 \end{pmatrix} \tag{310}$$

LogLik = 1992.60; Converged: Yes.

### Quadrature Method

$$\mu_1 = 0.0071 \tag{311}$$
$$\mu_2 = 0.0004 \tag{312}$$

$$\sigma_t^2(1) = 0.0001 + 0.3053\,y_{t-1}^2 + 0.5660\,\sigma_{t-1}^2 \tag{313}$$
$$\sigma_t^2(2) = 0.0001 + 0.0001\,y_{t-1}^2 + 0.0001\,\sigma_{t-1}^2 \tag{314}$$

$$P = \begin{pmatrix} 0.7077 & 0.2923 \\ 0.5343 & 0.4657 \end{pmatrix} \tag{315}$$

LogLik $= 1864.15$; Converged: Yes.

### Particle Filter Method

$$\mu_1 = 0.0000 \tag{316}$$
$$\mu_2 = 0.0000 \tag{317}$$

$$\sigma_t^2(1) = 0.1000 + 0.1000\,y_{t-1}^2 + 0.7000\,\sigma_{t-1}^2 \tag{318}$$
$$\sigma_t^2(2) = 0.2000 + 0.2000\,y_{t-1}^2 + 0.6000\,\sigma_{t-1}^2 \tag{319}$$

$$P = \begin{pmatrix} 0.9000 & 0.1000 \\ 0.1000 & 0.9000 \end{pmatrix} \tag{320}$$

LogLik $= -1463.24$; Converged: No.

### Exogenous Specification

**Table 17:** BNDX — Exogenous Specification

| **Method** | $c_1$ | $c_2$ | $\gamma_1$ | $\gamma_2$ | $\omega_1$ | $\alpha_1$ | $\beta_1$ | $\omega_2$ | $\alpha_2$ | $\beta_2$ | $p_{11}$ | $p_{22}$ | Conv. | LogL |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Gray's | 0.0072 | 0.0005 | -0.0099 | 0.0001 | 0.0249 | 0.2959 | 0.7019 | 0.0001 | 0.0001 | 0.0001 | 0.7255 | 0.4732 | Yes | 1874. |
| Plug-in | 0.0004 | 0.0098 | 0.0001 | -0.0050 | 0.0001 | 0.0001 | 0.0012 | 0.0103 | 0.3731 | 0.6265 | 0.4482 | 0.7258 | Yes | 1965. |
| Quadrature | 0.0082 | 0.0004 | -0.0099 | 0.0001 | 0.0001 | 0.2883 | 0.5664 | 0.0001 | 0.0001 | 0.0001 | 0.7055 | 0.4650 | Yes | 1871. |
| Particle Filter | 0.0000 | 0.0000 | 0.1000 | 0.1000 | 0.1000 | 0.1000 | 0.7000 | 0.2000 | 0.2000 | 0.6000 | 0.9000 | 0.9000 | No | -1707. |

Exogenous spec; pivot = $^{T}NX.Conv. = convergencestatus.$

### Fitted Equations (Exogenous):

### Gray's Method

$$\mu_1 = 0.0072 - 0.0099\,x_t \tag{321}$$
$$\mu_2 = 0.0005 + 0.0001\,x_t \tag{322}$$

$$\sigma_t^2(1) = 0.0249 + 0.2959\, y_{t-1}^2 + 0.7019\, \sigma_{t-1}^2 \tag{323}$$
$$\sigma_t^2(2) = 0.0001 + 0.0001\, y_{t-1}^2 + 0.0001\, \sigma_{t-1}^2 \tag{324}$$

$$P = \begin{pmatrix} 0.7255 & 0.2745 \\ 0.5268 & 0.4732 \end{pmatrix} \tag{325}$$

LogLik = 1874.11; Converged: Yes.

**Plug-in Method**

$$\mu_1 = 0.0004 + 0.0001\, x_t \tag{326}$$
$$\mu_2 = 0.0098 - 0.0050\, x_t \tag{327}$$

$$\sigma_t^2(1) = 0.0001 + 0.0001\, y_{t-1}^2 + 0.0012\, \sigma_{t-1}^2 \tag{328}$$
$$\sigma_t^2(2) = 0.0103 + 0.3731\, y_{t-1}^2 + 0.6265\, \sigma_{t-1}^2 \tag{329}$$

$$P = \begin{pmatrix} 0.4482 & 0.5518 \\ 0.2742 & 0.7258 \end{pmatrix} \tag{330}$$

LogLik = 1965.75; Converged: Yes.

**Quadrature Method**

$$\mu_1 = 0.0082 - 0.0099\, x_t \tag{331}$$
$$\mu_2 = 0.0004 + 0.0001\, x_t \tag{332}$$

$$\sigma_t^2(1) = 0.0001 + 0.2883\, y_{t-1}^2 + 0.5664\, \sigma_{t-1}^2 \tag{333}$$
$$\sigma_t^2(2) = 0.0001 + 0.0001\, y_{t-1}^2 + 0.0001\, \sigma_{t-1}^2 \tag{334}$$

$$P = \begin{pmatrix} 0.7055 & 0.2945 \\ 0.5350 & 0.4650 \end{pmatrix} \tag{335}$$

LogLik = 1871.45; Converged: Yes.

**Particle Filter Method**

$$\mu_1 = 0.0000 + 0.1000\, x_t \tag{336}$$
$$\mu_2 = 0.0000 + 0.1000\, x_t \tag{337}$$

$$\sigma_t^2(1) = 0.1000 + 0.1000\, y_{t-1}^2 + 0.7000\, \sigma_{t-1}^2 \tag{338}$$
$$\sigma_t^2(2) = 0.2000 + 0.2000\, y_{t-1}^2 + 0.6000\, \sigma_{t-1}^2 \tag{339}$$

$$P = \begin{pmatrix} 0.9000 & 0.1000 \\ 0.1000 & 0.9000 \end{pmatrix} \tag{340}$$

LogLik $= -1707.50$; Converged: No.

### 2.4. Asset: `LQD`

### Baseline Specification

**Table 18:** LQD — Baseline Specification

| Method | $c_1$ | $c_2$ | $\gamma_1$ | $\gamma_2$ | $\omega_1$ | $\alpha_1$ | $\beta_1$ | $\omega_2$ | $\alpha_2$ | $\beta_2$ | $p_{11}$ | $p_{22}$ | Conv. | LogLik |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Gray's | 0.0000 | 0.0001 | – | – | 0.0001 | 0.0001 | 0.0001 | 0.0791 | 0.3147 | 0.6852 | 0.3967 | 0.7547 | Yes | 310.69 |
| Plug-in | -0.0000 | 0.0325 | – | – | 0.0001 | 0.0001 | 0.0001 | 0.0295 | 0.2940 | 0.7060 | 0.5094 | 0.8025 | No | 482.71 |
| Quadrature | 0.0243 | -0.0003 | – | – | 0.0180 | 0.2731 | 0.7269 | 0.0001 | 0.0001 | 0.0001 | 0.7045 | 0.4639 | Yes | 293.74 |
| Particle Filter | 0.0000 | 0.0000 | – | – | 0.1000 | 0.1000 | 0.7000 | 0.2000 | 0.2000 | 0.6000 | 0.9000 | 0.9000 | No | -1996.75 |

Baseline spec: $\gamma_1 = \gamma_2 = 0$. Conv. = convergence status.

### Fitted Equations (Baseline):

### Gray's Method

$$\mu_1 = 0.0000 \tag{341}$$

$$\mu_2 = 0.0001 \tag{342}$$

$$\sigma_t^2(1) = 0.0001 + 0.0001\, y_{t-1}^2 + 0.0001\, \sigma_{t-1}^2 \tag{343}$$

$$\sigma_t^2(2) = 0.0791 + 0.3147\, y_{t-1}^2 + 0.6852\, \sigma_{t-1}^2 \tag{344}$$

$$P = \begin{pmatrix} 0.3967 & 0.6033 \\ 0.2453 & 0.7547 \end{pmatrix} \tag{345}$$

LogLik = 310.69; Converged: Yes.

### Plug-in Method

$$\mu_1 = -0.0000 \tag{346}$$

$$\mu_2 = 0.0325 \tag{347}$$

$$\sigma_t^2(1) = 0.0001 + 0.0001\, y_{t-1}^2 + 0.0001\, \sigma_{t-1}^2 \tag{348}$$

$$\sigma_t^2(2) = 0.0295 + 0.2940\, y_{t-1}^2 + 0.7060\, \sigma_{t-1}^2 \tag{349}$$

$$P = \begin{pmatrix} 0.5094 & 0.4906 \\ 0.1975 & 0.8025 \end{pmatrix} \tag{350}$$

LogLik = 482.71; Converged: No.

### Quadrature Method

$$\mu_1 = 0.0243 \tag{351}$$
$$\mu_2 = -0.0003 \tag{352}$$

$$\sigma_t^2(1) = 0.0180 + 0.2731\, y_{t-1}^2 + 0.7269\, \sigma_{t-1}^2 \tag{353}$$
$$\sigma_t^2(2) = 0.0001 + 0.0001\, y_{t-1}^2 + 0.0001\, \sigma_{t-1}^2 \tag{354}$$

$$P = \begin{pmatrix} 0.7045 & 0.2955 \\ 0.5361 & 0.4639 \end{pmatrix} \tag{355}$$

LogLik $= 293.74$; Converged: Yes.

### Particle Filter Method

$$\mu_1 = 0.0000 \tag{356}$$
$$\mu_2 = 0.0000 \tag{357}$$

$$\sigma_t^2(1) = 0.1000 + 0.1000\, y_{t-1}^2 + 0.7000\, \sigma_{t-1}^2 \tag{358}$$
$$\sigma_t^2(2) = 0.2000 + 0.2000\, y_{t-1}^2 + 0.6000\, \sigma_{t-1}^2 \tag{359}$$

$$P = \begin{pmatrix} 0.9000 & 0.1000 \\ 0.1000 & 0.9000 \end{pmatrix} \tag{360}$$

LogLik $= -1996.75$; Converged: No.

### Exogenous Specification

**Table 19:** LQD — Exogenous Specification

| **Method** | $c_1$ | $c_2$ | $\gamma_1$ | $\gamma_2$ | $\omega_1$ | $\alpha_1$ | $\beta_1$ | $\omega_2$ | $\alpha_2$ | $\beta_2$ | $p_{11}$ | $p_{22}$ | Conv. | LogL |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Gray's | 0.0044 | -0.0000 | 0.0111 | -0.0002 | 0.0895 | 0.3175 | 0.6825 | 0.0001 | 0.0001 | 0.0001 | 0.7551 | 0.4700 | Yes | 316. |
| Plug-in | 0.0153 | 0.0001 | 0.0110 | -0.0002 | 0.0286 | 0.3169 | 0.6831 | 0.0001 | 0.0001 | 0.0001 | 0.8154 | 0.5081 | Yes | 467. |
| Quadrature | 0.0475 | 0.0000 | 0.0230 | -0.0001 | 0.0231 | 0.3555 | 0.6445 | 0.0001 | 0.0001 | 0.0001 | 0.7441 | 0.3965 | Yes | 275. |
| Particle Filter | 0.0000 | 0.0000 | 0.1000 | 0.1000 | 0.1000 | 0.1000 | 0.7000 | 0.2000 | 0.2000 | 0.6000 | 0.9000 | 0.9000 | No | -2165. |

Exogenous spec; pivot = $^{T}NX.Conv. = convergencestatus.$

### Fitted Equations (Exogenous):

### Gray's Method

$$\mu_1 = 0.0044 + 0.0111\, x_t \tag{361}$$
$$\mu_2 = -0.0000 - 0.0002\, x_t \tag{362}$$

$$\sigma_t^2(1) = 0.0895 + 0.3175\, y_{t-1}^2 + 0.6825\, \sigma_{t-1}^2 \tag{363}$$
$$\sigma_t^2(2) = 0.0001 + 0.0001\, y_{t-1}^2 + 0.0001\, \sigma_{t-1}^2 \tag{364}$$

$$P = \begin{pmatrix} 0.7551 & 0.2449 \\ 0.5300 & 0.4700 \end{pmatrix} \tag{365}$$

LogLik = 316.53; Converged: Yes.

**Plug-in Method**

$$\mu_1 = 0.0153 + 0.0110\, x_t \tag{366}$$
$$\mu_2 = 0.0001 - 0.0002\, x_t \tag{367}$$

$$\sigma_t^2(1) = 0.0286 + 0.3169\, y_{t-1}^2 + 0.6831\, \sigma_{t-1}^2 \tag{368}$$
$$\sigma_t^2(2) = 0.0001 + 0.0001\, y_{t-1}^2 + 0.0001\, \sigma_{t-1}^2 \tag{369}$$

$$P = \begin{pmatrix} 0.8154 & 0.1846 \\ 0.4919 & 0.5081 \end{pmatrix} \tag{370}$$

LogLik = 467.55; Converged: Yes.

**Quadrature Method**

$$\mu_1 = 0.0475 + 0.0230\, x_t \tag{371}$$
$$\mu_2 = 0.0000 - 0.0001\, x_t \tag{372}$$

$$\sigma_t^2(1) = 0.0231 + 0.3555\, y_{t-1}^2 + 0.6445\, \sigma_{t-1}^2 \tag{373}$$
$$\sigma_t^2(2) = 0.0001 + 0.0001\, y_{t-1}^2 + 0.0001\, \sigma_{t-1}^2 \tag{374}$$

$$P = \begin{pmatrix} 0.7441 & 0.2559 \\ 0.6035 & 0.3965 \end{pmatrix} \tag{375}$$

LogLik = 275.63; Converged: Yes.

**Particle Filter Method**

$$\mu_1 = 0.0000 + 0.1000\, x_t \tag{376}$$
$$\mu_2 = 0.0000 + 0.1000\, x_t \tag{377}$$

$$\sigma_t^2(1) = 0.1000 + 0.1000\, y_{t-1}^2 + 0.7000\, \sigma_{t-1}^2 \tag{378}$$
$$\sigma_t^2(2) = 0.2000 + 0.2000\, y_{t-1}^2 + 0.6000\, \sigma_{t-1}^2 \tag{379}$$

$$P = \begin{pmatrix} 0.9000 & 0.1000 \\ 0.1000 & 0.9000 \end{pmatrix} \tag{380}$$

LogLik $= -2165.86$; Converged: No.

### 2.5. Asset: HYG

### Baseline Specification

**Table 20:** HYG — Baseline Specification

| Method | $c_1$ | $c_2$ | $\gamma_1$ | $\gamma_2$ | $\omega_1$ | $\alpha_1$ | $\beta_1$ | $\omega_2$ | $\alpha_2$ | $\beta_2$ | $p_{11}$ | $p_{22}$ | Conv. | LogLik |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Gray's | 0.0246 | -0.0015 | – | – | 0.0673 | 0.4066 | 0.5934 | 0.0001 | 0.0001 | 0.0001 | 0.8010 | 0.4364 | No | 597.67 |
| Plug-in | 0.0015 | 0.0408 | – | – | 0.0001 | 0.0001 | 0.2165 | 0.0300 | 0.3273 | 0.6727 | 0.6582 | 0.7240 | Yes | 697.38 |
| Quadrature | -0.0001 | 0.0526 | – | – | 0.0001 | 0.0001 | 0.0001 | 0.0272 | 0.5074 | 0.4926 | 0.4019 | 0.7372 | Yes | 579.36 |
| Particle Filter | 0.0000 | 0.0000 | – | – | 0.1000 | 0.1000 | 0.7000 | 0.2000 | 0.2000 | 0.6000 | 0.9000 | 0.9000 | Yes | -1889.53 |

Baseline spec: $\gamma_1 = \gamma_2 = 0$. Conv. = convergence status.

### Fitted Equations (Baseline):

### Gray's Method

$$\mu_1 = 0.0246 \tag{381}$$
$$\mu_2 = -0.0015 \tag{382}$$

$$\sigma_t^2(1) = 0.0673 + 0.4066\, y_{t-1}^2 + 0.5934\, \sigma_{t-1}^2 \tag{383}$$
$$\sigma_t^2(2) = 0.0001 + 0.0001\, y_{t-1}^2 + 0.0001\, \sigma_{t-1}^2 \tag{384}$$

$$P = \begin{pmatrix} 0.8010 & 0.1990 \\ 0.5636 & 0.4364 \end{pmatrix} \tag{385}$$

LogLik = 597.67; Converged: No.

### Plug-in Method

$$\mu_1 = 0.0015 \tag{386}$$
$$\mu_2 = 0.0408 \tag{387}$$

$$\sigma_t^2(1) = 0.0001 + 0.0001\, y_{t-1}^2 + 0.2165\, \sigma_{t-1}^2 \tag{388}$$
$$\sigma_t^2(2) = 0.0300 + 0.3273\, y_{t-1}^2 + 0.6727\, \sigma_{t-1}^2 \tag{389}$$

$$P = \begin{pmatrix} 0.6582 & 0.3418 \\ 0.2760 & 0.7240 \end{pmatrix} \tag{390}$$

LogLik = 697.38; Converged: Yes.

### Quadrature Method

$$\mu_1 = -0.0001 \tag{391}$$
$$\mu_2 = 0.0526 \tag{392}$$

$$\sigma_t^2(1) = 0.0001 + 0.0001\, y_{t-1}^2 + 0.0001\, \sigma_{t-1}^2 \tag{393}$$
$$\sigma_t^2(2) = 0.0272 + 0.5074\, y_{t-1}^2 + 0.4926\, \sigma_{t-1}^2 \tag{394}$$

$$P = \begin{pmatrix} 0.4019 & 0.5981 \\ 0.2628 & 0.7372 \end{pmatrix} \tag{395}$$

LogLik $= 579.36$; Converged: Yes.

### Particle Filter Method

$$\mu_1 = 0.0000 \tag{396}$$
$$\mu_2 = 0.0000 \tag{397}$$

$$\sigma_t^2(1) = 0.1000 + 0.1000\, y_{t-1}^2 + 0.7000\, \sigma_{t-1}^2 \tag{398}$$
$$\sigma_t^2(2) = 0.2000 + 0.2000\, y_{t-1}^2 + 0.6000\, \sigma_{t-1}^2 \tag{399}$$

$$P = \begin{pmatrix} 0.9000 & 0.1000 \\ 0.1000 & 0.9000 \end{pmatrix} \tag{400}$$

LogLik $= -1889.53$; Converged: Yes.

### Exogenous Specification

**Table 21:** HYG — Exogenous Specification

| **Method** | $c_1$ | $c_2$ | $\gamma_1$ | $\gamma_2$ | $\omega_1$ | $\alpha_1$ | $\beta_1$ | $\omega_2$ | $\alpha_2$ | $\beta_2$ | $p_{11}$ | $p_{22}$ | Conv. | Log |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Gray's | 0.0378 | -0.0003 | -0.0098 | 0.0000 | 0.0831 | 0.3436 | 0.6564 | 0.0001 | 0.3080 | 0.0001 | 0.4548 | 0.3933 | Yes | 270 |
| Plug-in | 0.0201 | -0.0001 | -0.0124 | -0.0000 | 0.0274 | 0.4540 | 0.5460 | 0.0001 | 0.0001 | 0.0001 | 0.6616 | 0.4027 | No | 745 |
| Quadrature | 0.0320 | -0.0001 | -0.0110 | 0.0000 | 0.0350 | 0.3466 | 0.6534 | 0.0001 | 0.4030 | 0.0001 | 0.4341 | 0.4188 | Yes | 260 |
| Particle Filter | 0.0000 | 0.0000 | 0.1000 | 0.1000 | 0.1000 | 0.1000 | 0.7000 | 0.2000 | 0.2000 | 0.6000 | 0.9000 | 0.9000 | Yes | -2060 |

Exogenous spec; pivot = $^{T}NX.Conv. = convergencestatus.$

### Fitted Equations (Exogenous):

### Gray's Method

$$\mu_1 = 0.0378 - 0.0098\, x_t \tag{401}$$
$$\mu_2 = -0.0003 + 0.0000\, x_t \tag{402}$$

$$\sigma_t^2(1) = 0.0831 + 0.3436\, y_{t-1}^2 + 0.6564\, \sigma_{t-1}^2 \tag{403}$$
$$\sigma_t^2(2) = 0.0001 + 0.3080\, y_{t-1}^2 + 0.0001\, \sigma_{t-1}^2 \tag{404}$$

$$P = \begin{pmatrix} 0.4548 & 0.5452 \\ 0.6067 & 0.3933 \end{pmatrix} \tag{405}$$

LogLik $= 270.36$; Converged: Yes.

**Plug-in Method**

$$\mu_1 = 0.0201 - 0.0124\, x_t \tag{406}$$
$$\mu_2 = -0.0001 - 0.0000\, x_t \tag{407}$$

$$\sigma_t^2(1) = 0.0274 + 0.4540\, y_{t-1}^2 + 0.5460\, \sigma_{t-1}^2 \tag{408}$$
$$\sigma_t^2(2) = 0.0001 + 0.0001\, y_{t-1}^2 + 0.0001\, \sigma_{t-1}^2 \tag{409}$$

$$P = \begin{pmatrix} 0.6616 & 0.3384 \\ 0.5973 & 0.4027 \end{pmatrix} \tag{410}$$

LogLik $= 745.60$; Converged: No.

**Quadrature Method**

$$\mu_1 = 0.0320 - 0.0110\, x_t \tag{411}$$
$$\mu_2 = -0.0001 + 0.0000\, x_t \tag{412}$$

$$\sigma_t^2(1) = 0.0350 + 0.3466\, y_{t-1}^2 + 0.6534\, \sigma_{t-1}^2 \tag{413}$$
$$\sigma_t^2(2) = 0.0001 + 0.4030\, y_{t-1}^2 + 0.0001\, \sigma_{t-1}^2 \tag{414}$$

$$P = \begin{pmatrix} 0.4341 & 0.5659 \\ 0.5812 & 0.4188 \end{pmatrix} \tag{415}$$

LogLik $= 260.54$; Converged: Yes.

**Particle Filter Method**

$$\mu_1 = 0.0000 + 0.1000\, x_t \tag{416}$$
$$\mu_2 = 0.0000 + 0.1000\, x_t \tag{417}$$

$$\sigma_t^2(1) = 0.1000 + 0.1000\, y_{t-1}^2 + 0.7000\, \sigma_{t-1}^2 \tag{418}$$
$$\sigma_t^2(2) = 0.2000 + 0.2000\, y_{t-1}^2 + 0.6000\, \sigma_{t-1}^2 \tag{419}$$

$$P = \begin{pmatrix} 0.9000 & 0.1000 \\ 0.1000 & 0.9000 \end{pmatrix} \tag{420}$$

LogLik $= -2060.93$; Converged: Yes.

### 2.6. Asset: `TNX`

### Baseline Specification

**Table 22:** TNX — Baseline Specification

| Method | $c_1$ | $c_2$ | $\gamma_1$ | $\gamma_2$ | $\omega_1$ | $\alpha_1$ | $\beta_1$ | $\omega_2$ | $\alpha_2$ | $\beta_2$ | $p_{11}$ | $p_{22}$ | Conv. | LogLik |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Gray's | -0.1184 | -0.1153 | – | – | 0.2283 | 0.1209 | 0.8788 | 0.0404 | 0.0723 | 0.7736 | 0.9332 | 0.8257 | No | -5744.47 |
| Plug-in | 0.0895 | 0.1516 | – | – | 0.1531 | 0.1622 | 0.8344 | 0.2504 | 0.2472 | 0.7193 | 0.9065 | 0.8745 | No | -5752.84 |
| Quadrature | -0.1040 | 0.0193 | – | – | 0.3746 | 0.0622 | 0.7342 | 0.0237 | 0.0999 | 0.9000 | 0.7449 | 0.9900 | Yes | -5708.41 |
| Particle Filter | -0.0000 | -0.0000 | – | – | 0.3000 | 0.0500 | 0.8500 | 0.1000 | 0.0800 | 0.7800 | 0.9500 | 0.8000 | Yes | -5821.83 |

Baseline spec: $\gamma_1 = \gamma_2 = 0$. Conv. = convergence status.

### Fitted Equations (Baseline):

### Gray's Method

$$\mu_1 = -0.1184 \tag{421}$$
$$\mu_2 = -0.1153 \tag{422}$$

$$\sigma_t^2(1) = 0.2283 + 0.1209\, y_{t-1}^2 + 0.8788\, \sigma_{t-1}^2 \tag{423}$$
$$\sigma_t^2(2) = 0.0404 + 0.0723\, y_{t-1}^2 + 0.7736\, \sigma_{t-1}^2 \tag{424}$$

$$P = \begin{pmatrix} 0.9332 & 0.0668 \\ 0.1743 & 0.8257 \end{pmatrix} \tag{425}$$

LogLik $= -5744.47$; Converged: No.

### Plug-in Method

$$\mu_1 = 0.0895 \tag{426}$$
$$\mu_2 = 0.1516 \tag{427}$$

$$\sigma_t^2(1) = 0.1531 + 0.1622\, y_{t-1}^2 + 0.8344\, \sigma_{t-1}^2 \tag{428}$$
$$\sigma_t^2(2) = 0.2504 + 0.2472\, y_{t-1}^2 + 0.7193\, \sigma_{t-1}^2 \tag{429}$$

$$P = \begin{pmatrix} 0.9065 & 0.0935 \\ 0.1255 & 0.8745 \end{pmatrix} \tag{430}$$

LogLik $= -5752.84$; Converged: No.

**Quadrature Method**

$$\mu_1 = -0.1040 \tag{431}$$
$$\mu_2 = 0.0193 \tag{432}$$

$$\sigma_t^2(1) = 0.3746 + 0.0622\, y_{t-1}^2 + 0.7342\, \sigma_{t-1}^2 \tag{433}$$
$$\sigma_t^2(2) = 0.0237 + 0.0999\, y_{t-1}^2 + 0.9000\, \sigma_{t-1}^2 \tag{434}$$

$$P = \begin{pmatrix} 0.7449 & 0.2551 \\ 0.0100 & 0.9900 \end{pmatrix} \tag{435}$$

LogLik $= -5708.41$; Converged: Yes.

**Particle Filter Method**

$$\mu_1 = -0.0000 \tag{436}$$
$$\mu_2 = -0.0000 \tag{437}$$

$$\sigma_t^2(1) = 0.3000 + 0.0500\, y_{t-1}^2 + 0.8500\, \sigma_{t-1}^2 \tag{438}$$
$$\sigma_t^2(2) = 0.1000 + 0.0800\, y_{t-1}^2 + 0.7800\, \sigma_{t-1}^2 \tag{439}$$

$$P = \begin{pmatrix} 0.9500 & 0.0500 \\ 0.2000 & 0.8000 \end{pmatrix} \tag{440}$$

LogLik $= -5821.83$; Converged: Yes.

## 3. Cryptocurrency Market

Pivot (exogenous conditioning variable): `BTC-USD`.

### 3.1. Asset: `ETH-USD`

#### Baseline Specification

**Table 23:** ETH-USD — Baseline Specification

| **Method** | $c_1$ | $c_2$ | $\gamma_1$ | $\gamma_2$ | $\omega_1$ | $\alpha_1$ | $\beta_1$ | $\omega_2$ | $\alpha_2$ | $\beta_2$ | $p_{11}$ | $p_{22}$ | Conv. | LogLik |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Gray's | 0.6126 | -0.6393 | – | – | 0.5422 | 0.0598 | 0.9000 | 0.3453 | 0.1928 | 0.8057 | 0.9045 | 0.7191 | Yes | -8106.68 |
| Plug-in | 0.3134 | 0.1208 | – | – | 0.4091 | 0.1081 | 0.8919 | 0.2186 | 0.1249 | 0.7828 | 0.9506 | 0.7508 | No | -8073.57 |
| Quadrature | 0.4393 | -0.4404 | – | – | 0.4501 | 0.1080 | 0.8919 | 0.2665 | 0.1473 | 0.7954 | 0.9554 | 0.7309 | Yes | -8120.74 |
| Particle Filter | 0.0000 | -0.0000 | – | – | 0.3000 | 0.0500 | 0.8500 | 0.1000 | 0.0800 | 0.7800 | 0.9500 | 0.8000 | Yes | -9075.70 |

Baseline spec: $\gamma_1 = \gamma_2 = 0$. Conv. = convergence status.

#### Fitted Equations (Baseline):

#### Gray's Method

$$\mu_1 = 0.6126 \tag{441}$$
$$\mu_2 = -0.6393 \tag{442}$$

$$\sigma_t^2(1) = 0.5422 + 0.0598\, y_{t-1}^2 + 0.9000\, \sigma_{t-1}^2 \tag{443}$$
$$\sigma_t^2(2) = 0.3453 + 0.1928\, y_{t-1}^2 + 0.8057\, \sigma_{t-1}^2 \tag{444}$$

$$P = \begin{pmatrix} 0.9045 & 0.0955 \\ 0.2809 & 0.7191 \end{pmatrix} \tag{445}$$

LogLik $= -8106.68$; Converged: Yes.

#### Plug-in Method

$$\mu_1 = 0.3134 \tag{446}$$
$$\mu_2 = 0.1208 \tag{447}$$

$$\sigma_t^2(1) = 0.4091 + 0.1081\, y_{t-1}^2 + 0.8919\, \sigma_{t-1}^2 \tag{448}$$
$$\sigma_t^2(2) = 0.2186 + 0.1249\, y_{t-1}^2 + 0.7828\, \sigma_{t-1}^2 \tag{449}$$

$$P = \begin{pmatrix} 0.9506 & 0.0494 \\ 0.2492 & 0.7508 \end{pmatrix} \tag{450}$$

LogLik $= -8073.57$; Converged: No.

### Quadrature Method

$$\mu_1 = 0.4393 \tag{451}$$
$$\mu_2 = -0.4404 \tag{452}$$

$$\sigma_t^2(1) = 0.4501 + 0.1080\, y_{t-1}^2 + 0.8919\, \sigma_{t-1}^2 \tag{453}$$
$$\sigma_t^2(2) = 0.2665 + 0.1473\, y_{t-1}^2 + 0.7954\, \sigma_{t-1}^2 \tag{454}$$

$$P = \begin{pmatrix} 0.9554 & 0.0446 \\ 0.2691 & 0.7309 \end{pmatrix} \tag{455}$$

LogLik $= -8120.74$; Converged: Yes.

### Particle Filter Method

$$\mu_1 = 0.0000 \tag{456}$$
$$\mu_2 = -0.0000 \tag{457}$$

$$\sigma_t^2(1) = 0.3000 + 0.0500\, y_{t-1}^2 + 0.8500\, \sigma_{t-1}^2 \tag{458}$$
$$\sigma_t^2(2) = 0.1000 + 0.0800\, y_{t-1}^2 + 0.7800\, \sigma_{t-1}^2 \tag{459}$$

$$P = \begin{pmatrix} 0.9500 & 0.0500 \\ 0.2000 & 0.8000 \end{pmatrix} \tag{460}$$

LogLik $= -9075.70$; Converged: Yes.

### Exogenous Specification

**Table 24:** ETH-USD — Exogenous Specification

| **Method** | $c_1$ | $c_2$ | $\gamma_1$ | $\gamma_2$ | $\omega_1$ | $\alpha_1$ | $\beta_1$ | $\omega_2$ | $\alpha_2$ | $\beta_2$ | $p_{11}$ | $p_{22}$ | Conv. | LogL |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Gray's | 0.4167 | -0.5453 | -0.0869 | 0.5502 | 0.4877 | 0.0668 | 0.8936 | 0.3127 | 0.1736 | 0.8255 | 0.9791 | 0.7073 | Yes | -8072. |
| Plug-in | 0.5905 | 0.5930 | -0.6630 | 0.7643 | 0.6762 | 0.2424 | 0.6657 | 0.5989 | 0.3196 | 0.6704 | 0.7854 | 0.8105 | No | -8295. |
| Quadrature | 0.2852 | -0.3721 | -0.0624 | 0.4278 | 0.4565 | 0.0608 | 0.9000 | 0.2452 | 0.1417 | 0.8569 | 0.9653 | 0.7370 | Yes | -8072. |
| Particle Filter | 0.0000 | 0.0000 | 0.0500 | 0.0500 | 0.3000 | 0.0500 | 0.8500 | 0.1000 | 0.0800 | 0.7800 | 0.9500 | 0.8000 | Yes | -9071. |

Exogenous spec; pivot = `BTC-USD`. Conv. = convergence status.

### Fitted Equations (Exogenous):

### Gray's Method

$$\mu_1 = 0.4167 - 0.0869\, x_t \tag{461}$$
$$\mu_2 = -0.5453 + 0.5502\, x_t \tag{462}$$

$$\sigma_t^2(1) = 0.4877 + 0.0668\, y_{t-1}^2 + 0.8936\, \sigma_{t-1}^2 \tag{463}$$
$$\sigma_t^2(2) = 0.3127 + 0.1736\, y_{t-1}^2 + 0.8255\, \sigma_{t-1}^2 \tag{464}$$

$$P = \begin{pmatrix} 0.9791 & 0.0209 \\ 0.2927 & 0.7073 \end{pmatrix} \tag{465}$$

LogLik $= -8072.60$; Converged: Yes.

**Plug-in Method**

$$\mu_1 = 0.5905 - 0.6630\, x_t \tag{466}$$
$$\mu_2 = 0.5930 + 0.7643\, x_t \tag{467}$$

$$\sigma_t^2(1) = 0.6762 + 0.2424\, y_{t-1}^2 + 0.6657\, \sigma_{t-1}^2 \tag{468}$$
$$\sigma_t^2(2) = 0.5989 + 0.3196\, y_{t-1}^2 + 0.6704\, \sigma_{t-1}^2 \tag{469}$$

$$P = \begin{pmatrix} 0.7854 & 0.2146 \\ 0.1895 & 0.8105 \end{pmatrix} \tag{470}$$

LogLik $= -8295.24$; Converged: No.

**Quadrature Method**

$$\mu_1 = 0.2852 - 0.0624\, x_t \tag{471}$$
$$\mu_2 = -0.3721 + 0.4278\, x_t \tag{472}$$

$$\sigma_t^2(1) = 0.4565 + 0.0608\, y_{t-1}^2 + 0.9000\, \sigma_{t-1}^2 \tag{473}$$
$$\sigma_t^2(2) = 0.2452 + 0.1417\, y_{t-1}^2 + 0.8569\, \sigma_{t-1}^2 \tag{474}$$

$$P = \begin{pmatrix} 0.9653 & 0.0347 \\ 0.2630 & 0.7370 \end{pmatrix} \tag{475}$$

LogLik $= -8072.25$; Converged: Yes.

**Particle Filter Method**

$$\mu_1 = 0.0000 + 0.0500\, x_t \tag{476}$$
$$\mu_2 = 0.0000 + 0.0500\, x_t \tag{477}$$

$$\sigma_t^2(1) = 0.3000 + 0.0500\, y_{t-1}^2 + 0.8500\, \sigma_{t-1}^2 \tag{478}$$
$$\sigma_t^2(2) = 0.1000 + 0.0800\, y_{t-1}^2 + 0.7800\, \sigma_{t-1}^2 \tag{479}$$

$$P = \begin{pmatrix} 0.9500 & 0.0500 \\ 0.2000 & 0.8000 \end{pmatrix} \tag{480}$$

LogLik $= -9071.62$; Converged: Yes.

### 3.2. Asset: `LTC-USD`

#### Baseline Specification

**Table 25:** LTC-USD — Baseline Specification

| **Method** | $c_1$ | $c_2$ | $\gamma_1$ | $\gamma_2$ | $\omega_1$ | $\alpha_1$ | $\beta_1$ | $\omega_2$ | $\alpha_2$ | $\beta_2$ | $p_{11}$ | $p_{22}$ | Conv. | LogLik |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Gray's | 0.4181 | -0.3490 | – | – | 0.4426 | 0.1128 | 0.8867 | 0.2586 | 0.1439 | 0.7944 | 0.9563 | 0.7337 | No | -8316.54 |
| Plug-in | -0.0339 | 0.1039 | – | – | 0.4205 | 0.1101 | 0.8898 | 0.2297 | 0.1314 | 0.7840 | 0.9508 | 0.7462 | No | -8269.21 |
| Quadrature | 0.4244 | -0.3186 | – | – | 0.4449 | 0.1151 | 0.8848 | 0.2611 | 0.1455 | 0.7943 | 0.9563 | 0.7327 | No | -8317.06 |
| Particle Filter | -0.0000 | 0.0000 | – | – | 0.3000 | 0.0500 | 0.8500 | 0.1000 | 0.0800 | 0.7800 | 0.9500 | 0.8000 | No | -9342.67 |

Baseline spec: $\gamma_1 = \gamma_2 = 0$. Conv. = convergence status.

#### Fitted Equations (Baseline):

#### Gray's Method

$$\mu_1 = 0.4181 \tag{481}$$
$$\mu_2 = -0.3490 \tag{482}$$

$$\sigma_t^2(1) = 0.4426 + 0.1128\, y_{t-1}^2 + 0.8867\, \sigma_{t-1}^2 \tag{483}$$
$$\sigma_t^2(2) = 0.2586 + 0.1439\, y_{t-1}^2 + 0.7944\, \sigma_{t-1}^2 \tag{484}$$

$$P = \begin{pmatrix} 0.9563 & 0.0437 \\ 0.2663 & 0.7337 \end{pmatrix} \tag{485}$$

LogLik $= -8316.54$; Converged: No.

#### Plug-in Method

$$\mu_1 = -0.0339 \tag{486}$$
$$\mu_2 = 0.1039 \tag{487}$$

$$\sigma_t^2(1) = 0.4205 + 0.1101\, y_{t-1}^2 + 0.8898\, \sigma_{t-1}^2 \tag{488}$$
$$\sigma_t^2(2) = 0.2297 + 0.1314\, y_{t-1}^2 + 0.7840\, \sigma_{t-1}^2 \tag{489}$$

$$P = \begin{pmatrix} 0.9508 & 0.0492 \\ 0.2538 & 0.7462 \end{pmatrix} \tag{490}$$

LogLik $= -8269.21$; Converged: No.

### Quadrature Method

$$\mu_1 = 0.4244 \tag{491}$$
$$\mu_2 = -0.3186 \tag{492}$$

$$\sigma_t^2(1) = 0.4449 + 0.1151\, y_{t-1}^2 + 0.8848\, \sigma_{t-1}^2 \tag{493}$$
$$\sigma_t^2(2) = 0.2611 + 0.1455\, y_{t-1}^2 + 0.7943\, \sigma_{t-1}^2 \tag{494}$$

$$P = \begin{pmatrix} 0.9563 & 0.0437 \\ 0.2673 & 0.7327 \end{pmatrix} \tag{495}$$

LogLik $= -8317.06$; Converged: No.

### Particle Filter Method

$$\mu_1 = -0.0000 \tag{496}$$
$$\mu_2 = 0.0000 \tag{497}$$

$$\sigma_t^2(1) = 0.3000 + 0.0500\, y_{t-1}^2 + 0.8500\, \sigma_{t-1}^2 \tag{498}$$
$$\sigma_t^2(2) = 0.1000 + 0.0800\, y_{t-1}^2 + 0.7800\, \sigma_{t-1}^2 \tag{499}$$

$$P = \begin{pmatrix} 0.9500 & 0.0500 \\ 0.2000 & 0.8000 \end{pmatrix} \tag{500}$$

LogLik $= -9342.67$; Converged: No.

### Exogenous Specification

**Table 26:** LTC-USD — Exogenous Specification

| Method | $c_1$ | $c_2$ | $\gamma_1$ | $\gamma_2$ | $\omega_1$ | $\alpha_1$ | $\beta_1$ | $\omega_2$ | $\alpha_2$ | $\beta_2$ | $p_{11}$ | $p_{22}$ | Conv. | Log |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Gray's | 0.2039 | -0.1757 | -0.1141 | 0.2751 | 0.4092 | 0.0845 | 0.9000 | 0.1869 | 0.1355 | 0.8606 | 0.9884 | 0.7561 | Yes | -8292 |
| Plug-in | -0.6148 | 0.6157 | -0.5239 | 0.6990 | 0.3360 | 0.2033 | 0.7488 | 0.4238 | 0.3039 | 0.6946 | 0.8040 | 0.8736 | No | -8471 |
| Quadrature | 0.1923 | -0.1625 | -0.1149 | 0.2713 | 0.4130 | 0.0865 | 0.9000 | 0.1897 | 0.1409 | 0.8585 | 0.9807 | 0.7488 | Yes | -8291 |
| Particle Filter | 0.0000 | 0.0000 | 0.0500 | 0.0500 | 0.3000 | 0.0500 | 0.8500 | 0.1000 | 0.0800 | 0.7800 | 0.9500 | 0.8000 | No | -9342 |

Exogenous spec; pivot = `BTC-USD`. Conv. = convergence status.

### Fitted Equations (Exogenous):

### Gray's Method

$$\mu_1 = 0.2039 - 0.1141\, x_t \tag{501}$$
$$\mu_2 = -0.1757 + 0.2751\, x_t \tag{502}$$

$$\sigma_t^2(1) = 0.4092 + 0.0845\, y_{t-1}^2 + 0.9000\, \sigma_{t-1}^2 \tag{503}$$
$$\sigma_t^2(2) = 0.1869 + 0.1355\, y_{t-1}^2 + 0.8606\, \sigma_{t-1}^2 \tag{504}$$

$$P = \begin{pmatrix} 0.9884 & 0.0116 \\ 0.2439 & 0.7561 \end{pmatrix} \tag{505}$$

LogLik $= -8292.09$; Converged: Yes.

**Plug-in Method**

$$\mu_1 = -0.6148 - 0.5239\, x_t \tag{506}$$
$$\mu_2 = 0.6157 + 0.6990\, x_t \tag{507}$$

$$\sigma_t^2(1) = 0.3360 + 0.2033\, y_{t-1}^2 + 0.7488\, \sigma_{t-1}^2 \tag{508}$$
$$\sigma_t^2(2) = 0.4238 + 0.3039\, y_{t-1}^2 + 0.6946\, \sigma_{t-1}^2 \tag{509}$$

$$P = \begin{pmatrix} 0.8040 & 0.1960 \\ 0.1264 & 0.8736 \end{pmatrix} \tag{510}$$

LogLik $= -8471.67$; Converged: No.

**Quadrature Method**

$$\mu_1 = 0.1923 - 0.1149\, x_t \tag{511}$$
$$\mu_2 = -0.1625 + 0.2713\, x_t \tag{512}$$

$$\sigma_t^2(1) = 0.4130 + 0.0865\, y_{t-1}^2 + 0.9000\, \sigma_{t-1}^2 \tag{513}$$
$$\sigma_t^2(2) = 0.1897 + 0.1409\, y_{t-1}^2 + 0.8585\, \sigma_{t-1}^2 \tag{514}$$

$$P = \begin{pmatrix} 0.9807 & 0.0193 \\ 0.2512 & 0.7488 \end{pmatrix} \tag{515}$$

LogLik $= -8291.69$; Converged: Yes.

**Particle Filter Method**

$$\mu_1 = 0.0000 + 0.0500\, x_t \tag{516}$$
$$\mu_2 = 0.0000 + 0.0500\, x_t \tag{517}$$

$$\sigma_t^2(1) = 0.3000 + 0.0500\, y_{t-1}^2 + 0.8500\, \sigma_{t-1}^2 \tag{518}$$
$$\sigma_t^2(2) = 0.1000 + 0.0800\, y_{t-1}^2 + 0.7800\, \sigma_{t-1}^2 \tag{519}$$

$$P = \begin{pmatrix} 0.9500 & 0.0500 \\ 0.2000 & 0.8000 \end{pmatrix} \tag{520}$$

LogLik $= -9342.78$; Converged: No.

### 3.3. Asset: `XRP-USD`

### Baseline Specification

**Table 27:** XRP-USD — Baseline Specification

| **Method** | $c_1$ | $c_2$ | $\gamma_1$ | $\gamma_2$ | $\omega_1$ | $\alpha_1$ | $\beta_1$ | $\omega_2$ | $\alpha_2$ | $\beta_2$ | $p_{11}$ | $p_{22}$ | Conv. | LogLik |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Gray's | 0.4434 | 0.6430 | – | – | 0.3379 | 0.1705 | 0.7626 | 0.4197 | 0.2863 | 0.7136 | 0.8160 | 0.8847 | No | -9143.45 |
| Plug-in | 0.4248 | -0.4242 | – | – | 0.4469 | 0.1223 | 0.8774 | 0.2607 | 0.1424 | 0.7849 | 0.9489 | 0.7341 | No | -8822.00 |
| Quadrature | 1.4539 | -1.4449 | – | – | 0.8731 | 0.1863 | 0.8134 | 0.6844 | 0.3883 | 0.5541 | 0.7274 | 0.7523 | No | -8514.52 |
| Particle Filter | -0.0000 | -0.0000 | – | – | 0.3000 | 0.0500 | 0.8500 | 0.1000 | 0.0800 | 0.7800 | 0.9500 | 0.8000 | Yes | -10373.34 |

Baseline spec: $\gamma_1 = \gamma_2 = 0$. Conv. = convergence status.

### Fitted Equations (Baseline):

### Gray's Method

$$\mu_1 = 0.4434 \tag{521}$$
$$\mu_2 = 0.6430 \tag{522}$$

$$\sigma_t^2(1) = 0.3379 + 0.1705\, y_{t-1}^2 + 0.7626\, \sigma_{t-1}^2 \tag{523}$$
$$\sigma_t^2(2) = 0.4197 + 0.2863\, y_{t-1}^2 + 0.7136\, \sigma_{t-1}^2 \tag{524}$$

$$P = \begin{pmatrix} 0.8160 & 0.1840 \\ 0.1153 & 0.8847 \end{pmatrix} \tag{525}$$

LogLik $= -9143.45$; Converged: No.

### Plug-in Method

$$\mu_1 = 0.4248 \tag{526}$$
$$\mu_2 = -0.4242 \tag{527}$$

$$\sigma_t^2(1) = 0.4469 + 0.1223\, y_{t-1}^2 + 0.8774\, \sigma_{t-1}^2 \tag{528}$$
$$\sigma_t^2(2) = 0.2607 + 0.1424\, y_{t-1}^2 + 0.7849\, \sigma_{t-1}^2 \tag{529}$$

$$P = \begin{pmatrix} 0.9489 & 0.0511 \\ 0.2659 & 0.7341 \end{pmatrix} \tag{530}$$

LogLik $= -8822.00$; Converged: No.

### Quadrature Method

$$\mu_1 = 1.4539 \tag{531}$$
$$\mu_2 = -1.4449 \tag{532}$$

$$\sigma_t^2(1) = 0.8731 + 0.1863\, y_{t-1}^2 + 0.8134\, \sigma_{t-1}^2 \tag{533}$$
$$\sigma_t^2(2) = 0.6844 + 0.3883\, y_{t-1}^2 + 0.5541\, \sigma_{t-1}^2 \tag{534}$$

$$P = \begin{pmatrix} 0.7274 & 0.2726 \\ 0.2477 & 0.7523 \end{pmatrix} \tag{535}$$

LogLik $= -8514.52$; Converged: No.

### Particle Filter Method

$$\mu_1 = -0.0000 \tag{536}$$
$$\mu_2 = -0.0000 \tag{537}$$

$$\sigma_t^2(1) = 0.3000 + 0.0500\, y_{t-1}^2 + 0.8500\, \sigma_{t-1}^2 \tag{538}$$
$$\sigma_t^2(2) = 0.1000 + 0.0800\, y_{t-1}^2 + 0.7800\, \sigma_{t-1}^2 \tag{539}$$

$$P = \begin{pmatrix} 0.9500 & 0.0500 \\ 0.2000 & 0.8000 \end{pmatrix} \tag{540}$$

LogLik $= -10373.34$; Converged: Yes.

### Exogenous Specification

**Table 28:** XRP-USD — Exogenous Specification

| **Method** | $c_1$ | $c_2$ | $\gamma_1$ | $\gamma_2$ | $\omega_1$ | $\alpha_1$ | $\beta_1$ | $\omega_2$ | $\alpha_2$ | $\beta_2$ | $p_{11}$ | $p_{22}$ | Conv. | Lo |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Gray's | 0.4513 | -0.4648 | -0.3028 | 0.4799 | 0.5408 | 0.0350 | 0.9000 | 0.2814 | 0.1950 | 0.8012 | 0.9489 | 0.7324 | No | -852 |
| Plug-in | 0.4094 | -0.4084 | -0.3651 | -0.3614 | 0.4399 | 0.1168 | 0.8829 | 0.2545 | 0.1348 | 0.7829 | 0.9479 | 0.7367 | No | -868 |
| Quadrature | 0.5024 | -0.5475 | -0.2190 | 0.3982 | 0.5735 | 0.0455 | 0.9000 | 0.3131 | 0.2130 | 0.7869 | 0.9438 | 0.7205 | Yes | -850 |
| Particle Filter | 0.0032 | 0.0032 | 0.0532 | 0.0467 | 0.3011 | 0.0505 | 0.8500 | 0.0999 | 0.0799 | 0.7801 | 0.9500 | 0.8001 | Yes | -1036 |

Exogenous spec; pivot = `BTC-USD`. Conv. = convergence status.

### Fitted Equations (Exogenous):

### Gray's Method

$$\mu_1 = 0.4513 - 0.3028\, x_t \tag{541}$$
$$\mu_2 = -0.4648 + 0.4799\, x_t \tag{542}$$

$$\sigma_t^2(1) = 0.5408 + 0.0350\,y_{t-1}^2 + 0.9000\,\sigma_{t-1}^2 \tag{543}$$

$$\sigma_t^2(2) = 0.2814 + 0.1950\,y_{t-1}^2 + 0.8012\,\sigma_{t-1}^2 \tag{544}$$

$$P = \begin{pmatrix} 0.9489 & 0.0511 \\ 0.2676 & 0.7324 \end{pmatrix} \tag{545}$$

LogLik $= -8520.39$; Converged: No.

**Plug-in Method**

$$\mu_1 = 0.4094 - 0.3651\,x_t \tag{546}$$

$$\mu_2 = -0.4084 - 0.3614\,x_t \tag{547}$$

$$\sigma_t^2(1) = 0.4399 + 0.1168\,y_{t-1}^2 + 0.8829\,\sigma_{t-1}^2 \tag{548}$$

$$\sigma_t^2(2) = 0.2545 + 0.1348\,y_{t-1}^2 + 0.7829\,\sigma_{t-1}^2 \tag{549}$$

$$P = \begin{pmatrix} 0.9479 & 0.0521 \\ 0.2633 & 0.7367 \end{pmatrix} \tag{550}$$

LogLik $= -8686.18$; Converged: No.

**Quadrature Method**

$$\mu_1 = 0.5024 - 0.2190\,x_t \tag{551}$$

$$\mu_2 = -0.5475 + 0.3982\,x_t \tag{552}$$

$$\sigma_t^2(1) = 0.5735 + 0.0455\,y_{t-1}^2 + 0.9000\,\sigma_{t-1}^2 \tag{553}$$

$$\sigma_t^2(2) = 0.3131 + 0.2130\,y_{t-1}^2 + 0.7869\,\sigma_{t-1}^2 \tag{554}$$

$$P = \begin{pmatrix} 0.9438 & 0.0562 \\ 0.2795 & 0.7205 \end{pmatrix} \tag{555}$$

LogLik $= -8507.94$; Converged: Yes.

**Particle Filter Method**

$$\mu_1 = 0.0032 + 0.0532\,x_t \tag{556}$$

$$\mu_2 = 0.0032 + 0.0467\,x_t \tag{557}$$

$$\sigma_t^2(1) = 0.3011 + 0.0505\,y_{t-1}^2 + 0.8500\,\sigma_{t-1}^2 \tag{558}$$

$$\sigma_t^2(2) = 0.0999 + 0.0799\,y_{t-1}^2 + 0.7801\,\sigma_{t-1}^2 \tag{559}$$

$$P = \begin{pmatrix} 0.9500 & 0.0500 \\ 0.1999 & 0.8001 \end{pmatrix} \tag{560}$$

LogLik $= -10360.16$; Converged: Yes.

### 3.4. Asset: `DOGE-USD`

### Baseline Specification

**Table 29:** DOGE-USD — Baseline Specification

| Method | $c_1$ | $c_2$ | $\gamma_1$ | $\gamma_2$ | $\omega_1$ | $\alpha_1$ | $\beta_1$ | $\omega_2$ | $\alpha_2$ | $\beta_2$ | $p_{11}$ | $p_{22}$ | Conv. | LogLik |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Gray's | 0.4167 | 0.0367 | – | – | 0.4397 | 0.1063 | 0.8935 | 0.2577 | 0.1609 | 0.8029 | 0.9631 | 0.7319 | No | -8752.64 |
| Plug-in | 0.2567 | -0.2505 | – | – | 0.3865 | 0.1072 | 0.8928 | 0.1955 | 0.1089 | 0.7798 | 0.9476 | 0.7612 | No | -8613.45 |
| Quadrature | 0.7571 | 0.7576 | – | – | 0.3856 | 0.2121 | 0.7378 | 0.4723 | 0.3139 | 0.6809 | 0.7704 | 0.9076 | No | -9069.31 |
| Particle Filter | 0.0000 | 0.0000 | – | – | 0.3000 | 0.0500 | 0.8500 | 0.1000 | 0.0800 | 0.7800 | 0.9500 | 0.8000 | No | -10260.15 |

Baseline spec: $\gamma_1 = \gamma_2 = 0$. Conv. = convergence status.

### Fitted Equations (Baseline):

### Gray's Method

$$\mu_1 = 0.4167 \tag{561}$$
$$\mu_2 = 0.0367 \tag{562}$$

$$\sigma_t^2(1) = 0.4397 + 0.1063\, y_{t-1}^2 + 0.8935\, \sigma_{t-1}^2 \tag{563}$$
$$\sigma_t^2(2) = 0.2577 + 0.1609\, y_{t-1}^2 + 0.8029\, \sigma_{t-1}^2 \tag{564}$$

$$P = \begin{pmatrix} 0.9631 & 0.0369 \\ 0.2681 & 0.7319 \end{pmatrix} \tag{565}$$

LogLik $= -8752.64$; Converged: No.

### Plug-in Method

$$\mu_1 = 0.2567 \tag{566}$$
$$\mu_2 = -0.2505 \tag{567}$$

$$\sigma_t^2(1) = 0.3865 + 0.1072\, y_{t-1}^2 + 0.8928\, \sigma_{t-1}^2 \tag{568}$$
$$\sigma_t^2(2) = 0.1955 + 0.1089\, y_{t-1}^2 + 0.7798\, \sigma_{t-1}^2 \tag{569}$$

$$P = \begin{pmatrix} 0.9476 & 0.0524 \\ 0.2388 & 0.7612 \end{pmatrix} \tag{570}$$

LogLik $= -8613.45$; Converged: No.

### Quadrature Method

$$\mu_1 = 0.7571 \tag{571}$$
$$\mu_2 = 0.7576 \tag{572}$$

$$\sigma_t^2(1) = 0.3856 + 0.2121\, y_{t-1}^2 + 0.7378\, \sigma_{t-1}^2 \tag{573}$$
$$\sigma_t^2(2) = 0.4723 + 0.3139\, y_{t-1}^2 + 0.6809\, \sigma_{t-1}^2 \tag{574}$$

$$P = \begin{pmatrix} 0.7704 & 0.2296 \\ 0.0924 & 0.9076 \end{pmatrix} \tag{575}$$

LogLik $= -9069.31$; Converged: No.

### Particle Filter Method

$$\mu_1 = 0.0000 \tag{576}$$
$$\mu_2 = 0.0000 \tag{577}$$

$$\sigma_t^2(1) = 0.3000 + 0.0500\, y_{t-1}^2 + 0.8500\, \sigma_{t-1}^2 \tag{578}$$
$$\sigma_t^2(2) = 0.1000 + 0.0800\, y_{t-1}^2 + 0.7800\, \sigma_{t-1}^2 \tag{579}$$

$$P = \begin{pmatrix} 0.9500 & 0.0500 \\ 0.2000 & 0.8000 \end{pmatrix} \tag{580}$$

LogLik $= -10260.15$; Converged: No.

### Exogenous Specification

**Table 30:** DOGE-USD — Exogenous Specification

| **Method** | $c_1$ | $c_2$ | $\gamma_1$ | $\gamma_2$ | $\omega_1$ | $\alpha_1$ | $\beta_1$ | $\omega_2$ | $\alpha_2$ | $\beta_2$ | $p_{11}$ | $p_{22}$ | Conv. | Log |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Gray's | 0.3375 | 0.1429 | -0.2900 | 0.3867 | 0.4231 | 0.1150 | 0.8845 | 0.2385 | 0.1427 | 0.7962 | 0.9397 | 0.7458 | No | -8743 |
| Plug-in | -0.6670 | 0.6699 | -0.5781 | 0.7548 | 0.3549 | 0.2127 | 0.7381 | 0.4377 | 0.3194 | 0.6804 | 0.7972 | 0.8755 | No | -8872 |
| Quadrature | 0.3266 | 0.1761 | -0.2776 | 0.3775 | 0.4121 | 0.1128 | 0.8872 | 0.2246 | 0.1395 | 0.7981 | 0.9400 | 0.7511 | Yes | -8741 |
| Particle Filter | 0.0000 | 0.0000 | 0.0500 | 0.0500 | 0.3000 | 0.0500 | 0.8500 | 0.1000 | 0.0800 | 0.7800 | 0.9500 | 0.8000 | Yes | -10266 |

Exogenous spec; pivot = `BTC-USD`. Conv. = convergence status.

### Fitted Equations (Exogenous):

### Gray's Method

$$\mu_1 = 0.3375 - 0.2900\, x_t \tag{581}$$
$$\mu_2 = 0.1429 + 0.3867\, x_t \tag{582}$$

$$\sigma_t^2(1) = 0.4231 + 0.1150\, y_{t-1}^2 + 0.8845\, \sigma_{t-1}^2 \tag{583}$$
$$\sigma_t^2(2) = 0.2385 + 0.1427\, y_{t-1}^2 + 0.7962\, \sigma_{t-1}^2 \tag{584}$$

$$P = \begin{pmatrix} 0.9397 & 0.0603 \\ 0.2542 & 0.7458 \end{pmatrix} \tag{585}$$

LogLik $= -8743.14$; Converged: No.

**Plug-in Method**

$$\mu_1 = -0.6670 - 0.5781\, x_t \tag{586}$$
$$\mu_2 = 0.6699 + 0.7548\, x_t \tag{587}$$

$$\sigma_t^2(1) = 0.3549 + 0.2127\, y_{t-1}^2 + 0.7381\, \sigma_{t-1}^2 \tag{588}$$
$$\sigma_t^2(2) = 0.4377 + 0.3194\, y_{t-1}^2 + 0.6804\, \sigma_{t-1}^2 \tag{589}$$

$$P = \begin{pmatrix} 0.7972 & 0.2028 \\ 0.1245 & 0.8755 \end{pmatrix} \tag{590}$$

LogLik $= -8872.36$; Converged: No.

**Quadrature Method**

$$\mu_1 = 0.3266 - 0.2776\, x_t \tag{591}$$
$$\mu_2 = 0.1761 + 0.3775\, x_t \tag{592}$$

$$\sigma_t^2(1) = 0.4121 + 0.1128\, y_{t-1}^2 + 0.8872\, \sigma_{t-1}^2 \tag{593}$$
$$\sigma_t^2(2) = 0.2246 + 0.1395\, y_{t-1}^2 + 0.7981\, \sigma_{t-1}^2 \tag{594}$$

$$P = \begin{pmatrix} 0.9400 & 0.0600 \\ 0.2489 & 0.7511 \end{pmatrix} \tag{595}$$

LogLik $= -8741.70$; Converged: Yes.

**Particle Filter Method**

$$\mu_1 = 0.0000 + 0.0500\, x_t \tag{596}$$
$$\mu_2 = 0.0000 + 0.0500\, x_t \tag{597}$$

$$\sigma_t^2(1) = 0.3000 + 0.0500\, y_{t-1}^2 + 0.8500\, \sigma_{t-1}^2 \tag{598}$$
$$\sigma_t^2(2) = 0.1000 + 0.0800\, y_{t-1}^2 + 0.7800\, \sigma_{t-1}^2 \tag{599}$$

$$P = \begin{pmatrix} 0.9500 & 0.0500 \\ 0.2000 & 0.8000 \end{pmatrix} \tag{600}$$

LogLik $= -10266.08$; Converged: Yes.

### 3.5. Asset: `NMC-USD`

#### Baseline Specification

**Table 31:** NMC-USD — Baseline Specification

| Method | $c_1$ | $c_2$ | $\gamma_1$ | $\gamma_2$ | $\omega_1$ | $\alpha_1$ | $\beta_1$ | $\omega_2$ | $\alpha_2$ | $\beta_2$ | $p_{11}$ | $p_{22}$ | Conv. | LogLik |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Gray's | 0.3516 | 0.3519 | – | – | 0.4205 | 0.1121 | 0.8861 | 0.2336 | 0.1310 | 0.7922 | 0.9638 | 0.7428 | No | -9126.19 |
| Plug-in | -0.3969 | 0.7938 | – | – | 0.4041 | 0.2430 | 0.7560 | 0.4814 | 0.2948 | 0.6342 | 0.7760 | 0.8951 | No | -9326.54 |
| Quadrature | 0.3394 | 0.3397 | – | – | 0.4165 | 0.1107 | 0.8879 | 0.2289 | 0.1276 | 0.7921 | 0.9659 | 0.7444 | No | -9121.91 |
| Particle Filter | -0.0000 | 0.0000 | – | – | 0.3000 | 0.0500 | 0.8500 | 0.1000 | 0.0800 | 0.7800 | 0.9500 | 0.8000 | Yes | -10525.65 |

Baseline spec: $\gamma_1 = \gamma_2 = 0$. Conv. = convergence status.

#### Fitted Equations (Baseline):

#### Gray's Method

$$\mu_1 = 0.3516 \tag{601}$$
$$\mu_2 = 0.3519 \tag{602}$$

$$\sigma_t^2(1) = 0.4205 + 0.1121\,y_{t-1}^2 + 0.8861\,\sigma_{t-1}^2 \tag{603}$$
$$\sigma_t^2(2) = 0.2336 + 0.1310\,y_{t-1}^2 + 0.7922\,\sigma_{t-1}^2 \tag{604}$$

$$P = \begin{pmatrix} 0.9638 & 0.0362 \\ 0.2572 & 0.7428 \end{pmatrix} \tag{605}$$

LogLik $= -9126.19$; Converged: No.

#### Plug-in Method

$$\mu_1 = -0.3969 \tag{606}$$
$$\mu_2 = 0.7938 \tag{607}$$

$$\sigma_t^2(1) = 0.4041 + 0.2430\,y_{t-1}^2 + 0.7560\,\sigma_{t-1}^2 \tag{608}$$
$$\sigma_t^2(2) = 0.4814 + 0.2948\,y_{t-1}^2 + 0.6342\,\sigma_{t-1}^2 \tag{609}$$

$$P = \begin{pmatrix} 0.7760 & 0.2240 \\ 0.1049 & 0.8951 \end{pmatrix} \tag{610}$$

LogLik $= -9326.54$; Converged: No.

### Quadrature Method

$$\mu_1 = 0.3394 \tag{611}$$
$$\mu_2 = 0.3397 \tag{612}$$

$$\sigma_t^2(1) = 0.4165 + 0.1107\, y_{t-1}^2 + 0.8879\, \sigma_{t-1}^2 \tag{613}$$
$$\sigma_t^2(2) = 0.2289 + 0.1276\, y_{t-1}^2 + 0.7921\, \sigma_{t-1}^2 \tag{614}$$

$$P = \begin{pmatrix} 0.9659 & 0.0341 \\ 0.2556 & 0.7444 \end{pmatrix} \tag{615}$$

LogLik $= -9121.91$; Converged: No.

### Particle Filter Method

$$\mu_1 = -0.0000 \tag{616}$$
$$\mu_2 = 0.0000 \tag{617}$$

$$\sigma_t^2(1) = 0.3000 + 0.0500\, y_{t-1}^2 + 0.8500\, \sigma_{t-1}^2 \tag{618}$$
$$\sigma_t^2(2) = 0.1000 + 0.0800\, y_{t-1}^2 + 0.7800\, \sigma_{t-1}^2 \tag{619}$$

$$P = \begin{pmatrix} 0.9500 & 0.0500 \\ 0.2000 & 0.8000 \end{pmatrix} \tag{620}$$

LogLik $= -10525.65$; Converged: Yes.

### Exogenous Specification

**Table 32:** NMC-USD — Exogenous Specification

| Method | $c_1$ | $c_2$ | $\gamma_1$ | $\gamma_2$ | $\omega_1$ | $\alpha_1$ | $\beta_1$ | $\omega_2$ | $\alpha_2$ | $\beta_2$ | $p_{11}$ | $p_{22}$ | Conv. | Lo |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Gray's | 0.2912 | 0.2977 | -0.2788 | 0.4511 | 0.4738 | 0.0678 | 0.9000 | 0.2611 | 0.2195 | 0.7801 | 0.8732 | 0.7470 | Yes | -902 |
| Plug-in | 0.2801 | 0.3165 | 0.3636 | -0.2702 | 0.4081 | 0.1090 | 0.8907 | 0.2200 | 0.1237 | 0.7828 | 0.9486 | 0.7509 | No | -900 |
| Quadrature | 0.1893 | -0.2462 | -0.2546 | 0.9347 | 0.7896 | 0.1000 | 0.9000 | 0.4749 | 0.3541 | 0.3659 | 0.9724 | 0.7735 | Yes | -898 |
| Particle Filter | 0.0000 | -0.0000 | 0.0500 | 0.0500 | 0.3000 | 0.0500 | 0.8500 | 0.1000 | 0.0800 | 0.7800 | 0.9500 | 0.8000 | Yes | -1052 |

Exogenous spec; pivot = `BTC-USD`. Conv. = convergence status.

### Fitted Equations (Exogenous):

### Gray's Method

$$\mu_1 = 0.2912 - 0.2788\, x_t \tag{621}$$
$$\mu_2 = 0.2977 + 0.4511\, x_t \tag{622}$$

$$\sigma_t^2(1) = 0.4738 + 0.0678\, y_{t-1}^2 + 0.9000\, \sigma_{t-1}^2 \quad (623)$$
$$\sigma_t^2(2) = 0.2611 + 0.2195\, y_{t-1}^2 + 0.7801\, \sigma_{t-1}^2 \quad (624)$$

$$P = \begin{pmatrix} 0.8732 & 0.1268 \\ 0.2530 & 0.7470 \end{pmatrix} \quad (625)$$

LogLik $= -9025.39$; Converged: Yes.

**Plug-in Method**

$$\mu_1 = 0.2801 + 0.3636\, x_t \quad (626)$$
$$\mu_2 = 0.3165 - 0.2702\, x_t \quad (627)$$

$$\sigma_t^2(1) = 0.4081 + 0.1090\, y_{t-1}^2 + 0.8907\, \sigma_{t-1}^2 \quad (628)$$
$$\sigma_t^2(2) = 0.2200 + 0.1237\, y_{t-1}^2 + 0.7828\, \sigma_{t-1}^2 \quad (629)$$

$$P = \begin{pmatrix} 0.9486 & 0.0514 \\ 0.2491 & 0.7509 \end{pmatrix} \quad (630)$$

LogLik $= -9005.87$; Converged: No.

**Quadrature Method**

$$\mu_1 = 0.1893 - 0.2546\, x_t \quad (631)$$
$$\mu_2 = -0.2462 + 0.9347\, x_t \quad (632)$$

$$\sigma_t^2(1) = 0.7896 + 0.1000\, y_{t-1}^2 + 0.9000\, \sigma_{t-1}^2 \quad (633)$$
$$\sigma_t^2(2) = 0.4749 + 0.3541\, y_{t-1}^2 + 0.3659\, \sigma_{t-1}^2 \quad (634)$$

$$P = \begin{pmatrix} 0.9724 & 0.0276 \\ 0.2265 & 0.7735 \end{pmatrix} \quad (635)$$

LogLik $= -8987.10$; Converged: Yes.

**Particle Filter Method**

$$\mu_1 = 0.0000 + 0.0500\, x_t \quad (636)$$
$$\mu_2 = -0.0000 + 0.0500\, x_t \quad (637)$$

$$\sigma_t^2(1) = 0.3000 + 0.0500\, y_{t-1}^2 + 0.8500\, \sigma_{t-1}^2 \quad (638)$$
$$\sigma_t^2(2) = 0.1000 + 0.0800\, y_{t-1}^2 + 0.7800\, \sigma_{t-1}^2 \quad (639)$$

$$P = \begin{pmatrix} 0.9500 & 0.0500 \\ 0.2000 & 0.8000 \end{pmatrix} \tag{640}$$

LogLik $= -10527.21$; Converged: Yes.

### 3.6. Asset: `BTC-USD`

#### Baseline Specification

**Table 33:** BTC-USD — Baseline Specification

| Method | $c_1$ | $c_2$ | $\gamma_1$ | $\gamma_2$ | $\omega_1$ | $\alpha_1$ | $\beta_1$ | $\omega_2$ | $\alpha_2$ | $\beta_2$ | $p_{11}$ | $p_{22}$ | Conv. | LogLik |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Gray's | 0.3250 | -0.1326 | – | – | 0.8259 | 0.0300 | 0.9000 | 0.3511 | 0.3121 | 0.6871 | 0.8265 | 0.7183 | Yes | -7297.71 |
| Plug-in | 0.2967 | 0.2183 | – | – | 0.4060 | 0.1119 | 0.8874 | 0.0915 | 0.1184 | 0.7786 | 0.9415 | 0.7555 | No | -7232.32 |
| Quadrature | 0.3018 | 0.0777 | – | – | 0.8128 | 0.0262 | 0.9000 | 0.3007 | 0.3465 | 0.6485 | 0.8635 | 0.7211 | No | -7297.61 |
| Particle Filter | 0.0000 | 0.0000 | – | – | 0.3000 | 0.0500 | 0.8500 | 0.1000 | 0.0800 | 0.7800 | 0.9500 | 0.8000 | Yes | -7853.80 |

Baseline spec: $\gamma_1 = \gamma_2 = 0$. Conv. = convergence status.

#### Fitted Equations (Baseline):

#### Gray's Method

$$\mu_1 = 0.3250 \tag{641}$$
$$\mu_2 = -0.1326 \tag{642}$$

$$\sigma_t^2(1) = 0.8259 + 0.0300\, y_{t-1}^2 + 0.9000\, \sigma_{t-1}^2 \tag{643}$$
$$\sigma_t^2(2) = 0.3511 + 0.3121\, y_{t-1}^2 + 0.6871\, \sigma_{t-1}^2 \tag{644}$$

$$P = \begin{pmatrix} 0.8265 & 0.1735 \\ 0.2817 & 0.7183 \end{pmatrix} \tag{645}$$

LogLik $= -7297.71$; Converged: Yes.

#### Plug-in Method

$$\mu_1 = 0.2967 \tag{646}$$
$$\mu_2 = 0.2183 \tag{647}$$

$$\sigma_t^2(1) = 0.4060 + 0.1119\, y_{t-1}^2 + 0.8874\, \sigma_{t-1}^2 \tag{648}$$
$$\sigma_t^2(2) = 0.0915 + 0.1184\, y_{t-1}^2 + 0.7786\, \sigma_{t-1}^2 \tag{649}$$

$$P = \begin{pmatrix} 0.9415 & 0.0585 \\ 0.2445 & 0.7555 \end{pmatrix} \tag{650}$$

LogLik $= -7232.32$; Converged: No.

### Quadrature Method

$$\mu_1 = 0.3018 \tag{651}$$
$$\mu_2 = 0.0777 \tag{652}$$

$$\sigma_t^2(1) = 0.8128 + 0.0262\, y_{t-1}^2 + 0.9000\, \sigma_{t-1}^2 \tag{653}$$
$$\sigma_t^2(2) = 0.3007 + 0.3465\, y_{t-1}^2 + 0.6485\, \sigma_{t-1}^2 \tag{654}$$

$$P = \begin{pmatrix} 0.8635 & 0.1365 \\ 0.2789 & 0.7211 \end{pmatrix} \tag{655}$$

LogLik $= -7297.61$; Converged: No.

### Particle Filter Method

$$\mu_1 = 0.0000 \tag{656}$$
$$\mu_2 = 0.0000 \tag{657}$$

$$\sigma_t^2(1) = 0.3000 + 0.0500\, y_{t-1}^2 + 0.8500\, \sigma_{t-1}^2 \tag{658}$$
$$\sigma_t^2(2) = 0.1000 + 0.0800\, y_{t-1}^2 + 0.7800\, \sigma_{t-1}^2 \tag{659}$$

$$P = \begin{pmatrix} 0.9500 & 0.0500 \\ 0.2000 & 0.8000 \end{pmatrix} \tag{660}$$

LogLik $= -7853.80$; Converged: Yes.

## 4. Derivatives Market

Pivot (exogenous conditioning variable): `BTC=F`.

### 4.1. Asset: `ES=F`

#### Baseline Specification

**Table 34:** ES=F — Baseline Specification

| Method | $c_1$ | $c_2$ | $\gamma_1$ | $\gamma_2$ | $\omega_1$ | $\alpha_1$ | $\beta_1$ | $\omega_2$ | $\alpha_2$ | $\beta_2$ | $p_{11}$ | $p_{22}$ | Conv. | LogLik |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Gray's | -0.2298 | 0.1249 | – | – | 0.3382 | 0.1647 | 0.8349 | 0.0001 | 0.0788 | 0.6385 | 0.8468 | 0.9184 | Yes | -3363.57 |
| Plug-in | 0.0393 | 0.0001 | – | – | 0.0990 | 0.2335 | 0.7656 | 0.0001 | 0.0001 | 0.0001 | 0.8017 | 0.4978 | Yes | -1095.35 |
| Quadrature | -0.0299 | -0.0005 | – | – | 0.3765 | 0.2055 | 0.7937 | 0.0001 | 0.0901 | 0.0001 | 0.6221 | 0.4310 | No | -2011.93 |
| Particle Filter | -0.0000 | 0.0000 | – | – | 0.1000 | 0.1000 | 0.7000 | 0.2000 | 0.2000 | 0.6000 | 0.9000 | 0.9000 | Yes | -3564.02 |

Baseline spec: $\gamma_1 = \gamma_2 = 0$. Conv. = convergence status.

#### Fitted Equations (Baseline):

#### Gray's Method

$$\mu_1 = -0.2298 \tag{661}$$
$$\mu_2 = 0.1249 \tag{662}$$

$$\sigma_t^2(1) = 0.3382 + 0.1647\, y_{t-1}^2 + 0.8349\, \sigma_{t-1}^2 \tag{663}$$
$$\sigma_t^2(2) = 0.0001 + 0.0788\, y_{t-1}^2 + 0.6385\, \sigma_{t-1}^2 \tag{664}$$

$$P = \begin{pmatrix} 0.8468 & 0.1532 \\ 0.0816 & 0.9184 \end{pmatrix} \tag{665}$$

LogLik $= -3363.57$; Converged: Yes.

#### Plug-in Method

$$\mu_1 = 0.0393 \tag{666}$$
$$\mu_2 = 0.0001 \tag{667}$$

$$\sigma_t^2(1) = 0.0990 + 0.2335\, y_{t-1}^2 + 0.7656\, \sigma_{t-1}^2 \tag{668}$$
$$\sigma_t^2(2) = 0.0001 + 0.0001\, y_{t-1}^2 + 0.0001\, \sigma_{t-1}^2 \tag{669}$$

$$P = \begin{pmatrix} 0.8017 & 0.1983 \\ 0.5022 & 0.4978 \end{pmatrix} \tag{670}$$

LogLik $= -1095.35$; Converged: Yes.

### Quadrature Method

$$\mu_1 = -0.0299 \tag{671}$$
$$\mu_2 = -0.0005 \tag{672}$$

$$\sigma_t^2(1) = 0.3765 + 0.2055\, y_{t-1}^2 + 0.7937\, \sigma_{t-1}^2 \tag{673}$$
$$\sigma_t^2(2) = 0.0001 + 0.0901\, y_{t-1}^2 + 0.0001\, \sigma_{t-1}^2 \tag{674}$$

$$P = \begin{pmatrix} 0.6221 & 0.3779 \\ 0.5690 & 0.4310 \end{pmatrix} \tag{675}$$

LogLik $= -2011.93$; Converged: No.

### Particle Filter Method

$$\mu_1 = -0.0000 \tag{676}$$
$$\mu_2 = 0.0000 \tag{677}$$

$$\sigma_t^2(1) = 0.1000 + 0.1000\, y_{t-1}^2 + 0.7000\, \sigma_{t-1}^2 \tag{678}$$
$$\sigma_t^2(2) = 0.2000 + 0.2000\, y_{t-1}^2 + 0.6000\, \sigma_{t-1}^2 \tag{679}$$

$$P = \begin{pmatrix} 0.9000 & 0.1000 \\ 0.1000 & 0.9000 \end{pmatrix} \tag{680}$$

LogLik $= -3564.02$; Converged: Yes.

### Exogenous Specification

**Table 35:** ES=F — Exogenous Specification

| **Method** | $c_1$ | $c_2$ | $\gamma_1$ | $\gamma_2$ | $\omega_1$ | $\alpha_1$ | $\beta_1$ | $\omega_2$ | $\alpha_2$ | $\beta_2$ | $p_{11}$ | $p_{22}$ | Conv. | Log |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Gray's | -0.0705 | 0.1069 | -0.0122 | 0.0036 | 0.5880 | 0.2359 | 0.7640 | 0.0234 | 0.1433 | 0.2650 | 0.7659 | 0.8125 | Yes | -3375 |
| Plug-in | -0.0002 | 0.0630 | -0.0001 | -0.0035 | 0.0001 | 0.0001 | 0.0001 | 0.0966 | 0.2883 | 0.7117 | 0.5117 | 0.7569 | Yes | -1108 |
| Quadrature | 0.0204 | 0.0655 | -0.0030 | 0.0038 | 0.6408 | 0.2333 | 0.7666 | 0.0001 | 0.0776 | 0.1548 | 0.7089 | 0.7513 | No | -3291 |
| Particle Filter | 0.0000 | 0.0000 | 0.0500 | 0.0500 | 0.3000 | 0.0500 | 0.8500 | 0.1000 | 0.0800 | 0.7800 | 0.9500 | 0.8000 | No | -3789 |

Exogenous spec; pivot = `BTC=F`. Conv. = convergence status.

### Fitted Equations (Exogenous):

### Gray's Method

$$\mu_1 = -0.0705 - 0.0122\, x_t \tag{681}$$
$$\mu_2 = 0.1069 + 0.0036\, x_t \tag{682}$$

$$\sigma_t^2(1) = 0.5880 + 0.2359\, y_{t-1}^2 + 0.7640\, \sigma_{t-1}^2 \tag{683}$$
$$\sigma_t^2(2) = 0.0234 + 0.1433\, y_{t-1}^2 + 0.2650\, \sigma_{t-1}^2 \tag{684}$$

$$P = \begin{pmatrix} 0.7659 & 0.2341 \\ 0.1875 & 0.8125 \end{pmatrix} \tag{685}$$

LogLik $= -3375.42$; Converged: Yes.

**Plug-in Method**

$$\mu_1 = -0.0002 - 0.0001\, x_t \tag{686}$$
$$\mu_2 = 0.0630 - 0.0035\, x_t \tag{687}$$

$$\sigma_t^2(1) = 0.0001 + 0.0001\, y_{t-1}^2 + 0.0001\, \sigma_{t-1}^2 \tag{688}$$
$$\sigma_t^2(2) = 0.0966 + 0.2883\, y_{t-1}^2 + 0.7117\, \sigma_{t-1}^2 \tag{689}$$

$$P = \begin{pmatrix} 0.5117 & 0.4883 \\ 0.2431 & 0.7569 \end{pmatrix} \tag{690}$$

LogLik $= -1108.91$; Converged: Yes.

**Quadrature Method**

$$\mu_1 = 0.0204 - 0.0030\, x_t \tag{691}$$
$$\mu_2 = 0.0655 + 0.0038\, x_t \tag{692}$$

$$\sigma_t^2(1) = 0.6408 + 0.2333\, y_{t-1}^2 + 0.7666\, \sigma_{t-1}^2 \tag{693}$$
$$\sigma_t^2(2) = 0.0001 + 0.0776\, y_{t-1}^2 + 0.1548\, \sigma_{t-1}^2 \tag{694}$$

$$P = \begin{pmatrix} 0.7089 & 0.2911 \\ 0.2487 & 0.7513 \end{pmatrix} \tag{695}$$

LogLik $= -3291.80$; Converged: No.

**Particle Filter Method**

$$\mu_1 = 0.0000 + 0.0500\, x_t \tag{696}$$
$$\mu_2 = 0.0000 + 0.0500\, x_t \tag{697}$$

$$\sigma_t^2(1) = 0.3000 + 0.0500\, y_{t-1}^2 + 0.8500\, \sigma_{t-1}^2 \tag{698}$$
$$\sigma_t^2(2) = 0.1000 + 0.0800\, y_{t-1}^2 + 0.7800\, \sigma_{t-1}^2 \tag{699}$$

$$P = \begin{pmatrix} 0.9500 & 0.0500 \\ 0.2000 & 0.8000 \end{pmatrix} \tag{700}$$

LogLik $= -3789.14$; Converged: No.

## 4.2. Asset: `NQ=F`

### Baseline Specification

**Table 36:** NQ=F — Baseline Specification

| **Method** | $c_1$ | $c_2$ | $\gamma_1$ | $\gamma_2$ | $\omega_1$ | $\alpha_1$ | $\beta_1$ | $\omega_2$ | $\alpha_2$ | $\beta_2$ | $p_{11}$ | $p_{22}$ | Conv. | LogLik |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Gray's | -0.4956 | 0.1470 | – | – | 0.7317 | 0.0746 | 0.9000 | 0.0001 | 0.0001 | 0.7779 | 0.8629 | 0.9609 | Yes | -4168.20 |
| Plug-in | -0.0002 | 0.1257 | – | – | 0.0001 | 0.0001 | 0.0001 | 0.2100 | 0.2449 | 0.7539 | 0.5117 | 0.7429 | Yes | -1708.53 |
| Quadrature | -0.2432 | 0.0004 | – | – | 0.9614 | 0.1800 | 0.8192 | 0.0352 | 0.0909 | 0.1859 | 0.5977 | 0.7402 | No | -4142.41 |
| Particle Filter | -0.0000 | 0.0000 | – | – | 0.3000 | 0.0500 | 0.8500 | 0.1000 | 0.0800 | 0.7800 | 0.9500 | 0.8000 | Yes | -4346.95 |

Baseline spec: $\gamma_1 = \gamma_2 = 0$. Conv. = convergence status.

### Fitted Equations (Baseline):

### Gray's Method

$$\mu_1 = -0.4956 \tag{701}$$
$$\mu_2 = 0.1470 \tag{702}$$

$$\sigma_t^2(1) = 0.7317 + 0.0746\, y_{t-1}^2 + 0.9000\, \sigma_{t-1}^2 \tag{703}$$
$$\sigma_t^2(2) = 0.0001 + 0.0001\, y_{t-1}^2 + 0.7779\, \sigma_{t-1}^2 \tag{704}$$

$$P = \begin{pmatrix} 0.8629 & 0.1371 \\ 0.0391 & 0.9609 \end{pmatrix} \tag{705}$$

LogLik $= -4168.20$; Converged: Yes.

### Plug-in Method

$$\mu_1 = -0.0002 \tag{706}$$
$$\mu_2 = 0.1257 \tag{707}$$

$$\sigma_t^2(1) = 0.0001 + 0.0001\, y_{t-1}^2 + 0.0001\, \sigma_{t-1}^2 \tag{708}$$
$$\sigma_t^2(2) = 0.2100 + 0.2449\, y_{t-1}^2 + 0.7539\, \sigma_{t-1}^2 \tag{709}$$

$$P = \begin{pmatrix} 0.5117 & 0.4883 \\ 0.2571 & 0.7429 \end{pmatrix} \tag{710}$$

LogLik $= -1708.53$; Converged: Yes.

### Quadrature Method

$$\mu_1 = -0.2432 \tag{711}$$
$$\mu_2 = 0.0004 \tag{712}$$

$$\sigma_t^2(1) = 0.9614 + 0.1800\, y_{t-1}^2 + 0.8192\, \sigma_{t-1}^2 \tag{713}$$
$$\sigma_t^2(2) = 0.0352 + 0.0909\, y_{t-1}^2 + 0.1859\, \sigma_{t-1}^2 \tag{714}$$

$$P = \begin{pmatrix} 0.5977 & 0.4023 \\ 0.2598 & 0.7402 \end{pmatrix} \tag{715}$$

LogLik $= -4142.41$; Converged: No.

### Particle Filter Method

$$\mu_1 = -0.0000 \tag{716}$$
$$\mu_2 = 0.0000 \tag{717}$$

$$\sigma_t^2(1) = 0.3000 + 0.0500\, y_{t-1}^2 + 0.8500\, \sigma_{t-1}^2 \tag{718}$$
$$\sigma_t^2(2) = 0.1000 + 0.0800\, y_{t-1}^2 + 0.7800\, \sigma_{t-1}^2 \tag{719}$$

$$P = \begin{pmatrix} 0.9500 & 0.0500 \\ 0.2000 & 0.8000 \end{pmatrix} \tag{720}$$

LogLik $= -4346.95$; Converged: Yes.

### Exogenous Specification

**Table 37:** NQ=F — Exogenous Specification

| **Method** | $c_1$ | $c_2$ | $\gamma_1$ | $\gamma_2$ | $\omega_1$ | $\alpha_1$ | $\beta_1$ | $\omega_2$ | $\alpha_2$ | $\beta_2$ | $p_{11}$ | $p_{22}$ | Conv. | Log |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Gray's | 0.0568 | 0.1104 | -0.0123 | -0.0051 | 0.2432 | 0.1515 | 0.8484 | 0.0019 | 0.0919 | 0.7138 | 0.8589 | 0.8545 | No | -4259 |
| Plug-in | -0.0001 | 0.0957 | -0.0003 | 0.0051 | 0.0001 | 0.0001 | 0.0001 | 0.2453 | 0.2590 | 0.7410 | 0.5016 | 0.7755 | Yes | -1717 |
| Quadrature | 0.0603 | 0.1124 | -0.0230 | 0.0022 | 0.2318 | 0.1811 | 0.8172 | 0.0001 | 0.1069 | 0.6669 | 0.8376 | 0.8715 | No | -4300 |
| Particle Filter | -0.0000 | -0.0000 | 0.0500 | 0.0500 | 0.3000 | 0.0500 | 0.8500 | 0.1000 | 0.0800 | 0.7800 | 0.9500 | 0.8000 | Yes | -4390 |

Exogenous spec; pivot = `BTC=F`. Conv. = convergence status.

### Fitted Equations (Exogenous):

### Gray's Method

$$\mu_1 = 0.0568 - 0.0123\, x_t \tag{721}$$
$$\mu_2 = 0.1104 - 0.0051\, x_t \tag{722}$$

$$\sigma_t^2(1) = 0.2432 + 0.1515\, y_{t-1}^2 + 0.8484\, \sigma_{t-1}^2 \tag{723}$$
$$\sigma_t^2(2) = 0.0019 + 0.0919\, y_{t-1}^2 + 0.7138\, \sigma_{t-1}^2 \tag{724}$$

$$P = \begin{pmatrix} 0.8589 & 0.1411 \\ 0.1455 & 0.8545 \end{pmatrix} \tag{725}$$

LogLik $= -4259.78$; Converged: No.

**Plug-in Method**

$$\mu_1 = -0.0001 - 0.0003\, x_t \tag{726}$$
$$\mu_2 = 0.0957 + 0.0051\, x_t \tag{727}$$

$$\sigma_t^2(1) = 0.0001 + 0.0001\, y_{t-1}^2 + 0.0001\, \sigma_{t-1}^2 \tag{728}$$
$$\sigma_t^2(2) = 0.2453 + 0.2590\, y_{t-1}^2 + 0.7410\, \sigma_{t-1}^2 \tag{729}$$

$$P = \begin{pmatrix} 0.5016 & 0.4984 \\ 0.2245 & 0.7755 \end{pmatrix} \tag{730}$$

LogLik $= -1717.33$; Converged: Yes.

**Quadrature Method**

$$\mu_1 = 0.0603 - 0.0230\, x_t \tag{731}$$
$$\mu_2 = 0.1124 + 0.0022\, x_t \tag{732}$$

$$\sigma_t^2(1) = 0.2318 + 0.1811\, y_{t-1}^2 + 0.8172\, \sigma_{t-1}^2 \tag{733}$$
$$\sigma_t^2(2) = 0.0001 + 0.1069\, y_{t-1}^2 + 0.6669\, \sigma_{t-1}^2 \tag{734}$$

$$P = \begin{pmatrix} 0.8376 & 0.1624 \\ 0.1285 & 0.8715 \end{pmatrix} \tag{735}$$

LogLik $= -4300.29$; Converged: No.

**Particle Filter Method**

$$\mu_1 = -0.0000 + 0.0500\, x_t \tag{736}$$
$$\mu_2 = -0.0000 + 0.0500\, x_t \tag{737}$$

$$\sigma_t^2(1) = 0.3000 + 0.0500\, y_{t-1}^2 + 0.8500\, \sigma_{t-1}^2 \tag{738}$$
$$\sigma_t^2(2) = 0.1000 + 0.0800\, y_{t-1}^2 + 0.7800\, \sigma_{t-1}^2 \tag{739}$$

$$P = \begin{pmatrix} 0.9500 & 0.0500 \\ 0.2000 & 0.8000 \end{pmatrix} \tag{740}$$

LogLik $= -4390.33$; Converged: Yes.

### 4.3. Asset: `VIX`

### Baseline Specification

**Table 38:** VIX — Baseline Specification

| Method | $c_1$ | $c_2$ | $\gamma_1$ | $\gamma_2$ | $\omega_1$ | $\alpha_1$ | $\beta_1$ | $\omega_2$ | $\alpha_2$ | $\beta_2$ | $p_{11}$ | $p_{22}$ | Conv. | LogLik |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Gray's | 0.3530 | 0.3507 | – | – | 0.4207 | 0.1111 | 0.8876 | 0.2340 | 0.1312 | 0.7930 | 0.9664 | 0.7423 | No | -9369.61 |
| Plug-in | 0.3015 | -0.3016 | – | – | 0.4032 | 0.1087 | 0.8894 | 0.2142 | 0.1188 | 0.7826 | 0.9513 | 0.7525 | No | -9248.85 |
| Quadrature | 0.3551 | 0.2797 | – | – | 0.4243 | 0.1126 | 0.8873 | 0.2380 | 0.1332 | 0.7932 | 0.9657 | 0.7407 | No | -9364.19 |
| Particle Filter | 0.0000 | 0.0000 | – | – | 0.3000 | 0.0500 | 0.8500 | 0.1000 | 0.0800 | 0.7800 | 0.9500 | 0.8000 | Yes | -11098.09 |

Baseline spec: $\gamma_1 = \gamma_2 = 0$. Conv. = convergence status.

### Fitted Equations (Baseline):

### Gray's Method

$$\mu_1 = 0.3530 \tag{741}$$
$$\mu_2 = 0.3507 \tag{742}$$

$$\sigma_t^2(1) = 0.4207 + 0.1111\, y_{t-1}^2 + 0.8876\, \sigma_{t-1}^2 \tag{743}$$
$$\sigma_t^2(2) = 0.2340 + 0.1312\, y_{t-1}^2 + 0.7930\, \sigma_{t-1}^2 \tag{744}$$

$$P = \begin{pmatrix} 0.9664 & 0.0336 \\ 0.2577 & 0.7423 \end{pmatrix} \tag{745}$$

LogLik $= -9369.61$; Converged: No.

### Plug-in Method

$$\mu_1 = 0.3015 \tag{746}$$
$$\mu_2 = -0.3016 \tag{747}$$

$$\sigma_t^2(1) = 0.4032 + 0.1087\, y_{t-1}^2 + 0.8894\, \sigma_{t-1}^2 \tag{748}$$
$$\sigma_t^2(2) = 0.2142 + 0.1188\, y_{t-1}^2 + 0.7826\, \sigma_{t-1}^2 \tag{749}$$

$$P = \begin{pmatrix} 0.9513 & 0.0487 \\ 0.2475 & 0.7525 \end{pmatrix} \tag{750}$$

LogLik $= -9248.85$; Converged: No.

### Quadrature Method

$$\mu_1 = 0.3551 \tag{751}$$
$$\mu_2 = 0.2797 \tag{752}$$

$$\sigma_t^2(1) = 0.4243 + 0.1126\, y_{t-1}^2 + 0.8873\, \sigma_{t-1}^2 \tag{753}$$
$$\sigma_t^2(2) = 0.2380 + 0.1332\, y_{t-1}^2 + 0.7932\, \sigma_{t-1}^2 \tag{754}$$

$$P = \begin{pmatrix} 0.9657 & 0.0343 \\ 0.2593 & 0.7407 \end{pmatrix} \tag{755}$$

LogLik $= -9364.19$; Converged: No.

### Particle Filter Method

$$\mu_1 = 0.0000 \tag{756}$$
$$\mu_2 = 0.0000 \tag{757}$$

$$\sigma_t^2(1) = 0.3000 + 0.0500\, y_{t-1}^2 + 0.8500\, \sigma_{t-1}^2 \tag{758}$$
$$\sigma_t^2(2) = 0.1000 + 0.0800\, y_{t-1}^2 + 0.7800\, \sigma_{t-1}^2 \tag{759}$$

$$P = \begin{pmatrix} 0.9500 & 0.0500 \\ 0.2000 & 0.8000 \end{pmatrix} \tag{760}$$

LogLik $= -11098.09$; Converged: Yes.

### Exogenous Specification

**Table 39:** VIX — Exogenous Specification

| **Method** | $c_1$ | $c_2$ | $\gamma_1$ | $\gamma_2$ | $\omega_1$ | $\alpha_1$ | $\beta_1$ | $\omega_2$ | $\alpha_2$ | $\beta_2$ | $p_{11}$ | $p_{22}$ | Conv. | Lo |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Gray's | 0.4237 | 0.3489 | 0.4686 | -0.3806 | 0.4444 | 0.1177 | 0.8811 | 0.2612 | 0.1498 | 0.7963 | 0.9416 | 0.7347 | No | -932 |
| Plug-in | 0.4222 | -0.4223 | 0.4677 | -0.3767 | 0.4437 | 0.1220 | 0.8776 | 0.2604 | 0.1473 | 0.7879 | 0.9459 | 0.7346 | No | -928 |
| Quadrature | 0.7128 | 0.7112 | -0.7926 | 0.8766 | 0.7166 | 0.2768 | 0.7224 | 0.6291 | 0.3050 | 0.6538 | 0.7899 | 0.8031 | No | -950 |
| Particle Filter | 0.0000 | 0.0000 | 0.0500 | 0.0500 | 0.3000 | 0.0500 | 0.8500 | 0.1000 | 0.0800 | 0.7800 | 0.9500 | 0.8000 | Yes | -1109 |

Exogenous spec; pivot = `BTC=F`. Conv. = convergence status.

### Fitted Equations (Exogenous):

### Gray's Method

$$\mu_1 = 0.4237 + 0.4686\, x_t \tag{761}$$
$$\mu_2 = 0.3489 - 0.3806\, x_t \tag{762}$$

$$\sigma_t^2(1) = 0.4444 + 0.1177\, y_{t-1}^2 + 0.8811\, \sigma_{t-1}^2 \tag{763}$$
$$\sigma_t^2(2) = 0.2612 + 0.1498\, y_{t-1}^2 + 0.7963\, \sigma_{t-1}^2 \tag{764}$$

$$P = \begin{pmatrix} 0.9416 & 0.0584 \\ 0.2653 & 0.7347 \end{pmatrix} \tag{765}$$

LogLik $= -9321.98$; Converged: No.

**Plug-in Method**

$$\mu_1 = 0.4222 + 0.4677\, x_t \tag{766}$$
$$\mu_2 = -0.4223 - 0.3767\, x_t \tag{767}$$

$$\sigma_t^2(1) = 0.4437 + 0.1220\, y_{t-1}^2 + 0.8776\, \sigma_{t-1}^2 \tag{768}$$
$$\sigma_t^2(2) = 0.2604 + 0.1473\, y_{t-1}^2 + 0.7879\, \sigma_{t-1}^2 \tag{769}$$

$$P = \begin{pmatrix} 0.9459 & 0.0541 \\ 0.2654 & 0.7346 \end{pmatrix} \tag{770}$$

LogLik $= -9284.58$; Converged: No.

**Quadrature Method**

$$\mu_1 = 0.7128 - 0.7926\, x_t \tag{771}$$
$$\mu_2 = 0.7112 + 0.8766\, x_t \tag{772}$$

$$\sigma_t^2(1) = 0.7166 + 0.2768\, y_{t-1}^2 + 0.7224\, \sigma_{t-1}^2 \tag{773}$$
$$\sigma_t^2(2) = 0.6291 + 0.3050\, y_{t-1}^2 + 0.6538\, \sigma_{t-1}^2 \tag{774}$$

$$P = \begin{pmatrix} 0.7899 & 0.2101 \\ 0.1969 & 0.8031 \end{pmatrix} \tag{775}$$

LogLik $= -9503.13$; Converged: No.

**Particle Filter Method**

$$\mu_1 = 0.0000 + 0.0500\, x_t \tag{776}$$
$$\mu_2 = 0.0000 + 0.0500\, x_t \tag{777}$$

$$\sigma_t^2(1) = 0.3000 + 0.0500\, y_{t-1}^2 + 0.8500\, \sigma_{t-1}^2 \tag{778}$$
$$\sigma_t^2(2) = 0.1000 + 0.0800\, y_{t-1}^2 + 0.7800\, \sigma_{t-1}^2 \tag{779}$$

$$P = \begin{pmatrix} 0.9500 & 0.0500 \\ 0.2000 & 0.8000 \end{pmatrix} \tag{780}$$

LogLik $= -11097.16$; Converged: Yes.

### 4.4. Asset: `GC=F`

### Baseline Specification

**Table 40:** GC=F — Baseline Specification

| **Method** | $c_1$ | $c_2$ | $\gamma_1$ | $\gamma_2$ | $\omega_1$ | $\alpha_1$ | $\beta_1$ | $\omega_2$ | $\alpha_2$ | $\beta_2$ | $p_{11}$ | $p_{22}$ | Conv. | LogLik |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Gray's | 0.0482 | 0.0002 | – | – | 0.8257 | 0.0482 | 0.0001 | 0.0001 | 0.0001 | 0.0001 | 0.7494 | 0.5474 | Yes | -1105.49 |
| Plug-in | 0.0357 | 0.0001 | – | – | 0.0416 | 0.0848 | 0.9000 | 0.0001 | 0.0001 | 0.0001 | 0.7598 | 0.5127 | Yes | -884.73 |
| Quadrature | 0.0497 | 0.0002 | – | – | 0.8321 | 0.0493 | 0.0001 | 0.0001 | 0.0001 | 0.0001 | 0.7491 | 0.5464 | Yes | -1130.01 |
| Particle Filter | 0.0000 | 0.0000 | – | – | 0.1000 | 0.1000 | 0.7000 | 0.2000 | 0.2000 | 0.6000 | 0.9000 | 0.9000 | Yes | -3267.32 |

Baseline spec: $\gamma_1 = \gamma_2 = 0$. Conv. = convergence status.

### Fitted Equations (Baseline):

### Gray's Method

$$\mu_1 = 0.0482 \tag{781}$$
$$\mu_2 = 0.0002 \tag{782}$$

$$\sigma_t^2(1) = 0.8257 + 0.0482\, y_{t-1}^2 + 0.0001\, \sigma_{t-1}^2 \tag{783}$$
$$\sigma_t^2(2) = 0.0001 + 0.0001\, y_{t-1}^2 + 0.0001\, \sigma_{t-1}^2 \tag{784}$$

$$P = \begin{pmatrix} 0.7494 & 0.2506 \\ 0.4526 & 0.5474 \end{pmatrix} \tag{785}$$

LogLik $= -1105.49$; Converged: Yes.

### Plug-in Method

$$\mu_1 = 0.0357 \tag{786}$$
$$\mu_2 = 0.0001 \tag{787}$$

$$\sigma_t^2(1) = 0.0416 + 0.0848\, y_{t-1}^2 + 0.9000\, \sigma_{t-1}^2 \tag{788}$$
$$\sigma_t^2(2) = 0.0001 + 0.0001\, y_{t-1}^2 + 0.0001\, \sigma_{t-1}^2 \tag{789}$$

$$P = \begin{pmatrix} 0.7598 & 0.2402 \\ 0.4873 & 0.5127 \end{pmatrix} \tag{790}$$

LogLik $= -884.73$; Converged: Yes.

### Quadrature Method

$$\mu_1 = 0.0497 \tag{791}$$
$$\mu_2 = 0.0002 \tag{792}$$

$$\sigma_t^2(1) = 0.8321 + 0.0493\, y_{t-1}^2 + 0.0001\, \sigma_{t-1}^2 \tag{793}$$
$$\sigma_t^2(2) = 0.0001 + 0.0001\, y_{t-1}^2 + 0.0001\, \sigma_{t-1}^2 \tag{794}$$

$$P = \begin{pmatrix} 0.7491 & 0.2509 \\ 0.4536 & 0.5464 \end{pmatrix} \tag{795}$$

LogLik $= -1130.01$; Converged: Yes.

### Particle Filter Method

$$\mu_1 = 0.0000 \tag{796}$$
$$\mu_2 = 0.0000 \tag{797}$$

$$\sigma_t^2(1) = 0.1000 + 0.1000\, y_{t-1}^2 + 0.7000\, \sigma_{t-1}^2 \tag{798}$$
$$\sigma_t^2(2) = 0.2000 + 0.2000\, y_{t-1}^2 + 0.6000\, \sigma_{t-1}^2 \tag{799}$$

$$P = \begin{pmatrix} 0.9000 & 0.1000 \\ 0.1000 & 0.9000 \end{pmatrix} \tag{800}$$

LogLik $= -3267.32$; Converged: Yes.

### Exogenous Specification

**Table 41:** GC=F — Exogenous Specification

| **Method** | $c_1$ | $c_2$ | $\gamma_1$ | $\gamma_2$ | $\omega_1$ | $\alpha_1$ | $\beta_1$ | $\omega_2$ | $\alpha_2$ | $\beta_2$ | $p_{11}$ | $p_{22}$ | Conv. | Log |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Gray's | 0.0467 | 0.0002 | -0.0008 | 0.0002 | 0.8268 | 0.0481 | 0.0001 | 0.0001 | 0.0001 | 0.0001 | 0.7484 | 0.5471 | Yes | -1105 |
| Plug-in | 0.0356 | 0.0001 | -0.0023 | -0.0000 | 0.0415 | 0.0850 | 0.9000 | 0.0001 | 0.0001 | 0.0001 | 0.7598 | 0.5129 | Yes | -884 |
| Quadrature | 0.0479 | 0.0003 | -0.0010 | 0.0002 | 0.8316 | 0.0497 | 0.0001 | 0.0001 | 0.0001 | 0.0001 | 0.7491 | 0.5468 | Yes | -1129 |
| Particle Filter | 0.0000 | -0.0000 | 0.1000 | 0.1000 | 0.1000 | 0.1000 | 0.7000 | 0.2000 | 0.2000 | 0.6000 | 0.9000 | 0.9000 | Yes | -3551 |

Exogenous spec; pivot = `BTC=F`. Conv. = convergence status.

### Fitted Equations (Exogenous):

### Gray's Method

$$\mu_1 = 0.0467 - 0.0008\, x_t \tag{801}$$
$$\mu_2 = 0.0002 + 0.0002\, x_t \tag{802}$$

$$\sigma_t^2(1) = 0.8268 + 0.0481\, y_{t-1}^2 + 0.0001\, \sigma_{t-1}^2 \tag{803}$$
$$\sigma_t^2(2) = 0.0001 + 0.0001\, y_{t-1}^2 + 0.0001\, \sigma_{t-1}^2 \tag{804}$$

$$P = \begin{pmatrix} 0.7484 & 0.2516 \\ 0.4529 & 0.5471 \end{pmatrix} \tag{805}$$

LogLik $= -1105.22$; Converged: Yes.

**Plug-in Method**

$$\mu_1 = 0.0356 - 0.0023\, x_t \tag{806}$$
$$\mu_2 = 0.0001 - 0.0000\, x_t \tag{807}$$

$$\sigma_t^2(1) = 0.0415 + 0.0850\, y_{t-1}^2 + 0.9000\, \sigma_{t-1}^2 \tag{808}$$
$$\sigma_t^2(2) = 0.0001 + 0.0001\, y_{t-1}^2 + 0.0001\, \sigma_{t-1}^2 \tag{809}$$

$$P = \begin{pmatrix} 0.7598 & 0.2402 \\ 0.4871 & 0.5129 \end{pmatrix} \tag{810}$$

LogLik $= -884.64$; Converged: Yes.

**Quadrature Method**

$$\mu_1 = 0.0479 - 0.0010\, x_t \tag{811}$$
$$\mu_2 = 0.0003 + 0.0002\, x_t \tag{812}$$

$$\sigma_t^2(1) = 0.8316 + 0.0497\, y_{t-1}^2 + 0.0001\, \sigma_{t-1}^2 \tag{813}$$
$$\sigma_t^2(2) = 0.0001 + 0.0001\, y_{t-1}^2 + 0.0001\, \sigma_{t-1}^2 \tag{814}$$

$$P = \begin{pmatrix} 0.7491 & 0.2509 \\ 0.4532 & 0.5468 \end{pmatrix} \tag{815}$$

LogLik $= -1129.58$; Converged: Yes.

**Particle Filter Method**

$$\mu_1 = 0.0000 + 0.1000\, x_t \tag{816}$$
$$\mu_2 = -0.0000 + 0.1000\, x_t \tag{817}$$

$$\sigma_t^2(1) = 0.1000 + 0.1000\, y_{t-1}^2 + 0.7000\, \sigma_{t-1}^2 \tag{818}$$
$$\sigma_t^2(2) = 0.2000 + 0.2000\, y_{t-1}^2 + 0.6000\, \sigma_{t-1}^2 \tag{819}$$

$$P = \begin{pmatrix} 0.9000 & 0.1000 \\ 0.1000 & 0.9000 \end{pmatrix} \tag{820}$$

LogLik $= -3551.41$; Converged: Yes.

## 4.5. Asset: `CL=F`

### Baseline Specification

**Table 42:** CL=F — Baseline Specification

| Method | $c_1$ | $c_2$ | $\gamma_1$ | $\gamma_2$ | $\omega_1$ | $\alpha_1$ | $\beta_1$ | $\omega_2$ | $\alpha_2$ | $\beta_2$ | $p_{11}$ | $p_{22}$ | Conv. | LogLik |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Gray's | -0.1029 | 0.4738 | – | – | 0.2855 | 0.0985 | 0.9000 | 0.0001 | 0.1613 | 0.5101 | 0.9353 | 0.6840 | Yes | -5807.49 |
| Plug-in | -0.0540 | 0.0125 | – | – | 0.5635 | 0.1940 | 0.8059 | 0.3725 | 0.2065 | 0.6159 | 0.7590 | 0.9173 | No | -5767.89 |
| Quadrature | 0.0587 | 0.0703 | – | – | 0.0846 | 0.0764 | 0.9000 | 0.2353 | 0.2164 | 0.7820 | 0.9900 | 0.8262 | Yes | -5819.91 |
| Particle Filter | 0.0000 | 0.0000 | – | – | 0.3000 | 0.0500 | 0.8500 | 0.1000 | 0.0800 | 0.7800 | 0.9500 | 0.8000 | Yes | -5874.11 |

Baseline spec: $\gamma_1 = \gamma_2 = 0$. Conv. = convergence status.

### Fitted Equations (Baseline):

### Gray's Method

$$\mu_1 = -0.1029 \tag{821}$$
$$\mu_2 = 0.4738 \tag{822}$$

$$\sigma_t^2(1) = 0.2855 + 0.0985\, y_{t-1}^2 + 0.9000\, \sigma_{t-1}^2 \tag{823}$$
$$\sigma_t^2(2) = 0.0001 + 0.1613\, y_{t-1}^2 + 0.5101\, \sigma_{t-1}^2 \tag{824}$$

$$P = \begin{pmatrix} 0.9353 & 0.0647 \\ 0.3160 & 0.6840 \end{pmatrix} \tag{825}$$

LogLik $= -5807.49$; Converged: Yes.

### Plug-in Method

$$\mu_1 = -0.0540 \tag{826}$$
$$\mu_2 = 0.0125 \tag{827}$$

$$\sigma_t^2(1) = 0.5635 + 0.1940\, y_{t-1}^2 + 0.8059\, \sigma_{t-1}^2 \tag{828}$$
$$\sigma_t^2(2) = 0.3725 + 0.2065\, y_{t-1}^2 + 0.6159\, \sigma_{t-1}^2 \tag{829}$$

$$P = \begin{pmatrix} 0.7590 & 0.2410 \\ 0.0827 & 0.9173 \end{pmatrix} \tag{830}$$

LogLik $= -5767.89$; Converged: No.

### Quadrature Method

$$\mu_1 = 0.0587 \tag{831}$$
$$\mu_2 = 0.0703 \tag{832}$$

$$\sigma_t^2(1) = 0.0846 + 0.0764\, y_{t-1}^2 + 0.9000\, \sigma_{t-1}^2 \tag{833}$$
$$\sigma_t^2(2) = 0.2353 + 0.2164\, y_{t-1}^2 + 0.7820\, \sigma_{t-1}^2 \tag{834}$$

$$P = \begin{pmatrix} 0.9900 & 0.0100 \\ 0.1738 & 0.8262 \end{pmatrix} \tag{835}$$

LogLik $= -5819.91$; Converged: Yes.

### Particle Filter Method

$$\mu_1 = 0.0000 \tag{836}$$
$$\mu_2 = 0.0000 \tag{837}$$

$$\sigma_t^2(1) = 0.3000 + 0.0500\, y_{t-1}^2 + 0.8500\, \sigma_{t-1}^2 \tag{838}$$
$$\sigma_t^2(2) = 0.1000 + 0.0800\, y_{t-1}^2 + 0.7800\, \sigma_{t-1}^2 \tag{839}$$

$$P = \begin{pmatrix} 0.9500 & 0.0500 \\ 0.2000 & 0.8000 \end{pmatrix} \tag{840}$$

LogLik $= -5874.11$; Converged: Yes.

### Exogenous Specification

**Table 43:** CL=F — Exogenous Specification

| **Method** | $c_1$ | $c_2$ | $\gamma_1$ | $\gamma_2$ | $\omega_1$ | $\alpha_1$ | $\beta_1$ | $\omega_2$ | $\alpha_2$ | $\beta_2$ | $p_{11}$ | $p_{22}$ | Conv. | LogL |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Gray's | 0.0456 | 0.0593 | -0.0001 | -0.0137 | 0.2848 | 0.1052 | 0.8945 | 0.1006 | 0.0789 | 0.7595 | 0.9007 | 0.8103 | No | -5823. |
| Plug-in | 0.0656 | 0.0005 | 0.0761 | 0.0005 | 0.2946 | 0.1629 | 0.8371 | 0.0001 | 0.0001 | 0.1781 | 0.7478 | 0.4320 | No | -3123. |
| Quadrature | 0.0781 | 0.1144 | 0.0027 | -0.0241 | 0.1253 | 0.0606 | 0.9000 | 0.2606 | 0.2353 | 0.7647 | 0.9884 | 0.8334 | Yes | -5823. |
| Particle Filter | 0.0000 | 0.0000 | 0.0500 | 0.0500 | 0.3000 | 0.0500 | 0.8500 | 0.1000 | 0.0800 | 0.7800 | 0.9500 | 0.8000 | No | -5884. |

Exogenous spec; pivot = `BTC=F`. Conv. = convergence status.

### Fitted Equations (Exogenous):

### Gray's Method

$$\mu_1 = 0.0456 - 0.0001\, x_t \tag{841}$$
$$\mu_2 = 0.0593 - 0.0137\, x_t \tag{842}$$

$$\sigma_t^2(1) = 0.2848 + 0.1052\,y_{t-1}^2 + 0.8945\,\sigma_{t-1}^2 \tag{843}$$
$$\sigma_t^2(2) = 0.1006 + 0.0789\,y_{t-1}^2 + 0.7595\,\sigma_{t-1}^2 \tag{844}$$

$$P = \begin{pmatrix} 0.9007 & 0.0993 \\ 0.1897 & 0.8103 \end{pmatrix} \tag{845}$$

LogLik $= -5823.05$; Converged: No.

**Plug-in Method**

$$\mu_1 = 0.0656 + 0.0761\,x_t \tag{846}$$
$$\mu_2 = 0.0005 + 0.0005\,x_t \tag{847}$$

$$\sigma_t^2(1) = 0.2946 + 0.1629\,y_{t-1}^2 + 0.8371\,\sigma_{t-1}^2 \tag{848}$$
$$\sigma_t^2(2) = 0.0001 + 0.0001\,y_{t-1}^2 + 0.1781\,\sigma_{t-1}^2 \tag{849}$$

$$P = \begin{pmatrix} 0.7478 & 0.2522 \\ 0.5680 & 0.4320 \end{pmatrix} \tag{850}$$

LogLik $= -3123.80$; Converged: No.

**Quadrature Method**

$$\mu_1 = 0.0781 + 0.0027\,x_t \tag{851}$$
$$\mu_2 = 0.1144 - 0.0241\,x_t \tag{852}$$

$$\sigma_t^2(1) = 0.1253 + 0.0606\,y_{t-1}^2 + 0.9000\,\sigma_{t-1}^2 \tag{853}$$
$$\sigma_t^2(2) = 0.2606 + 0.2353\,y_{t-1}^2 + 0.7647\,\sigma_{t-1}^2 \tag{854}$$

$$P = \begin{pmatrix} 0.9884 & 0.0116 \\ 0.1666 & 0.8334 \end{pmatrix} \tag{855}$$

LogLik $= -5823.02$; Converged: Yes.

**Particle Filter Method**

$$\mu_1 = 0.0000 + 0.0500\,x_t \tag{856}$$
$$\mu_2 = 0.0000 + 0.0500\,x_t \tag{857}$$

$$\sigma_t^2(1) = 0.3000 + 0.0500\,y_{t-1}^2 + 0.8500\,\sigma_{t-1}^2 \tag{858}$$
$$\sigma_t^2(2) = 0.1000 + 0.0800\,y_{t-1}^2 + 0.7800\,\sigma_{t-1}^2 \tag{859}$$

$$P = \begin{pmatrix} 0.9500 & 0.0500 \\ 0.2000 & 0.8000 \end{pmatrix} \tag{860}$$

LogLik $= -5884.73$; Converged: No.

## 4.6. Asset: BTC=F

### Baseline Specification

**Table 44:** BTC=F — Baseline Specification

| Method | $c_1$ | $c_2$ | $\gamma_1$ | $\gamma_2$ | $\omega_1$ | $\alpha_1$ | $\beta_1$ | $\omega_2$ | $\alpha_2$ | $\beta_2$ | $p_{11}$ | $p_{22}$ | Conv. | LogLik |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Gray's | 0.4107 | -0.1779 | – | – | 0.6992 | 0.0041 | 0.9000 | 0.5715 | 0.1156 | 0.8784 | 0.8169 | 0.7175 | Yes | -7441.63 |
| Plug-in | 0.7002 | 0.6944 | – | – | 0.7114 | 0.2689 | 0.7137 | 0.6196 | 0.2778 | 0.5905 | 0.7869 | 0.8226 | No | -7692.15 |
| Quadrature | 0.4257 | -0.1560 | – | – | 0.6956 | 0.0037 | 0.9000 | 0.5464 | 0.1097 | 0.8836 | 0.8764 | 0.8495 | Yes | -7441.91 |
| Particle Filter | -0.0000 | 0.0000 | – | – | 0.3000 | 0.0500 | 0.8500 | 0.1000 | 0.0800 | 0.7800 | 0.9500 | 0.8000 | No | -8163.70 |

Baseline spec: $\gamma_1 = \gamma_2 = 0$. Conv. = convergence status.

### Fitted Equations (Baseline):

### Gray's Method

$$\mu_1 = 0.4107 \tag{861}$$
$$\mu_2 = -0.1779 \tag{862}$$

$$\sigma_t^2(1) = 0.6992 + 0.0041\, y_{t-1}^2 + 0.9000\, \sigma_{t-1}^2 \tag{863}$$
$$\sigma_t^2(2) = 0.5715 + 0.1156\, y_{t-1}^2 + 0.8784\, \sigma_{t-1}^2 \tag{864}$$

$$P = \begin{pmatrix} 0.8169 & 0.1831 \\ 0.2825 & 0.7175 \end{pmatrix} \tag{865}$$

LogLik $= -7441.63$; Converged: Yes.

### Plug-in Method

$$\mu_1 = 0.7002 \tag{866}$$
$$\mu_2 = 0.6944 \tag{867}$$

$$\sigma_t^2(1) = 0.7114 + 0.2689\, y_{t-1}^2 + 0.7137\, \sigma_{t-1}^2 \tag{868}$$
$$\sigma_t^2(2) = 0.6196 + 0.2778\, y_{t-1}^2 + 0.5905\, \sigma_{t-1}^2 \tag{869}$$

$$P = \begin{pmatrix} 0.7869 & 0.2131 \\ 0.1774 & 0.8226 \end{pmatrix} \tag{870}$$

LogLik $= -7692.15$; Converged: No.

**Quadrature Method**

$$\mu_1 = 0.4257 \tag{871}$$
$$\mu_2 = -0.1560 \tag{872}$$

$$\sigma_t^2(1) = 0.6956 + 0.0037\, y_{t-1}^2 + 0.9000\, \sigma_{t-1}^2 \tag{873}$$
$$\sigma_t^2(2) = 0.5464 + 0.1097\, y_{t-1}^2 + 0.8836\, \sigma_{t-1}^2 \tag{874}$$

$$P = \begin{pmatrix} 0.8764 & 0.1236 \\ 0.1505 & 0.8495 \end{pmatrix} \tag{875}$$

LogLik $= -7441.91$; Converged: Yes.

**Particle Filter Method**

$$\mu_1 = -0.0000 \tag{876}$$
$$\mu_2 = 0.0000 \tag{877}$$

$$\sigma_t^2(1) = 0.3000 + 0.0500\, y_{t-1}^2 + 0.8500\, \sigma_{t-1}^2 \tag{878}$$
$$\sigma_t^2(2) = 0.1000 + 0.0800\, y_{t-1}^2 + 0.7800\, \sigma_{t-1}^2 \tag{879}$$

$$P = \begin{pmatrix} 0.9500 & 0.0500 \\ 0.2000 & 0.8000 \end{pmatrix} \tag{880}$$

LogLik $= -8163.70$; Converged: No.

# 5. Foreign Exchange Market

Pivot (exogenous conditioning variable): `EURUSD=X`.

## 5.1. Asset: `GBPUSD=X`

### Baseline Specification

**Table 45:** GBPUSD=X — Baseline Specification

| Method | $c_1$ | $c_2$ | $\gamma_1$ | $\gamma_2$ | $\omega_1$ | $\alpha_1$ | $\beta_1$ | $\omega_2$ | $\alpha_2$ | $\beta_2$ | $p_{11}$ | $p_{22}$ | Conv. | LogLik |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Gray's | 0.0002 | -0.0002 | – | – | 0.0948 | 0.1729 | 0.8264 | 0.0001 | 0.0001 | 0.0001 | 0.7522 | 0.4401 | Yes | -51.61 |
| Plug-in | -0.0015 | -0.0001 | – | – | 0.0120 | 0.0905 | 0.9000 | 0.0001 | 0.0001 | 0.0009 | 0.7643 | 0.4823 | Yes | 63.64 |
| Quadrature | -0.0079 | -0.0002 | – | – | 0.0001 | 0.1098 | 0.8891 | 0.0001 | 0.0001 | 0.0001 | 0.7627 | 0.4483 | Yes | -76.25 |
| Particle Filter | 0.0000 | 0.0000 | – | – | 0.1000 | 0.1000 | 0.7000 | 0.2000 | 0.2000 | 0.6000 | 0.9000 | 0.9000 | Yes | -2116.49 |

Baseline spec: $\gamma_1 = \gamma_2 = 0$. Conv. = convergence status.

### Fitted Equations (Baseline):

### Gray's Method

$$\mu_1 = 0.0002 \tag{881}$$
$$\mu_2 = -0.0002 \tag{882}$$

$$\sigma_t^2(1) = 0.0948 + 0.1729\, y_{t-1}^2 + 0.8264\, \sigma_{t-1}^2 \tag{883}$$
$$\sigma_t^2(2) = 0.0001 + 0.0001\, y_{t-1}^2 + 0.0001\, \sigma_{t-1}^2 \tag{884}$$

$$P = \begin{pmatrix} 0.7522 & 0.2478 \\ 0.5599 & 0.4401 \end{pmatrix} \tag{885}$$

LogLik $= -51.61$; Converged: Yes.

### Plug-in Method

$$\mu_1 = -0.0015 \tag{886}$$
$$\mu_2 = -0.0001 \tag{887}$$

$$\sigma_t^2(1) = 0.0120 + 0.0905\, y_{t-1}^2 + 0.9000\, \sigma_{t-1}^2 \tag{888}$$
$$\sigma_t^2(2) = 0.0001 + 0.0001\, y_{t-1}^2 + 0.0009\, \sigma_{t-1}^2 \tag{889}$$

$$P = \begin{pmatrix} 0.7643 & 0.2357 \\ 0.5177 & 0.4823 \end{pmatrix} \tag{890}$$

LogLik $= 63.64$; Converged: Yes.

### Quadrature Method

$$\mu_1 = -0.0079 \tag{891}$$
$$\mu_2 = -0.0002 \tag{892}$$

$$\sigma_t^2(1) = 0.0001 + 0.1098\, y_{t-1}^2 + 0.8891\, \sigma_{t-1}^2 \tag{893}$$
$$\sigma_t^2(2) = 0.0001 + 0.0001\, y_{t-1}^2 + 0.0001\, \sigma_{t-1}^2 \tag{894}$$

$$P = \begin{pmatrix} 0.7627 & 0.2373 \\ 0.5517 & 0.4483 \end{pmatrix} \tag{895}$$

LogLik $= -76.25$; Converged: Yes.

### Particle Filter Method

$$\mu_1 = 0.0000 \tag{896}$$
$$\mu_2 = 0.0000 \tag{897}$$

$$\sigma_t^2(1) = 0.1000 + 0.1000\, y_{t-1}^2 + 0.7000\, \sigma_{t-1}^2 \tag{898}$$
$$\sigma_t^2(2) = 0.2000 + 0.2000\, y_{t-1}^2 + 0.6000\, \sigma_{t-1}^2 \tag{899}$$

$$P = \begin{pmatrix} 0.9000 & 0.1000 \\ 0.1000 & 0.9000 \end{pmatrix} \tag{900}$$

LogLik $= -2116.49$; Converged: Yes.

### Exogenous Specification

**Table 46:** GBPUSD=X — Exogenous Specification

| **Method** | $c_1$ | $c_2$ | $\gamma_1$ | $\gamma_2$ | $\omega_1$ | $\alpha_1$ | $\beta_1$ | $\omega_2$ | $\alpha_2$ | $\beta_2$ | $p_{11}$ | $p_{22}$ | Conv. | LogL |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Gray's | -0.0002 | -0.0019 | 0.0007 | 0.0401 | 0.0001 | 0.0001 | 0.0001 | 0.0987 | 0.1496 | 0.8503 | 0.4826 | 0.7470 | Yes | -54. |
| Plug-in | -0.0015 | -0.0001 | 0.0086 | 0.0005 | 0.0120 | 0.0905 | 0.9000 | 0.0001 | 0.0001 | 0.0009 | 0.7644 | 0.4822 | Yes | 63. |
| Quadrature | 0.0014 | 0.0001 | 0.0177 | 0.0007 | 0.0189 | 0.2002 | 0.7965 | 0.0001 | 0.0001 | 0.0001 | 0.7483 | 0.4411 | Yes | -78. |
| Particle Filter | 0.0000 | -0.0000 | 0.1000 | 0.1000 | 0.1000 | 0.1000 | 0.7000 | 0.2000 | 0.2000 | 0.6000 | 0.9000 | 0.9000 | Yes | -2120. |

Exogenous spec; pivot = `EURUSD=X`. Conv. = convergence status.

### Fitted Equations (Exogenous):

### Gray's Method

$$\mu_1 = -0.0002 + 0.0007\, x_t \tag{901}$$
$$\mu_2 = -0.0019 + 0.0401\, x_t \tag{902}$$

$$\sigma_t^2(1) = 0.0001 + 0.0001\, y_{t-1}^2 + 0.0001\, \sigma_{t-1}^2 \tag{903}$$
$$\sigma_t^2(2) = 0.0987 + 0.1496\, y_{t-1}^2 + 0.8503\, \sigma_{t-1}^2 \tag{904}$$

$$P = \begin{pmatrix} 0.4826 & 0.5174 \\ 0.2530 & 0.7470 \end{pmatrix} \tag{905}$$

LogLik $= -54.16$; Converged: Yes.

**Plug-in Method**

$$\mu_1 = -0.0015 + 0.0086\, x_t \tag{906}$$
$$\mu_2 = -0.0001 + 0.0005\, x_t \tag{907}$$

$$\sigma_t^2(1) = 0.0120 + 0.0905\, y_{t-1}^2 + 0.9000\, \sigma_{t-1}^2 \tag{908}$$
$$\sigma_t^2(2) = 0.0001 + 0.0001\, y_{t-1}^2 + 0.0009\, \sigma_{t-1}^2 \tag{909}$$

$$P = \begin{pmatrix} 0.7644 & 0.2356 \\ 0.5178 & 0.4822 \end{pmatrix} \tag{910}$$

LogLik $= 63.75$; Converged: Yes.

**Quadrature Method**

$$\mu_1 = 0.0014 + 0.0177\, x_t \tag{911}$$
$$\mu_2 = 0.0001 + 0.0007\, x_t \tag{912}$$

$$\sigma_t^2(1) = 0.0189 + 0.2002\, y_{t-1}^2 + 0.7965\, \sigma_{t-1}^2 \tag{913}$$
$$\sigma_t^2(2) = 0.0001 + 0.0001\, y_{t-1}^2 + 0.0001\, \sigma_{t-1}^2 \tag{914}$$

$$P = \begin{pmatrix} 0.7483 & 0.2517 \\ 0.5589 & 0.4411 \end{pmatrix} \tag{915}$$

LogLik $= -78.17$; Converged: Yes.

**Particle Filter Method**

$$\mu_1 = 0.0000 + 0.1000\, x_t \tag{916}$$
$$\mu_2 = -0.0000 + 0.1000\, x_t \tag{917}$$

$$\sigma_t^2(1) = 0.1000 + 0.1000\, y_{t-1}^2 + 0.7000\, \sigma_{t-1}^2 \tag{918}$$
$$\sigma_t^2(2) = 0.2000 + 0.2000\, y_{t-1}^2 + 0.6000\, \sigma_{t-1}^2 \tag{919}$$

$$P = \begin{pmatrix} 0.9000 & 0.1000 \\ 0.1000 & 0.9000 \end{pmatrix} \tag{920}$$

LogLik $= -2120.49$; Converged: Yes.

### 5.2. Asset: JPY=X

### Baseline Specification

**Table 47:** JPY=X — Baseline Specification

| Method | $c_1$ | $c_2$ | $\gamma_1$ | $\gamma_2$ | $\omega_1$ | $\alpha_1$ | $\beta_1$ | $\omega_2$ | $\alpha_2$ | $\beta_2$ | $p_{11}$ | $p_{22}$ | Conv. | LogLik |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Gray's | 0.0452 | 0.0000 | – | – | 0.0919 | 0.1881 | 0.8119 | 0.0001 | 0.0001 | 0.0001 | 0.7805 | 0.4230 | No | -64.72 |
| Plug-in | -0.0002 | 0.0195 | – | – | 0.0001 | 0.0001 | 0.0001 | 0.0362 | 0.3056 | 0.6944 | 0.4976 | 0.7621 | No | 71.36 |
| Quadrature | 0.0234 | 0.0004 | – | – | 0.0010 | 0.2142 | 0.7854 | 0.0001 | 0.0001 | 0.0001 | 0.7634 | 0.3640 | Yes | -78.14 |
| Particle Filter | 0.0000 | 0.0000 | – | – | 0.1000 | 0.1000 | 0.7000 | 0.2000 | 0.2000 | 0.6000 | 0.9000 | 0.9000 | No | -2108.91 |

Baseline spec: $\gamma_1 = \gamma_2 = 0$. Conv. = convergence status.

### Fitted Equations (Baseline):

### Gray's Method

$$\mu_1 = 0.0452 \tag{921}$$
$$\mu_2 = 0.0000 \tag{922}$$

$$\sigma_t^2(1) = 0.0919 + 0.1881\, y_{t-1}^2 + 0.8119\, \sigma_{t-1}^2 \tag{923}$$
$$\sigma_t^2(2) = 0.0001 + 0.0001\, y_{t-1}^2 + 0.0001\, \sigma_{t-1}^2 \tag{924}$$

$$P = \begin{pmatrix} 0.7805 & 0.2195 \\ 0.5770 & 0.4230 \end{pmatrix} \tag{925}$$

LogLik $= -64.72$; Converged: No.

### Plug-in Method

$$\mu_1 = -0.0002 \tag{926}$$
$$\mu_2 = 0.0195 \tag{927}$$

$$\sigma_t^2(1) = 0.0001 + 0.0001\, y_{t-1}^2 + 0.0001\, \sigma_{t-1}^2 \tag{928}$$
$$\sigma_t^2(2) = 0.0362 + 0.3056\, y_{t-1}^2 + 0.6944\, \sigma_{t-1}^2 \tag{929}$$

$$P = \begin{pmatrix} 0.4976 & 0.5024 \\ 0.2379 & 0.7621 \end{pmatrix} \tag{930}$$

LogLik $= 71.36$; Converged: No.

### Quadrature Method

$$\mu_1 = 0.0234 \tag{931}$$
$$\mu_2 = 0.0004 \tag{932}$$

$$\sigma_t^2(1) = 0.0010 + 0.2142\, y_{t-1}^2 + 0.7854\, \sigma_{t-1}^2 \tag{933}$$
$$\sigma_t^2(2) = 0.0001 + 0.0001\, y_{t-1}^2 + 0.0001\, \sigma_{t-1}^2 \tag{934}$$

$$P = \begin{pmatrix} 0.7634 & 0.2366 \\ 0.6360 & 0.3640 \end{pmatrix} \tag{935}$$

LogLik $= -78.14$; Converged: Yes.

### Particle Filter Method

$$\mu_1 = 0.0000 \tag{936}$$
$$\mu_2 = 0.0000 \tag{937}$$

$$\sigma_t^2(1) = 0.1000 + 0.1000\, y_{t-1}^2 + 0.7000\, \sigma_{t-1}^2 \tag{938}$$
$$\sigma_t^2(2) = 0.2000 + 0.2000\, y_{t-1}^2 + 0.6000\, \sigma_{t-1}^2 \tag{939}$$

$$P = \begin{pmatrix} 0.9000 & 0.1000 \\ 0.1000 & 0.9000 \end{pmatrix} \tag{940}$$

LogLik $= -2108.91$; Converged: No.

### Exogenous Specification

**Table 48:** JPY=X — Exogenous Specification

| **Method** | $c_1$ | $c_2$ | $\gamma_1$ | $\gamma_2$ | $\omega_1$ | $\alpha_1$ | $\beta_1$ | $\omega_2$ | $\alpha_2$ | $\beta_2$ | $p_{11}$ | $p_{22}$ | Conv. | LogL |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Gray's | -0.0001 | 0.0309 | 0.0025 | -0.0809 | 0.0001 | 0.0001 | 0.0001 | 0.0875 | 0.2330 | 0.7670 | 0.4197 | 0.7736 | No | -66. |
| Plug-in | 0.0001 | 0.0682 | 0.0024 | -0.0378 | 0.0001 | 0.0001 | 0.0001 | 0.0358 | 0.3058 | 0.6942 | 0.4986 | 0.7820 | No | 63. |
| Quadrature | 0.0000 | 0.0095 | 0.0014 | -0.1335 | 0.0001 | 0.0001 | 0.0001 | 0.0085 | 0.1774 | 0.8223 | 0.4331 | 0.7419 | No | -74. |
| Particle Filter | 0.0000 | 0.0000 | 0.1000 | 0.1000 | 0.1000 | 0.1000 | 0.7000 | 0.2000 | 0.2000 | 0.6000 | 0.9000 | 0.9000 | No | -2116. |

Exogenous spec; pivot = `EURUSD=X`. Conv. = convergence status.

### Fitted Equations (Exogenous):

### Gray's Method

$$\mu_1 = -0.0001 + 0.0025\, x_t \tag{941}$$
$$\mu_2 = 0.0309 - 0.0809\, x_t \tag{942}$$

$$\sigma_t^2(1) = 0.0001 + 0.0001\, y_{t-1}^2 + 0.0001\, \sigma_{t-1}^2 \tag{943}$$
$$\sigma_t^2(2) = 0.0875 + 0.2330\, y_{t-1}^2 + 0.7670\, \sigma_{t-1}^2 \tag{944}$$

$$P = \begin{pmatrix} 0.4197 & 0.5803 \\ 0.2264 & 0.7736 \end{pmatrix} \tag{945}$$

LogLik $= -66.56$; Converged: No.

**Plug-in Method**

$$\mu_1 = 0.0001 + 0.0024\, x_t \tag{946}$$
$$\mu_2 = 0.0682 - 0.0378\, x_t \tag{947}$$

$$\sigma_t^2(1) = 0.0001 + 0.0001\, y_{t-1}^2 + 0.0001\, \sigma_{t-1}^2 \tag{948}$$
$$\sigma_t^2(2) = 0.0358 + 0.3058\, y_{t-1}^2 + 0.6942\, \sigma_{t-1}^2 \tag{949}$$

$$P = \begin{pmatrix} 0.4986 & 0.5014 \\ 0.2180 & 0.7820 \end{pmatrix} \tag{950}$$

LogLik $= 63.15$; Converged: No.

**Quadrature Method**

$$\mu_1 = 0.0000 + 0.0014\, x_t \tag{951}$$
$$\mu_2 = 0.0095 - 0.1335\, x_t \tag{952}$$

$$\sigma_t^2(1) = 0.0001 + 0.0001\, y_{t-1}^2 + 0.0001\, \sigma_{t-1}^2 \tag{953}$$
$$\sigma_t^2(2) = 0.0085 + 0.1774\, y_{t-1}^2 + 0.8223\, \sigma_{t-1}^2 \tag{954}$$

$$P = \begin{pmatrix} 0.4331 & 0.5669 \\ 0.2581 & 0.7419 \end{pmatrix} \tag{955}$$

LogLik $= -74.43$; Converged: No.

**Particle Filter Method**

$$\mu_1 = 0.0000 + 0.1000\, x_t \tag{956}$$
$$\mu_2 = 0.0000 + 0.1000\, x_t \tag{957}$$

$$\sigma_t^2(1) = 0.1000 + 0.1000\, y_{t-1}^2 + 0.7000\, \sigma_{t-1}^2 \tag{958}$$
$$\sigma_t^2(2) = 0.2000 + 0.2000\, y_{t-1}^2 + 0.6000\, \sigma_{t-1}^2 \tag{959}$$

$$P = \begin{pmatrix} 0.9000 & 0.1000 \\ 0.1000 & 0.9000 \end{pmatrix} \tag{960}$$

LogLik $= -2116.35$; Converged: No.

### 5.3. Asset: `AUDUSD=X`

### Baseline Specification

**Table 49:** AUDUSD=X — Baseline Specification

| **Method** | $c_1$ | $c_2$ | $\gamma_1$ | $\gamma_2$ | $\omega_1$ | $\alpha_1$ | $\beta_1$ | $\omega_2$ | $\alpha_2$ | $\beta_2$ | $p_{11}$ | $p_{22}$ | Conv. | LogLik |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Gray's | -0.0004 | 0.0125 | – | – | 0.0001 | 0.0001 | 0.0001 | 0.1190 | 0.1000 | 0.9000 | 0.4328 | 0.7540 | Yes | -510.71 |
| Plug-in | -0.0000 | -0.0158 | – | – | 0.0001 | 0.0001 | 0.0001 | 0.0174 | 0.0866 | 0.9000 | 0.4713 | 0.7732 | Yes | -376.14 |
| Quadrature | -0.0126 | 0.0003 | – | – | 0.0350 | 0.1698 | 0.8276 | 0.0001 | 0.0001 | 0.0001 | 0.7515 | 0.4062 | Yes | -540.09 |
| Particle Filter | 0.0000 | -0.0000 | – | – | 0.1000 | 0.1000 | 0.7000 | 0.2000 | 0.2000 | 0.6000 | 0.9000 | 0.9000 | Yes | -2384.22 |

Baseline spec: $\gamma_1 = \gamma_2 = 0$. Conv. = convergence status.

### Fitted Equations (Baseline):

### Gray's Method

$$\mu_1 = -0.0004 \tag{961}$$
$$\mu_2 = 0.0125 \tag{962}$$

$$\sigma_t^2(1) = 0.0001 + 0.0001\, y_{t-1}^2 + 0.0001\, \sigma_{t-1}^2 \tag{963}$$
$$\sigma_t^2(2) = 0.1190 + 0.1000\, y_{t-1}^2 + 0.9000\, \sigma_{t-1}^2 \tag{964}$$

$$P = \begin{pmatrix} 0.4328 & 0.5672 \\ 0.2460 & 0.7540 \end{pmatrix} \tag{965}$$

LogLik $= -510.71$; Converged: Yes.

### Plug-in Method

$$\mu_1 = -0.0000 \tag{966}$$
$$\mu_2 = -0.0158 \tag{967}$$

$$\sigma_t^2(1) = 0.0001 + 0.0001\, y_{t-1}^2 + 0.0001\, \sigma_{t-1}^2 \tag{968}$$
$$\sigma_t^2(2) = 0.0174 + 0.0866\, y_{t-1}^2 + 0.9000\, \sigma_{t-1}^2 \tag{969}$$

$$P = \begin{pmatrix} 0.4713 & 0.5287 \\ 0.2268 & 0.7732 \end{pmatrix} \tag{970}$$

LogLik $= -376.14$; Converged: Yes.

### Quadrature Method

$$\mu_1 = -0.0126 \tag{971}$$
$$\mu_2 = 0.0003 \tag{972}$$

$$\sigma_t^2(1) = 0.0350 + 0.1698\, y_{t-1}^2 + 0.8276\, \sigma_{t-1}^2 \tag{973}$$
$$\sigma_t^2(2) = 0.0001 + 0.0001\, y_{t-1}^2 + 0.0001\, \sigma_{t-1}^2 \tag{974}$$

$$P = \begin{pmatrix} 0.7515 & 0.2485 \\ 0.5938 & 0.4062 \end{pmatrix} \tag{975}$$

LogLik $= -540.09$; Converged: Yes.

### Particle Filter Method

$$\mu_1 = 0.0000 \tag{976}$$
$$\mu_2 = -0.0000 \tag{977}$$

$$\sigma_t^2(1) = 0.1000 + 0.1000\, y_{t-1}^2 + 0.7000\, \sigma_{t-1}^2 \tag{978}$$
$$\sigma_t^2(2) = 0.2000 + 0.2000\, y_{t-1}^2 + 0.6000\, \sigma_{t-1}^2 \tag{979}$$

$$P = \begin{pmatrix} 0.9000 & 0.1000 \\ 0.1000 & 0.9000 \end{pmatrix} \tag{980}$$

LogLik $= -2384.22$; Converged: Yes.

### Exogenous Specification

**Table 50:** AUDUSD=X — Exogenous Specification

| Method | $c_1$ | $c_2$ | $\gamma_1$ | $\gamma_2$ | $\omega_1$ | $\alpha_1$ | $\beta_1$ | $\omega_2$ | $\alpha_2$ | $\beta_2$ | $p_{11}$ | $p_{22}$ | Conv. | Log |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Gray's | -0.0068 | 0.0002 | -0.0928 | -0.0012 | 0.1200 | 0.0997 | 0.9000 | 0.0001 | 0.0001 | 0.0001 | 0.7479 | 0.4420 | Yes | -509 |
| Plug-in | -0.0163 | -0.0000 | -0.0526 | -0.0007 | 0.0173 | 0.0869 | 0.9000 | 0.0001 | 0.0001 | 0.0001 | 0.7731 | 0.4712 | Yes | -37 |
| Quadrature | -0.0015 | 0.0000 | -0.0135 | -0.0012 | 0.0226 | 0.1004 | 0.8987 | 0.0001 | 0.0001 | 0.0001 | 0.7533 | 0.4427 | Yes | -53 |
| Particle Filter | 0.0000 | -0.0000 | 0.1000 | 0.1000 | 0.1000 | 0.1000 | 0.7000 | 0.2000 | 0.2000 | 0.6000 | 0.9000 | 0.9000 | Yes | -239 |

Exogenous spec; pivot = `EURUSD=X`. Conv. = convergence status.

### Fitted Equations (Exogenous):

### Gray's Method

$$\mu_1 = -0.0068 - 0.0928\, x_t \tag{981}$$
$$\mu_2 = 0.0002 - 0.0012\, x_t \tag{982}$$

$$\sigma_t^2(1) = 0.1200 + 0.0997\, y_{t-1}^2 + 0.9000\, \sigma_{t-1}^2 \tag{983}$$
$$\sigma_t^2(2) = 0.0001 + 0.0001\, y_{t-1}^2 + 0.0001\, \sigma_{t-1}^2 \tag{984}$$

$$P = \begin{pmatrix} 0.7479 & 0.2521 \\ 0.5580 & 0.4420 \end{pmatrix} \tag{985}$$

LogLik $= -509.76$; Converged: Yes.

**Plug-in Method**

$$\mu_1 = -0.0163 - 0.0526\, x_t \tag{986}$$
$$\mu_2 = -0.0000 - 0.0007\, x_t \tag{987}$$

$$\sigma_t^2(1) = 0.0173 + 0.0869\, y_{t-1}^2 + 0.9000\, \sigma_{t-1}^2 \tag{988}$$
$$\sigma_t^2(2) = 0.0001 + 0.0001\, y_{t-1}^2 + 0.0001\, \sigma_{t-1}^2 \tag{989}$$

$$P = \begin{pmatrix} 0.7731 & 0.2269 \\ 0.5288 & 0.4712 \end{pmatrix} \tag{990}$$

LogLik $= -375.10$; Converged: Yes.

**Quadrature Method**

$$\mu_1 = -0.0015 - 0.0135\, x_t \tag{991}$$
$$\mu_2 = 0.0000 - 0.0012\, x_t \tag{992}$$

$$\sigma_t^2(1) = 0.0226 + 0.1004\, y_{t-1}^2 + 0.8987\, \sigma_{t-1}^2 \tag{993}$$
$$\sigma_t^2(2) = 0.0001 + 0.0001\, y_{t-1}^2 + 0.0001\, \sigma_{t-1}^2 \tag{994}$$

$$P = \begin{pmatrix} 0.7533 & 0.2467 \\ 0.5573 & 0.4427 \end{pmatrix} \tag{995}$$

LogLik $= -532.90$; Converged: Yes.

**Particle Filter Method**

$$\mu_1 = 0.0000 + 0.1000\, x_t \tag{996}$$
$$\mu_2 = -0.0000 + 0.1000\, x_t \tag{997}$$

$$\sigma_t^2(1) = 0.1000 + 0.1000\, y_{t-1}^2 + 0.7000\, \sigma_{t-1}^2 \tag{998}$$
$$\sigma_t^2(2) = 0.2000 + 0.2000\, y_{t-1}^2 + 0.6000\, \sigma_{t-1}^2 \tag{999}$$

$$P = \begin{pmatrix} 0.9000 & 0.1000 \\ 0.1000 & 0.9000 \end{pmatrix} \tag{1000}$$

LogLik $= -2390.26$; Converged: Yes.

## 5.4. Asset: CHF=X

### Baseline Specification

**Table 51:** CHF=X — Baseline Specification

| Method | $c_1$ | $c_2$ | $\gamma_1$ | $\gamma_2$ | $\omega_1$ | $\alpha_1$ | $\beta_1$ | $\omega_2$ | $\alpha_2$ | $\beta_2$ | $p_{11}$ | $p_{22}$ | Conv. | LogLik |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Gray's | 0.0094 | 0.0003 | – | – | 0.0687 | 0.2163 | 0.7800 | 0.0001 | 0.0001 | 0.0001 | 0.7638 | 0.4721 | Yes | 302.30 |
| Plug-in | -0.0005 | 0.0002 | – | – | 0.0244 | 0.1622 | 0.7876 | 0.0001 | 0.0001 | 0.0001 | 0.7655 | 0.4830 | Yes | 372.12 |
| Quadrature | 0.0008 | 0.0002 | – | – | 0.0001 | 0.1670 | 0.8114 | 0.0001 | 0.0001 | 0.0001 | 0.7539 | 0.4700 | Yes | 286.83 |
| Particle Filter | 0.0000 | 0.0000 | – | – | 0.1000 | 0.1000 | 0.7000 | 0.2000 | 0.2000 | 0.6000 | 0.9000 | 0.9000 | No | -1913.41 |

Baseline spec: $\gamma_1 = \gamma_2 = 0$. Conv. = convergence status.

### Fitted Equations (Baseline):

### Gray's Method

$$\mu_1 = 0.0094 \tag{1001}$$
$$\mu_2 = 0.0003 \tag{1002}$$

$$\sigma_t^2(1) = 0.0687 + 0.2163\, y_{t-1}^2 + 0.7800\, \sigma_{t-1}^2 \tag{1003}$$
$$\sigma_t^2(2) = 0.0001 + 0.0001\, y_{t-1}^2 + 0.0001\, \sigma_{t-1}^2 \tag{1004}$$

$$P = \begin{pmatrix} 0.7638 & 0.2362 \\ 0.5279 & 0.4721 \end{pmatrix} \tag{1005}$$

LogLik = 302.30; Converged: Yes.

### Plug-in Method

$$\mu_1 = -0.0005 \tag{1006}$$
$$\mu_2 = 0.0002 \tag{1007}$$

$$\sigma_t^2(1) = 0.0244 + 0.1622\, y_{t-1}^2 + 0.7876\, \sigma_{t-1}^2 \tag{1008}$$
$$\sigma_t^2(2) = 0.0001 + 0.0001\, y_{t-1}^2 + 0.0001\, \sigma_{t-1}^2 \tag{1009}$$

$$P = \begin{pmatrix} 0.7655 & 0.2345 \\ 0.5170 & 0.4830 \end{pmatrix} \tag{1010}$$

LogLik = 372.12; Converged: Yes.

### Quadrature Method

$$\mu_1 = 0.0008 \tag{1011}$$
$$\mu_2 = 0.0002 \tag{1012}$$

$$\sigma_t^2(1) = 0.0001 + 0.1670\, y_{t-1}^2 + 0.8114\, \sigma_{t-1}^2 \tag{1013}$$
$$\sigma_t^2(2) = 0.0001 + 0.0001\, y_{t-1}^2 + 0.0001\, \sigma_{t-1}^2 \tag{1014}$$

$$P = \begin{pmatrix} 0.7539 & 0.2461 \\ 0.5300 & 0.4700 \end{pmatrix} \tag{1015}$$

LogLik $= 286.83$; Converged: Yes.

### Particle Filter Method

$$\mu_1 = 0.0000 \tag{1016}$$
$$\mu_2 = 0.0000 \tag{1017}$$

$$\sigma_t^2(1) = 0.1000 + 0.1000\, y_{t-1}^2 + 0.7000\, \sigma_{t-1}^2 \tag{1018}$$
$$\sigma_t^2(2) = 0.2000 + 0.2000\, y_{t-1}^2 + 0.6000\, \sigma_{t-1}^2 \tag{1019}$$

$$P = \begin{pmatrix} 0.9000 & 0.1000 \\ 0.1000 & 0.9000 \end{pmatrix} \tag{1020}$$

LogLik $= -1913.41$; Converged: No.

### Exogenous Specification

**Table 52:** CHF=X — Exogenous Specification

| **Method** | $c_1$ | $c_2$ | $\gamma_1$ | $\gamma_2$ | $\omega_1$ | $\alpha_1$ | $\beta_1$ | $\omega_2$ | $\alpha_2$ | $\beta_2$ | $p_{11}$ | $p_{22}$ | Conv. | Log |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Gray's | -0.0039 | 0.0000 | 0.0172 | 0.0007 | 0.0715 | 0.2024 | 0.7974 | 0.0001 | 0.0001 | 0.0001 | 0.7560 | 0.4638 | Yes | 303 |
| Plug-in | 0.0002 | -0.0005 | 0.0004 | -0.0164 | 0.0001 | 0.0001 | 0.0001 | 0.0244 | 0.1615 | 0.7882 | 0.4831 | 0.7654 | Yes | 372 |
| Quadrature | 0.0002 | 0.0006 | 0.0006 | -0.0223 | 0.0001 | 0.0001 | 0.0001 | 0.0001 | 0.1667 | 0.8117 | 0.4697 | 0.7537 | Yes | 287 |
| Particle Filter | -0.0000 | 0.0000 | 0.1000 | 0.1000 | 0.1000 | 0.1000 | 0.7000 | 0.2000 | 0.2000 | 0.6000 | 0.9000 | 0.9000 | Yes | -1921 |

Exogenous spec; pivot = `EURUSD=X`. Conv. = convergence status.

### Fitted Equations (Exogenous):

### Gray's Method

$$\mu_1 = -0.0039 + 0.0172\, x_t \tag{1021}$$
$$\mu_2 = 0.0000 + 0.0007\, x_t \tag{1022}$$

$$\sigma_t^2(1) = 0.0715 + 0.2024\, y_{t-1}^2 + 0.7974\, \sigma_{t-1}^2 \tag{1023}$$
$$\sigma_t^2(2) = 0.0001 + 0.0001\, y_{t-1}^2 + 0.0001\, \sigma_{t-1}^2 \tag{1024}$$

$$P = \begin{pmatrix} 0.7560 & 0.2440 \\ 0.5362 & 0.4638 \end{pmatrix} \tag{1025}$$

LogLik = 303.39; Converged: Yes.

**Plug-in Method**

$$\mu_1 = 0.0002 + 0.0004\, x_t \tag{1026}$$
$$\mu_2 = -0.0005 - 0.0164\, x_t \tag{1027}$$

$$\sigma_t^2(1) = 0.0001 + 0.0001\, y_{t-1}^2 + 0.0001\, \sigma_{t-1}^2 \tag{1028}$$
$$\sigma_t^2(2) = 0.0244 + 0.1615\, y_{t-1}^2 + 0.7882\, \sigma_{t-1}^2 \tag{1029}$$

$$P = \begin{pmatrix} 0.4831 & 0.5169 \\ 0.2346 & 0.7654 \end{pmatrix} \tag{1030}$$

LogLik = 372.35; Converged: Yes.

**Quadrature Method**

$$\mu_1 = 0.0002 + 0.0006\, x_t \tag{1031}$$
$$\mu_2 = 0.0006 - 0.0223\, x_t \tag{1032}$$

$$\sigma_t^2(1) = 0.0001 + 0.0001\, y_{t-1}^2 + 0.0001\, \sigma_{t-1}^2 \tag{1033}$$
$$\sigma_t^2(2) = 0.0001 + 0.1667\, y_{t-1}^2 + 0.8117\, \sigma_{t-1}^2 \tag{1034}$$

$$P = \begin{pmatrix} 0.4697 & 0.5303 \\ 0.2463 & 0.7537 \end{pmatrix} \tag{1035}$$

LogLik = 287.22; Converged: Yes.

**Particle Filter Method**

$$\mu_1 = -0.0000 + 0.1000\, x_t \tag{1036}$$
$$\mu_2 = 0.0000 + 0.1000\, x_t \tag{1037}$$

$$\sigma_t^2(1) = 0.1000 + 0.1000\, y_{t-1}^2 + 0.7000\, \sigma_{t-1}^2 \tag{1038}$$
$$\sigma_t^2(2) = 0.2000 + 0.2000\, y_{t-1}^2 + 0.6000\, \sigma_{t-1}^2 \tag{1039}$$

$$P = \begin{pmatrix} 0.9000 & 0.1000 \\ 0.1000 & 0.9000 \end{pmatrix} \tag{1040}$$

LogLik $= -1921.41$; Converged: Yes.

## 5.5. Asset: `EURGBP=X`

### Baseline Specification

**Table 53:** EURGBP=X — Baseline Specification

| Method | $c_1$ | $c_2$ | $\gamma_1$ | $\gamma_2$ | $\omega_1$ | $\alpha_1$ | $\beta_1$ | $\omega_2$ | $\alpha_2$ | $\beta_2$ | $p_{11}$ | $p_{22}$ | Conv. | LogLik |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Gray's | 0.0046 | 0.0003 | – | – | 0.2237 | 0.0801 | 0.0036 | 0.0001 | 0.0001 | 0.0001 | 0.7494 | 0.4941 | Yes | 274.11 |
| Plug-in | 0.0114 | 0.0002 | – | – | 0.0057 | 0.1000 | 0.9000 | 0.0001 | 0.0001 | 0.0001 | 0.8046 | 0.6121 | Yes | 557.11 |
| Quadrature | 0.0003 | 0.0044 | – | – | 0.0001 | 0.0001 | 0.0001 | 0.2239 | 0.0797 | 0.0001 | 0.4959 | 0.7477 | Yes | 252.26 |
| Particle Filter | -0.0000 | 0.0000 | – | – | 0.1000 | 0.1000 | 0.7000 | 0.2000 | 0.2000 | 0.6000 | 0.9000 | 0.9000 | Yes | -1878.74 |

Baseline spec: $\gamma_1 = \gamma_2 = 0$. Conv. = convergence status.

### Fitted Equations (Baseline):

### Gray's Method

$$\mu_1 = 0.0046 \tag{1041}$$
$$\mu_2 = 0.0003 \tag{1042}$$

$$\sigma_t^2(1) = 0.2237 + 0.0801\, y_{t-1}^2 + 0.0036\, \sigma_{t-1}^2 \tag{1043}$$
$$\sigma_t^2(2) = 0.0001 + 0.0001\, y_{t-1}^2 + 0.0001\, \sigma_{t-1}^2 \tag{1044}$$

$$P = \begin{pmatrix} 0.7494 & 0.2506 \\ 0.5059 & 0.4941 \end{pmatrix} \tag{1045}$$

LogLik = 274.11; Converged: Yes.

### Plug-in Method

$$\mu_1 = 0.0114 \tag{1046}$$
$$\mu_2 = 0.0002 \tag{1047}$$

$$\sigma_t^2(1) = 0.0057 + 0.1000\, y_{t-1}^2 + 0.9000\, \sigma_{t-1}^2 \tag{1048}$$
$$\sigma_t^2(2) = 0.0001 + 0.0001\, y_{t-1}^2 + 0.0001\, \sigma_{t-1}^2 \tag{1049}$$

$$P = \begin{pmatrix} 0.8046 & 0.1954 \\ 0.3879 & 0.6121 \end{pmatrix} \tag{1050}$$

LogLik = 557.11; Converged: Yes.

### Quadrature Method

$$\mu_1 = 0.0003 \tag{1051}$$
$$\mu_2 = 0.0044 \tag{1052}$$

$$\sigma_t^2(1) = 0.0001 + 0.0001\, y_{t-1}^2 + 0.0001\, \sigma_{t-1}^2 \tag{1053}$$
$$\sigma_t^2(2) = 0.2239 + 0.0797\, y_{t-1}^2 + 0.0001\, \sigma_{t-1}^2 \tag{1054}$$

$$P = \begin{pmatrix} 0.4959 & 0.5041 \\ 0.2523 & 0.7477 \end{pmatrix} \tag{1055}$$

LogLik = 252.26; Converged: Yes.

### Particle Filter Method

$$\mu_1 = -0.0000 \tag{1056}$$
$$\mu_2 = 0.0000 \tag{1057}$$

$$\sigma_t^2(1) = 0.1000 + 0.1000\, y_{t-1}^2 + 0.7000\, \sigma_{t-1}^2 \tag{1058}$$
$$\sigma_t^2(2) = 0.2000 + 0.2000\, y_{t-1}^2 + 0.6000\, \sigma_{t-1}^2 \tag{1059}$$

$$P = \begin{pmatrix} 0.9000 & 0.1000 \\ 0.1000 & 0.9000 \end{pmatrix} \tag{1060}$$

LogLik = −1878.74; Converged: Yes.

### Exogenous Specification

**Table 54:** EURGBP=X — Exogenous Specification

| **Method** | $c_1$ | $c_2$ | $\gamma_1$ | $\gamma_2$ | $\omega_1$ | $\alpha_1$ | $\beta_1$ | $\omega_2$ | $\alpha_2$ | $\beta_2$ | $p_{11}$ | $p_{22}$ | Conv. | Log |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Gray's | 0.0045 | 0.0003 | -0.0208 | -0.0002 | 0.2235 | 0.0805 | 0.0035 | 0.0001 | 0.0001 | 0.0001 | 0.7494 | 0.4944 | Yes | 274 |
| Plug-in | 0.0003 | -0.0408 | -0.0009 | 0.0103 | 0.0001 | 0.0001 | 0.0001 | 0.0190 | 0.2387 | 0.7612 | 0.3547 | 0.7669 | Yes | 498 |
| Quadrature | 0.0046 | 0.0003 | -0.0210 | -0.0002 | 0.2237 | 0.0797 | 0.0001 | 0.0001 | 0.0001 | 0.0001 | 0.7479 | 0.4958 | Yes | 252 |
| Particle Filter | 0.0000 | -0.0000 | 0.1000 | 0.1000 | 0.1000 | 0.1000 | 0.7000 | 0.2000 | 0.2000 | 0.6000 | 0.9000 | 0.9000 | Yes | -1883 |

Exogenous spec; pivot = `EURUSD=X`. Conv. = convergence status.

### Fitted Equations (Exogenous):

### Gray's Method

$$\mu_1 = 0.0045 - 0.0208\, x_t \tag{1061}$$
$$\mu_2 = 0.0003 - 0.0002\, x_t \tag{1062}$$

$$\sigma_t^2(1) = 0.2235 + 0.0805\, y_{t-1}^2 + 0.0035\, \sigma_{t-1}^2 \tag{1063}$$
$$\sigma_t^2(2) = 0.0001 + 0.0001\, y_{t-1}^2 + 0.0001\, \sigma_{t-1}^2 \tag{1064}$$

$$P = \begin{pmatrix} 0.7494 & 0.2506 \\ 0.5056 & 0.4944 \end{pmatrix} \tag{1065}$$

LogLik = 274.41; Converged: Yes.

**Plug-in Method**

$$\mu_1 = 0.0003 - 0.0009\, x_t \tag{1066}$$
$$\mu_2 = -0.0408 + 0.0103\, x_t \tag{1067}$$

$$\sigma_t^2(1) = 0.0001 + 0.0001\, y_{t-1}^2 + 0.0001\, \sigma_{t-1}^2 \tag{1068}$$
$$\sigma_t^2(2) = 0.0190 + 0.2387\, y_{t-1}^2 + 0.7612\, \sigma_{t-1}^2 \tag{1069}$$

$$P = \begin{pmatrix} 0.3547 & 0.6453 \\ 0.2331 & 0.7669 \end{pmatrix} \tag{1070}$$

LogLik = 498.56; Converged: Yes.

**Quadrature Method**

$$\mu_1 = 0.0046 - 0.0210\, x_t \tag{1071}$$
$$\mu_2 = 0.0003 - 0.0002\, x_t \tag{1072}$$

$$\sigma_t^2(1) = 0.2237 + 0.0797\, y_{t-1}^2 + 0.0001\, \sigma_{t-1}^2 \tag{1073}$$
$$\sigma_t^2(2) = 0.0001 + 0.0001\, y_{t-1}^2 + 0.0001\, \sigma_{t-1}^2 \tag{1074}$$

$$P = \begin{pmatrix} 0.7479 & 0.2521 \\ 0.5042 & 0.4958 \end{pmatrix} \tag{1075}$$

LogLik = 252.55; Converged: Yes.

**Particle Filter Method**

$$\mu_1 = 0.0000 + 0.1000\, x_t \tag{1076}$$
$$\mu_2 = -0.0000 + 0.1000\, x_t \tag{1077}$$

$$\sigma_t^2(1) = 0.1000 + 0.1000\, y_{t-1}^2 + 0.7000\, \sigma_{t-1}^2 \tag{1078}$$
$$\sigma_t^2(2) = 0.2000 + 0.2000\, y_{t-1}^2 + 0.6000\, \sigma_{t-1}^2 \tag{1079}$$

$$P = \begin{pmatrix} 0.9000 & 0.1000 \\ 0.1000 & 0.9000 \end{pmatrix} \tag{1080}$$

LogLik $= -1883.37$; Converged: Yes.

### 5.6. Asset: `EURUSD=X`

### Baseline Specification

**Table 55:** EURUSD=X — Baseline Specification

| Method | $c_1$ | $c_2$ | $\gamma_1$ | $\gamma_2$ | $\omega_1$ | $\alpha_1$ | $\beta_1$ | $\omega_2$ | $\alpha_2$ | $\beta_2$ | $p_{11}$ | $p_{22}$ | Conv. | LogLik |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Gray's | -0.0041 | -0.0005 | – | – | 0.0697 | 0.0991 | 0.9000 | 0.0001 | 0.0001 | 0.0001 | 0.7488 | 0.4934 | Yes | 308.28 |
| Plug-in | -0.0085 | -0.0001 | – | – | 0.0086 | 0.0902 | 0.9000 | 0.0001 | 0.0001 | 0.0001 | 0.7669 | 0.4815 | Yes | 399.99 |
| Quadrature | -0.0044 | -0.0001 | – | – | 0.0001 | 0.1158 | 0.8263 | 0.0001 | 0.0001 | 0.0001 | 0.7519 | 0.4557 | Yes | 291.55 |
| Particle Filter | -0.0000 | -0.0000 | – | – | 0.1000 | 0.1000 | 0.7000 | 0.2000 | 0.2000 | 0.6000 | 0.9000 | 0.9000 | Yes | -1893.04 |

Baseline spec: $\gamma_1 = \gamma_2 = 0$. Conv. = convergence status.

### Fitted Equations (Baseline):

### Gray's Method

$$\mu_1 = -0.0041 \tag{1081}$$
$$\mu_2 = -0.0005 \tag{1082}$$

$$\sigma_t^2(1) = 0.0697 + 0.0991\, y_{t-1}^2 + 0.9000\, \sigma_{t-1}^2 \tag{1083}$$
$$\sigma_t^2(2) = 0.0001 + 0.0001\, y_{t-1}^2 + 0.0001\, \sigma_{t-1}^2 \tag{1084}$$

$$P = \begin{pmatrix} 0.7488 & 0.2512 \\ 0.5066 & 0.4934 \end{pmatrix} \tag{1085}$$

LogLik = 308.28; Converged: Yes.

### Plug-in Method

$$\mu_1 = -0.0085 \tag{1086}$$
$$\mu_2 = -0.0001 \tag{1087}$$

$$\sigma_t^2(1) = 0.0086 + 0.0902\, y_{t-1}^2 + 0.9000\, \sigma_{t-1}^2 \tag{1088}$$
$$\sigma_t^2(2) = 0.0001 + 0.0001\, y_{t-1}^2 + 0.0001\, \sigma_{t-1}^2 \tag{1089}$$

$$P = \begin{pmatrix} 0.7669 & 0.2331 \\ 0.5185 & 0.4815 \end{pmatrix} \tag{1090}$$

LogLik = 399.99; Converged: Yes.

**Quadrature Method**

$$\mu_1 = -0.0044 \tag{1091}$$

$$\mu_2 = -0.0001 \tag{1092}$$

$$\sigma_t^2(1) = 0.0001 + 0.1158\, y_{t-1}^2 + 0.8263\, \sigma_{t-1}^2 \tag{1093}$$

$$\sigma_t^2(2) = 0.0001 + 0.0001\, y_{t-1}^2 + 0.0001\, \sigma_{t-1}^2 \tag{1094}$$

$$P = \begin{pmatrix} 0.7519 & 0.2481 \\ 0.5443 & 0.4557 \end{pmatrix} \tag{1095}$$

LogLik $= 291.55$; Converged: Yes.

**Particle Filter Method**

$$\mu_1 = -0.0000 \tag{1096}$$

$$\mu_2 = -0.0000 \tag{1097}$$

$$\sigma_t^2(1) = 0.1000 + 0.1000\, y_{t-1}^2 + 0.7000\, \sigma_{t-1}^2 \tag{1098}$$

$$\sigma_t^2(2) = 0.2000 + 0.2000\, y_{t-1}^2 + 0.6000\, \sigma_{t-1}^2 \tag{1099}$$

$$P = \begin{pmatrix} 0.9000 & 0.1000 \\ 0.1000 & 0.9000 \end{pmatrix} \tag{1100}$$

LogLik $= -1893.04$; Converged: Yes.